Gradient Descent Program: Universality and Landscape

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Abstract

How efficiently can gradient descent execute an algorithm, and when does a higher-dimensional parametrization help it optimize? We show that every bounded-fan-in Boolean circuit compiles into a strongly convex objective whose unique minimizer encodes the circuit's gate values. Starting from an encoded Boolean state, gradient descent reaches this minimizer in a number of steps proportional to the circuit depth. A reduction from gradient descent on a single fixed strongly convex objective shows that these dynamics can simulate polynomial-space computation, yielding a surprising consequence: despite the minimizer that the dynamic converges to is known, determining the T-th iterate is PSPACE-complete when T is given in binary. This is true both in exact arithmetic and for a specified finite-precision implementation. We also provide a more efficient, clock-free construction that evaluates finite-delay Boolean circuits by projected gradient descent with linear total work. Having encoded circuit execution in gradient dynamics, we next examine how overparameterization shapes the resulting optimization landscape. Specifically, we lift Boolean circuits to bounded-trace Gram matrices, embedding discrete circuit states in a larger, continuous space. This connects circuit execution to semidefinite relaxations, a standard approach that brings tools from continuous optimization to bear on discrete problems. For these relaxations, we design fixed loss functions that let ordinary gradient descent either find an approximately feasible solution or certify infeasibility in polynomially many (exact-arithmetic) steps. We also show that, with enough overparameterization, every point whose loss exceeds the best relaxation value by more than a chosen tolerance has many negative curvature directions for descent to escape. Applications include matrix factorization and completion, sensing, clustering, cuts, constraint satisfaction, moment hierarchies, and neural parameterizations.

1 Introduction

Gradient methods are widely used in machine learning and continuous optimization, supported by hardware and software that are increasingly organized around backpropagation through neural models (Jouppi et al., 2023; Baydin et al., 2018). Despite this infrastructure, their use in combinatorial optimization remains less direct, because discrete choices obstruct differentiation and a continuous surrogate must still allow feasible solutions of controlled quality to be recovered (Bengio et al., 2021). Promising examples are scattered across routing, satisfiability, integer programming, and graph optimization (Kool et al., 2019; Selsam et al., 2019; Gasse et al., 2019; Khalil et al., 2017).

These examples suggest three complementary directions. First, networks learn solution constructions or search decisions from labeled instances, solver traces, or rewards (Kool et al., 2019; Selsam

et al., 2019; Selsam and Bjørner, 2019; Gasse et al., 2019). Second, gradient methods optimize continuous representations, including probability distributions and *relaxations* that enlarge the discrete feasible set (Karalias and Loukas, 2020; Kyrillidis et al., 2021). An important class of relaxations consists of semidefinite programs (SDPs), which optimize a linear objective over positive semidefinite (PSD) matrices subject to linear constraints. Such a matrix is the Gram matrix of a set of vectors and records their inner products. For a maximization problem, the relaxed optimum upper-bounds the discrete optimum, and a rounding procedure converts the matrix solution into a discrete assignment. An approximation guarantee then bounds the returned score relative to the optimum, in expectation when the rounding is randomized (Goemans and Williamson, 1995). SATNet and OptGNN incorporate these relaxations into trainable models (Wang et al., 2019; Yau et al., 2024). Third, neural algorithmic models can learn to reproduce the intermediate steps of known algorithms (Veličković et al., 2020). Together, these directions motivate the following question.

How efficiently can gradient descent execute an algorithm, and when can richer representations recover the guarantees of continuous relaxations?

This question separates an *execution picture* from a *landscape picture*. In the execution picture, the variables encode working memory and descent implements algorithmic updates, so the trajectory reproduces the computation with controlled memory, work, and precision. In the landscape picture, the variables represent candidate solutions, and the question is whether the objective's geometry excludes poor local optima and supports a guarantee after rounding. Figure 1 contrasts these two roles of gradient descent.

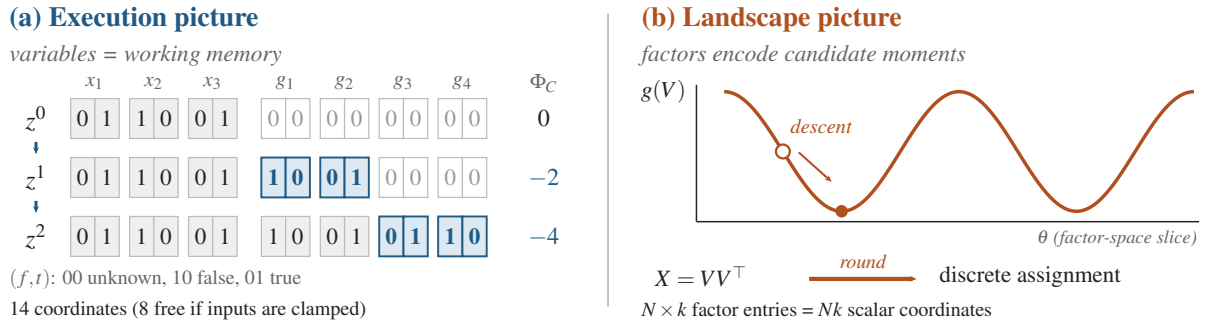


Figure 1: Execution and landscape. (a) Seven wires use 14 false/true rails; two gates resolve per step. (b) The convex matrix loss X_{12}^2 becomes the nonconvex factor slice $\cos^2\theta$, where θ is the angle between two planar unit rows. Both wells yield the same Gram matrix. Rounding is separate.

Our contributions. We address these questions through the following results.

1. We demonstrate the expressive power of gradient descent by compiling Boolean circuits into gradient-descent computations using memory and total work linear in the circuit size, provided unchanged updates are cached (Theorem 2.1). We also give a strongly convex construction in which the unique minimizer encodes the circuit's output. Starting from the prescribed Boolean initialization, gradient descent reaches this minimizer in at most as many exact-arithmetic steps as the depth of the circuit.

2. We extend the compiler from finite circuit evaluation to computations that repeatedly update and reuse their state. This lets a fixed gradient-descent trajectory simulate polynomial-space

computation for arbitrarily many steps. As a consequence, even for a strongly convex energy with a known minimizer, predicting a designated output bit after T exact gradient-descent steps is PSPACE-complete when T is given in binary (Theorem 3.1).

3. For smooth semidefinite relaxations, we prescribe a regularized matrix factorization and initialization for which ordinary gradient descent approaches the convex optimum to any prescribed accuracy in exact arithmetic, with a computable certificate (Theorem 4.1). We also show how increasing the factor dimension limits the gap between local minima and the convex optimum (Theorem 4.3).

1.1 Related work

The execution picture builds on computational universality for gradient and neural systems. Abbe and Sandon (2023) establish polynomial-time universality of SGD on specially constructed networks and describe it as P-completeness, and the same line of work identifies precision conditions under which minibatch SGD and full-batch GD can simulate efficient sample-based learning (Abbe et al., 2021). Their analysis crucially depends on careful initialization in a nonconvex landscape. Chatziafratis et al. (2019) study eventual coordinate positivity under online gradient descent. Fearnley and Savani (2018) prove PSPACE-completeness for all-switches strategy improvement through iterated-circuit problems, and Balaji et al. (2019) prove that computing an optimal policy of a finite-horizon Markov decision process is EXPTIME-complete when the horizon is given in binary. Our prediction result adds a fixed strongly convex energy in exact and rounded arithmetic. Dissipative systems, recurrent graph networks, and resistive networks give further comparisons (Šíma and Orponen, 2001; Ahvonen et al., 2024; Rosenbluth and Grohe, 2026; Scellier and Mishra, 2025).

The landscape picture builds on SDP factorization and gradient-based barrier methods (Burer et al., 2002, 2003; Journée et al., 2010). Existing guarantees cover linear elliptope objectives (Mei et al., 2017), smooth SDP manifolds (Boumal et al., 2020), and quantitative factor certificates (Zhang et al., 2023). Global GD convergence is also known for positive definite matrix square roots (Jain et al., 2017) and for regularized Bures–Wasserstein barycenters (Altschuler et al., 2021, Theorem 4, corrected version), whose congruence update lifts to ordinary GD on matrix factors. Polynomial approximations support stochastic spectral gradients (Han et al., 2018). Low-rank recovery relies on assumptions about the measurements or observations (Ge et al., 2017), while factored optimization has local guarantees under additional conditions (Bhojanapalli et al., 2016). Our square-factor construction uses one fixed regularized loss and a scaled-identity initialization to approximate the bounded convex optimum, and its matrix-fitting guarantee concerns objective value rather than recovery. OptGNN and the Lasserre hierarchy supply structured comparisons for constraint satisfaction (Yau et al., 2024; Lasserre, 2001a; Laurent, 2003).

2 Algorithms as gradient descent programs

We first show how gradient descent can carry out a computation by encoding working memory as the state of a descent trajectory. A simulation specifies a differentiable objective $\Phi\colon \mathbb{R}^d \to \mathbb{R}$ on d state coordinates, a product box $K \subseteq \mathbb{R}^d$, an encoded initial state $z^0 \in K$, and a decoder for the final state. We then run ordinary simultaneous projected gradient descent (PGD),

$$z^{t+1} = \Pi_K(z^t - \nabla\Phi(z^t)), \tag{1}$$

with every derivative evaluated at the old state. The objective measures progress through the encoded computation.

2.1 Making a gradient execute an update

The first obstacle is feedback: in a circuit, a gate reads its inputs and writes its output, leaving the input unchanged. However, gradient descent on, for example, $(y - g(x))^2/2$ generally updates both y and x, so an output mismatch can alter previously computed input values. We suppress this feedback at encoded states with the polynomials

$$q(u) = 3u^2 - 2u^3, \qquad r(u) = u - q(u),$$

which satisfy $q(b) = b$, $q'(b) = 0$, $r(b) = 0$, and $r'(b) = 1$ at $b \in \{0, 1\}$. Thus q reads a bit without passing a derivative through it, while r directs the derivative to the coordinate being written. If R_i is a polynomial update rule for coordinate i, the factor $r(z_i)[q(z_i) - R_i(q(z))]$ has gradient $(b_i - R_i(b))e_i$ at every Boolean state b. Summing these factors makes one unit gradient step execute all rules at once. An additional term $V(q(z))$ tracks the objective's Boolean values as the gates get updated, without disturbing the updates (Lemma A.1).

2.2 Progress from resolved wires

To make the objective measure computational progress, we count the wires whose values have been resolved. Consider a circuit with B input wires and S two-input AND and XOR gates, possibly with a constant-one wire. We represent each wire by two coordinates. One indicating false and the other true. Setting both to zero indicates that the wire's value is unknown. This is called (*dual-rail* encoding),

$$\underbrace{(0,0)}_{\text{unknown}} \longrightarrow \underbrace{(1,0)}_{\text{false}} \quad \text{or} \quad \underbrace{(0,1)}_{\text{true}},$$

with inputs initialized to their given values and internal rails to zero. An AND gate resolves to false as soon as one input is known to be false, whereas an XOR gate waits for both inputs. The resulting rail rules R_i have degree at most four and read at most four parent coordinates. The circuit may contain cycles, provided that repeated simultaneous updates determine every wire's value within D rounds. This *finite-delay* condition holds for acyclic circuits and for the cyclic circuits that simulate bounded-word parallel programs (Heath, 2023). With I indexing the $2S$ internal rails, define

$$\Phi_C(z) = -\sum_{i \in I} q(z_i) + \sum_{i \in I} r(z_i)\,[q(z_i) - R_i(q(z))]. \tag{2}$$

At each state reached by the simulation, the second sum is zero. When a gate's value becomes known, one of its two coordinates changes from zero to one, so the objective decreases by one. Each gate's value is determined at most once. The number of known gate values therefore measures progress without an auxiliary clock, as the next theorem makes precise.

Theorem 2.1 (Descent without an additional clock)**.** *For a finite-delay circuit as above, Φ_C is an explicit polynomial of degree at most 15 with $O(B + S)$ coordinates and factors, each involving at most five coordinates. Unit-step PGD on the unit cube from the specified rail initialization reproduces the circuit output within D iterations and then remains fixed, and every reached step satisfies*

$$\Phi_C(z^{t+1}) - \Phi_C(z^t) = -\|z^{t+1} - z^t\|_2^2 = -\|\nabla \Phi_C(z^t)\|_2^2. \tag{3}$$

An implementation that caches unchanged factor derivatives uses $O(B + S)$ total work and storage, including initialization and readout, with words of $O(\log(B + S))$ bits.

Proof sketch. The scalar lift turns unit PGD into the exact rail update. Known information only grows, so each gate changes one rail once. At every reached vertex the correction factors vanish and Φ_C equals minus the number of resolved gates, giving (3). Each factor reads constantly many coordinates and therefore needs constantly many refreshes, giving the work bound (Appendix A, which includes a worked circuit with a feedback edge). Each R_i has degree ≤ 4; q, r have degree 3, so Φ_C has degree ≤ 15.

The work bound depends on caching: scanning every factor in every round would cost $O((B + S)(D + 1))$. The descent identity holds along the specified Boolean trajectory, so fractional initializations need a separate analysis. A small change to the value term makes the terminal state a strict local minimum (Proposition A.5).

2.3 Strongly convex constructions

Theorem 2.1 gives universality but stores the computation in the trajectory of a nonconvex objective. To place the circuit's answer at an optimum, we instead use a strongly convex objective with coordinate scales that shrink with depth.

Proposition 2.2 (Circuit values at a strongly convex minimizer)**.** *Consider an acyclic AND/OR/NOT/COPY circuit of total size S_{tot} and depth $D \geq 1$ with $r \geq 1$ sink outputs and at most two distinct parents per gate. For each clamped input there is an explicit energy on $O(S_{\text{tot}})$ coordinates with $O(S_{\text{tot}})$ local factors that is globally $C^{1,1}$, 3/4-strongly convex, and 5/4-smooth, so that unit-step GD is globally 1/4-contractive, and from every scaled Boolean start GD reaches the unique minimizer, which encodes all gate values, within D steps. The local numerical data use $O(D + \log r)$ bits, and dense execution costs $O(S_{\text{tot}} D)$ operations.*

Proof sketch. Gates later in the circuit use progressively smaller powers-of-two scales, limiting their effect on earlier gates. At encoded bit values, the gate functions have zero slope in their inputs, allowing gradient descent to reproduce the intended Boolean updates.. A Hessian bound then gives strong convexity, and induction on depth identifies the minimizer (Appendix B).

For computations that repeatedly rewrite their state, we use a third construction in which the intermediate gradient-descent iterates encode the computation. For a synchronous Boolean map $F\colon \{0,1\}^m \to \{0,1\}^m$ with at most two inputs per register, one fixed $C^{1,1}$ energy on $2m$ coordinates is 3/4-strongly convex and 5/4-smooth. Unit-step GD from $c_i^0 = 1$, $x_i^0 = b_i$ has the exact orbit

$$c_i^t = \tau^t, \qquad x_i^t = \tau^t (F^t(b))_i, \tag{4}$$

where $\tau = 2^{-O(\log(2+\Delta))}$ and Δ is the maximum number of factors reading one register (Theorem C.1). The number of coordinates stays fixed as the horizon grows, while the decoding margin $\tau^t/2$ shrinks. Heath's compiler connects these constructions to general programs: it turns a bounded-word program with input-size parameter n, work budget W and parallel time T into a circuit with $S = O(W \log^4 n)$ and $D = O(T \log^3 n)$ (Heath, 2023) (Appendix A.8). The recurrent construction and its rounded version will let us study the difficulty of predicting these computations.

3 Prediction complexity under contraction

The strongly convex constructions of Section 2 converge rapidly because their conditioning is fixed. In the recurrent construction, the minimizer is even known in advance, but the computation is encoded in the trajectory approaching it. Rapid convergence does not make that computation easy to predict, even when the recurrent circuit consists of constant-size blocks arranged along a

line, each communicating only with its two neighbors. To make this distinction precise, we use the *prediction problem* familiar from cellular automata: given an initial tape over a fixed finite alphabet and an integer $T \geq 0$ in binary, decide whether a designated output bit equals one after T complete recurrent blocks, with fixed modules and initialization rule. The theorem below classifies this problem both in exact arithmetic and for a specified rounded implementation with constant precision and $O(\log(T+2))$ bits per exponent (Appendix D).

Theorem 3.1 (PSPACE-complete prediction with communication along a path)**.** *There is a fixed finite family of ReLU message-passing modules operating on the path $1-2-\cdots-M$ with marked endpoints and left and right ports, such that each vertex stores a constant number of real coordinates and each recurrent block uses a constant number of nearest-neighbor communication stages. For each M, a fixed local rule defines an energy E_M that does not depend on the input tape or on the iteration count. This energy is globally $C^{1,1}$, $3/4$-strongly convex, and $5/4$-smooth, with unique minimizer zero, and its unit-step gradient map is $1/4$-contractive. In exact arithmetic, along trajectories from the prescribed encoded initializations, each recurrent block implements one gradient-descent step on E_M.*

Given an initial tape and an integer $T \geq 0$ in binary, deciding whether the output bit at vertex 1 equals one after T recurrent blocks is PSPACE*-complete under polynomial-time many-one reductions, and the same classification holds for the specified finite-precision rounded implementation.*

Proof sketch. For hardness, we turn a fixed polynomial-space machine into a local tape rule, which the rounded compiler simulates through an exponentially long run. The tape and the binary horizon still have polynomial descriptions. For membership, the Boolean registers and an iteration counter fit in polynomial space (Appendix E).

After T steps, the two possible bit values are separated by only about τ^T, while T itself can be specified using only $O(\log T)$ bits. Therefore, rapid convergence of the trajectory to zero does not make the encoded bit easy to predict. In fact, a polynomial-time algorithm for predicting that bit would imply P = PSPACE. This shows that a strongly convex objective can be easy to optimize while its intermediate gradient-descent trajectory remains hard to predict.

Two lines of prior work are closely related. Universality results for gradient-based dynamics show that computation can be encoded in optimization trajectories, but use carefully chosen initializations on nonconvex landscapes (Abbe and Sandon, 2023; Abbe et al., 2021). Separately, hardness results for iterated circuits show that predicting a computation at a time T given in binary can be PSPACE-hard; we review these in Section 1. Our result combines these two perspectives: the trajectory of ordinary gradient descent is PSPACE-hard to predict even for a fixed local scalar energy that is uniformly strongly convex.

4 Representations that support gradient descent

The execution results start from a supplied algorithm, so the quality of the answer is inherited from that algorithm. To derive a solver from the task itself, we first write down the constraints that the circuit's wire values must satisfy. For a finite circuit, or a recurrent circuit unrolled for a specified number of iterations, these constraints encode the gate relations together with prescribed inputs and outputs (e.g. $z = xy$ for an AND gate with x, y input and z output). We then lift this constraint system to a matrix representation and study the landscape of the resulting optimization problem. After encoding wires by signs, we relax these relations to affine constraints on their moments, which are the expected products of wire values. This gives a PSD matrix called *moment matrix* which has diagonal one because each represented product of signs squares to one.

These moments record correlations that determine what a gradient can see. For intuition, consider the cut problem. On a single edge, independently uniformly sampling labels obtains a cut with probability $1/2$, whereas sampling a uniform label and assigning the opposite label to the other vertex always cuts the edge. Both sampling rules have zero marginal means, so those means alone do not distinguish their scores. The expected cut score has zero gradient when independently sampled labels have zero means, while a correlation coordinate exposes an improvement to first order (Appendix F).

Vector embeddings, also called *factors*, $V \in \mathbb{R}^{d\times k}$ represent these correlations through a Gram matrix $X = VV^\top \in \mathbb{R}^{d\times d}$. The number of columns k is the *search rank*; the matrix rank may be smaller. Below, we give two theorems that quantify how the factor dimension affects optimization of smooth convex matrix losses, including smoothed affine violations.

For any prescribed accuracy, the first theorem specifies a square factor $V \in \mathbb{R}^{d\times d}$, a fixed regularized loss, and a scaled-identity initialization $V_0 = I_d/\sqrt{d}$ for which ordinary gradient descent reaches that accuracy in finitely many exact-arithmetic steps. The algorithm stops when the gradient norm falls below an explicit threshold guaranteeing that the objective value is at most some ε above the global minimum of the matrix relaxation.

The second gives an explicit bound on the gap to the convex optimum. For a normalized factor with search rank $k \geq 2$, this gap is at most $M/(k-1) + \eta/2$, provided the Hessian along norm-preserving directions has no eigenvalue below $-\eta$. Here M bounds the curvature of the convex matrix loss. Increasing the number of columns therefore reduces the first term, while the second measures the remaining negative curvature.

We then recover and extend many known results as corollaries of these two theorems (Table 1). For Boolean tasks, the size and tightness of the relaxation, together with rounding, determine the resulting guarantee.

4.1 Constant-step gradient descent on a square factor

Throughout, $\mathcal{D}_d$ denotes the set of *density matrices*, which are $d \times d$ PSD matrices of unit trace.

We first use a square factor of rank d to optimize a smooth convex objective over $\mathcal{D}_d$ to prescribed accuracy using an explicit stopping rule based on the gradient norm. Let ϕ be convex and C^2 near $\mathcal{D}_d$ with $\|\nabla\phi\|_{\mathrm{op}} \leq L$ and $0 \leq \nabla^2\phi[H,H] \leq M\|H\|_F^2$ there. For a nonzero square factor $V \in \mathbb{R}^{d\times d}$, we normalize the Gram matrix to impose unit trace. To keep the iterates uniformly positive definite, we add a regularizer with weight $\mu > 0$ and degree p, obtained by truncating the series of the barrier $-\mu \log\det X$:

$$X(V) = \frac{VV^\top}{\|V\|_F^2}, \qquad \mathcal{L}_p(V) = \phi(X(V)) + \mu \sum_{j=1}^{p} \frac{\mathrm{tr}[(I_d - X(V))^j]}{j}. \tag{5}$$

Log-determinant barriers have been used in gradient-based methods for semidefinite optimization (Burer et al., 2002, 2003). Here we use a finite polynomial truncation of this barrier, yielding a regularizer that is polynomial in X, remains finite even at singular matrices, and has derivatives that can be evaluated using matrix products. The theorem shows that this finite regularizer preserves a uniform lower bound on the eigenvalues along the entire run.

Theorem 4.1 (Ordinary GD for convex objectives on density matrices)**.** *Set* $\Lambda = L+\mu d$, $r = \mu/(4\Lambda)$, *and* $p = \lceil r^{-1}\log(64\Lambda/\mu)\rceil$, *and let* $M_F \geq 0$ *bound the Hessian of* $U \mapsto \phi(UU^\top)$ *by* $2L + M_F$ *at unit factors with* $UU^\top \succeq rI_d$ *(one may take* $M_F = 4M$*). Run ordinary GD as:*

$$V_{t+1} = V_t - \eta\nabla\mathcal{L}_p(V_t), \qquad V_0 = I_d/\sqrt{d}, \qquad 0 < \eta \leq [64\Lambda + M_F]^{-1}. \tag{6}$$

Then every iterate satisfies $X(V_t) \succeq rI_d$ *and* $1 \leq \|V_t\|_F^2 \leq 1 + 4\eta L$*; the regularized objective has a unique minimizer* $X_* \in \mathcal{D}_d$ *and* $\mathcal{L}_p(V_t)$ *converges to its value at the geometric rate* $2L(1 - 4\mu\eta/(1 + 4\eta L))^t$*; and with* $G_t = \|V_t\|_F\nabla\mathcal{L}_p(V_t)$*, every iterate certifies*

$$\phi(X(V_t)) - \min_{\mathcal{D}_d}\phi \leq \mu d + \frac{\|G_t\|_F^2}{8\mu}. \tag{7}$$

Proof sketch. The polynomial preserves a lower bound on the smallest eigenvalue along every trial step, so the loss is smooth where the iterates live. In this region, a weighted log-determinant inequality bounds the loss gap by the squared gradient, giving geometric convergence and the gradient certificate. Scale invariance controls the growth of $\|V_t\|_F$, so the step-size coefficient $\eta/\|V_t\|_F^2$ for the normalized factor stays bounded away from 0. □

The proof is in Appendix G. To use the certificate, choose $\mu = \varepsilon/(2d)$ and stop when $\|G_t\|_F^2 \leq 4\mu\varepsilon$. This occurs within $\widetilde{O}(d(L+M)/\varepsilon)$ steps and leaves ϕ within ε of its minimum. Apart from evaluating $\nabla\phi$, each gradient requires $O(\log p)$ matrix products and no matrix solves or spectral subroutines.

In perspective, gradient-based SDP barriers already read feasibility and complementarity from transformed gradients (Burer et al., 2003, Corollary 1), with global convergence through a sequence of barrier subproblems (Burer et al., 2003, Algorithm GLB and Theorem 5). Factored GD already has finite convergence bounds under a suitable initialization and matching search rank (Bhojanapalli et al., 2016, Theorem 5 and Assumption (A1)). The contribution here is the joint prescription of a fixed polynomial regularizer, scaled-identity initialization, constant step, uniform lower bound on the eigenvalues, and finite certificate for smooth convex objectives on $\mathcal{D}_d$. In comparison, the certificate of Zhang et al. (2023, Proposition 10) also combines gradient, Hessian, and rank-deficiency terms; while (7) uses only the regularized gradient along the full-rank trajectory. Projection-free solvers already provide finite accuracy certificates using an eigenvector oracle at every step (Hazan, 2008; Jaggi, 2013), so this construction makes no claim to a faster solver.

Example: semidefinite feasibility. We use bounded semidefinite feasibility as a running example for both theorems. The goal is to find a positive semidefinite matrix $Z \in \mathbb{R}^{n\times n}$ satisfying given affine constraints and $\operatorname{tr} Z \leq R$; or to certify that no such matrix exists. For $R > 0$, we write $Z = RX_{1:n,1:n}$, where $X_{1:n,1:n}$ is the submatrix of the first n rows and columns of a density matrix $X \in \mathcal{D}_d$ with $d = n + 1$. This representation enforces positive semidefiniteness and the trace bound, with the extra diagonal entry absorbing the unused trace.

Write the remaining constraints as $a_i(X) = b_i + \langle B_i, X\rangle \leq 0$, $1 \leq i \leq m$. A positive residual $a_i(X)$ measures how much constraint i is violated. Introducing $a_0(X) = 0$, we define the largest violation and its smallest achievable value by

$$v(X) = \max_{0\leq i\leq m} a_i(X), \qquad \gamma_* = \min_{X\in\mathcal{D}_d} v(X).$$

Thus $v(X) = 0$ exactly when X satisfies all constraints, while $\gamma_* > 0$ certifies infeasibility.

To apply the smooth optimization theorem, we replace the nonsmooth maximum by

$$\phi_\tau(X) = \tau\log\sum_{i=0}^{m} e^{a_i(X)/\tau}, \qquad v(X) \leq \phi_\tau(X) \leq v(X) + \tau\log q,$$

Table 1: Selected applications of Theorems 4.1 (first row) and 4.3(b) (last two); full list and comparisons in Tables 2–3. **A**: new prescribed-GD guarantee; **E**: new quantitative extension. The landscape gap is at most $(R + C/2)/(k-1) + N\zeta/2$ when $\operatorname{Hess} F \preceq \zeta I$; the listed constants are $R + C/2$. Here ρ is the feasibility penalty and $\tau > 0$ the contrastive temperature.

Problem and precedent	Coordinates	Guarantee	Status / result
Matrix completion (Candès and Recht, 2009)	$(a+b+1)^2$	observed entries exact, nuclear norm within ε of the minimum via norm-bound feasibility tests and entry repair	**A** Cor. H.6
Lovász theta (Lovász, 1979)	$(n+1)k$	$1 + 21\rho/4$: certified interval for $\vartheta(G)/n$ after an explicit repair	**E** Prop. L.7
Contrastive embeddings (Khosla et al., 2020; Behnia and Thrampoulidis, 2024; Vock et al., 2026)	Nk	$4/\tau + 2/\tau^2$: in-sample loss within the bound of the convex optimum	**E** App. L

where $q = m + 1$. The following corollary bounds γ_* from both sides: the current matrix gives an upper bound, while the gradient certificate gives a lower bound. Set $L = \max_{1 \le i \le m} \|B_i\|_{\mathrm{op}} > 0$.

Corollary 4.2 (Bounded semidefinite feasibility)**.** *For $0 < \varepsilon \le L$, apply Theorem 4.1 to $\phi = \phi_\tau$ with*

$$\tau = \frac{\varepsilon}{4 \log q}, \qquad \mu = \frac{\varepsilon}{4d}, \qquad \eta = [64\Lambda + 4L^2/\tau]^{-1}, \quad \Lambda = L + \mu d.$$

At each iterate, write $X = X(V)$, $G = \|V\|_F \nabla \mathcal{L}_p(V)$, and $\pi_i = e^{a_i(X)/\tau} / \sum_{j=0}^m e^{a_j(X)/\tau}$. Then

$$\underline{\gamma}(V) := \sum_{i=0}^m \pi_i a_i(X) - \mu d - \frac{\|G\|_F^2}{8\mu} \le \gamma_* \le v(X).$$

After $\widetilde{O}(d(L/\varepsilon)^2 \log q)$ exact-arithmetic gradient steps, these bounds differ by at most ε. If $v(X) \le \varepsilon$, the returned matrix violates no constraint by more than ε. Otherwise, $\underline{\gamma}(V) > 0$ certifies that no matrix within the trace bound satisfies all constraints.

The same theorem also gives guarantees for matrix fitting under a nuclear-norm constraint (Mazumder et al., 2010), using the standard reformulation over density matrices (Jaggi and Sulovský, 2010); see Corollary H.3. Appendix H gives the iteration bound and specializes the construction to positive semidefinite sensing and matrix completion with exact observed entries. Appendix I treats the Peng–Wei clustering relaxation. For the matrix-fitting problems, recovering a planted low-rank matrix additionally requires assumptions on the measurements (Ge et al., 2017).

4.2 What smaller ranks guarantee

The square factor gives a specified descent algorithm. Smaller factors use fewer coordinates, but can have stationary points with poor objective values. Our second theorem bounds this suboptimality in two representations. The first fixes the trace, while the second fixes every diagonal entry. For the second form, define the elliptope $\mathcal{E}_N = \{X \succeq 0 : X_{ii} = 1\}, \mathcal{M}_{N,k} = (\mathbb{S}^{k-1})^N$, and let $R(A) = \max_{Z \in \mathcal{E}_N} \langle A, Z \rangle - \min_{Z \in \mathcal{E}_N} \langle A, Z \rangle$.

Theorem 4.3 (Landscape bounds at fixed rank)**.** *Let $k \geq 2$ and $\eta \geq 0$. All gradients and Hessians below are Riemannian, with the metric induced by the Frobenius inner product, except for the matrix derivatives of f.*

(a) Unit trace. *Let f be convex and C^2 near $\mathcal{D}_d$, let $W \in \mathbb{R}^{d\times k}$ satisfy $\|W\|_F = 1$, and set $X = WW^\top$, $A = \nabla f(X)$, and $g(W) = f(WW^\top)$. If $\operatorname{Hess} g(W) \succeq -\eta I$ and $\nabla^2 f(X)[H,H] \leq M\|H\|_F^2$ for every symmetric H, where $M \geq 0$, then*

$$f(X) - \min_{\mathcal{D}_d} f \leq \Delta(X) := \langle A, X\rangle - \lambda_{\min}(A) \leq \frac{M}{k-1} + \frac{\eta}{2}. \tag{8}$$

Here Δ is the Frank–Wolfe gap (Jaggi, 2013). If also $\operatorname{grad} g(W) = 0$, the denominator improves to k. The resulting M/k coefficient is sharp for Δ.

(b) Unit diagonal. *Let f be concave and C^2 near $\mathcal{E}_N$, let $V \in \mathcal{M}_{N,k}$ have rows $v_i^\top$, and set $F(V) = f(VV^\top)$ and $A = \nabla f(VV^\top)$. Suppose $\operatorname{Hess} F(V) \preceq \eta I$. For a comparator $X_* = YY^\top \in \mathcal{E}_N$, write the rows of $Y \in \mathbb{R}^{N\times s}$ as $y_i^\top$, and draw $G \in \mathbb{R}^{k\times s}$ with independent $\mathcal{N}(0, 1/k)$ entries. Let U have rows $u_i^\top$, where $u_i = (I - v_i v_i^\top)Gy_i$, and put $H = VU^\top + UV^\top$. If $C \geq 0$ satisfies the curvature condition*

$$-k\,\mathbb{E}\,\nabla^2 f(VV^\top)[H,H] \leq C, \tag{9}$$

where the expectation is over G, then

$$f(X_*) - F(V) \leq \frac{R(A) + C/2}{k-1} + \frac{N\eta}{2}. \tag{10}$$

Taking X_* to be a maximizer bounds the gap to the relaxation optimum. Neither bound requires first-order stationarity.

Proof sketch. For part (a), take a unit bottom eigenvector u of A and a unit vector $z \perp W^\top u$. The tangent direction $uz^\top$ has curvature $-2\Delta(X) + \nabla^2 f(X)[H,H]$, where $H = Wzu^\top + uz^\top W^\top$ and $\|H\|_F^2 = 2\|Wz\|^2$. Averaging over an orthonormal basis of $(W^\top u)^\perp$ gives the stated bound (Appendix J). Part (b) averages over Gaussian directions projected onto the tangent space of each row sphere (Appendix K). □

Example: semidefinite feasibility at smaller search ranks. We return to the semidefinite feasibility problem of Corollary 4.2 and show that with high enough rank, every local minimum has a largest constraint violation within ϵ of the best possible value over the relaxation.

Corollary 4.4 (Rank supplies many negative-curvature directions)**.** *Using notation from Corollary 4.2, for the smoothed feasibility objective ϕ_τ with $\tau = \varepsilon/(4\log q)$ and $k - 1 \geq 32L^2 \log q/\varepsilon^2$, every factor W either has $\Delta(WW^\top) \leq 3\varepsilon/4$, in which case $\max_i a_i(WW^\top) - \gamma_* \leq \varepsilon$, or the sphere Hessian of $\phi_\tau(WW^\top)$ has at least $\lceil (k-1)/2 \rceil$ eigenvalues below $-\varepsilon/2$.*

Thus, at any point with largest constraint violation exceeding its minimum γ_* over the relaxation by more than ε, the number of negative-curvature directions grows linearly with the rank. Reaching an approximate second-order point with Hessian tolerance below $\varepsilon/2$ therefore guarantees that every constraint violation is at most $\gamma_* + \varepsilon$. For a feasible instance, $\gamma_* = 0$. Gradient descent with small random perturbations can reach such points (Jin et al., 2017; Criscitiello and Boumal, 2019). More escape directions help it leave saddles faster locally (Appendix J), but whether this improves the overall convergence rate remains open.

The new guarantees are the prescribed GD construction (Theorem 4.1), the trace-sphere certificate and negative-curvature count (Theorem 4.3(a) and Corollary 4.4), and the nonlinear unit-row bound (Theorem 4.3(b)).

4.3 Transfer to neural parameterizations

The landscape bound transfers to a neural encoder when its output head can realize every tangent direction of the factor. For the concave function f in Theorem 4.3(b), write $F(V) = f(VV^\top)$ and let $\Psi = [\psi_1, \dots, \psi_N] \in \mathbb{R}^{h \times N}$ collect the encoder features and let B be a linear head. The rows of V are the normalized head outputs $v_i = z_i / \|z_i\|$ with $z_i = B\psi_i$.

Corollary 4.5 (Normalized neural head)**.** *Suppose $\sigma_N(\Psi) > 0$ and all $z_i \neq 0$. If the Hessian of the head loss satisfies $\nabla^2_B[-F] \succeq -\gamma I$ with the features held fixed and $\gamma \geq 0$, then the bound in Theorem 4.3(b) holds with $\eta = \gamma(\max_i \|z_i\|/\sigma_N(\Psi))^2$, and an exact second-order stationary parameter point maps to an exact second-order stationary factor point.*

The proof is in Appendix M. The appendix also recovers the known prediction-loss certificates for full-rank feature matrices and regularized homogeneous networks (Nguyen and Hein, 2017; Haeffele and Vidal, 2017).

5 Discussion

We asked how efficiently gradient descent can execute an algorithm and when a richer representation recovers the guarantees of a relaxation. On the first question, we gave three constructions. The objective of Theorem 2.1 executes a circuit and measures progress by resolved wires. The static strongly convex compiler stores the answer at its minimizer, while the recurrent construction carries it in a transient whose decoding margin shrinks with time. A trajectory can therefore approach a known minimizer at a fixed geometric rate while its shrinking coordinates encode a PSPACE-hard answer, making the numerical representation and decoder part of the guarantee.

On the second question, additional state supports both an algorithmic guarantee and a landscape bound. With a square factor, the polynomial regularizer keeps the smallest eigenvalue bounded below and gives ordinary constant-step GD a finite stopping guarantee from one scaled-identity start on every instance. This guarantee charges the trace radius, coefficient scale, and matrix products, and certifies feasibility to additive tolerance. At smaller search ranks, extra columns provide many negative-curvature directions, although converting their number into an improved global rate remains open. Transferring this geometry to neural parameterizations requires the encoder to expose the tangent directions used by the factor argument. The transfer corollary therefore identifies the conditioning of the feature matrix as the quantity to monitor during training.

Use of generative AI

Under our guidance, OpenAI Codex and Anthropic Claude helped develop the theoretical models, formulate hypotheses and mathematical claims, contribute proof ideas, write proofs, and provide feedback on research methodology. They also assisted with drafting and editing the prose, finding relevant literature, and formatting references. We reviewed the AI-assisted text, claims, proofs, and references, and take responsibility for the final content.

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Appendices

The appendices follow the main paper and record the supporting constructions, proofs, and additional guarantees. Appendix A proves the circuit simulation, including its work bounds, terminal minimum, message-passing realization, and computational-model extensions. Appendix B constructs the strongly convex circuit evaluator, while Appendix C treats recurrent computation in a fixed convex energy. Appendix D supplies the rounded local implementation and robustness to shrinking data errors with exact clocks; Appendix E proves the exact- and finite-precision prediction complexity results. Appendix F motivates correlation coordinates and distinguishes moment representations from their semidefinite relaxations. Appendix G proves the polynomially regularized gradient-descent guarantees, finite stopping bounds, and arithmetic costs, and gives initialization and search-rank counterexamples. Appendix H specializes these guarantees to matrix factorization, sensing, and completion. Appendix I develops the alternative exact log-determinant method, including its certificates, conditioning limitation, and applications; this method requires matrix inverses or linear solves. Appendix J proves the trace-constrained landscape certificates, negative-curvature multiplicity bounds, and quantitative local escape guarantee. Appendix K proves the unit-diagonal landscape bound, exact optimality at square search rank, and asymptotic normalized-descent convergence under analyticity. Appendix L develops the applications to cuts, constraint satisfaction, moment hierarchies, affine feasibility, Lovász theta, continuous objectives, and clustering, separating relaxation optimization from feasibility repair and rounding. Finally, Appendix M proves the neural-head transfer, gives a conditional finite-horizon feature-conditioning guarantee, and records classical neural optimality certificates.

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Circuit execution and prediction complexity

A Proofs for the simulation construction

We prove the finite-delay simulation, its work bound, and its terminal-state and message-passing variants, then record the resource bounds for general programs. All trajectories use the specified encoded initialization and exact arithmetic.

A.1 The scalar lift

The scalar functions below separate reading from writing: q reads a bit with zero derivative, while r passes a derivative to the target coordinate.

Lemma A.1 (Boolean update lift)**.** *Let $A\colon \{0,1\}^N \to \{0,1\}^N$ have polynomial update rules $A_i(b) = R_i(b)$. Define $q(u) = 3u^2 - 2u^3$ and $r(u) = u - q(u)$, with q applied coordinatewise. For any polynomial V, let*

$$\Phi(z) = V(q(z)) + \sum_{i=1}^{N} r(z_i)\big[q(z_i) - R_i(q(z))\big].$$

At every Boolean vertex b,

$$\Phi(b) = V(b), \qquad \nabla\Phi(b) = b - A(b).$$

Unit-step PGD on $[0,1]^N$ therefore maps b to $A(b)$.

Proof. At a Boolean vertex, $q(b_i) = b_i$, $q'(b_i) = 0$, $r(b_i) = 0$, and $r'(b_i) = 1$. Thus every correction term vanishes, giving $\Phi(b) = V(b)$, and the value term has zero gradient. In correction term i, only differentiation of its leading $r(z_i)$ survives, contributing $b_i - R_i(b)$ to coordinate i. This remains true for self-inputs and repeated parents. Hence $b - \nabla\Phi(b) = A(b) \in [0,1]^N$, so projection leaves the update unchanged. □

An input coordinate has the identity rule, whose correction term is zero; its derivatives through other factors vanish because $q' = 0$. It may therefore be free or clamped without changing the trajectory.

A.2 Rails and finite-delay circuits

Theorem 2.1 applies the lift to the dual-rail encoding of Section 2, whose update rules we now write out in full. Every wire w has two coordinates, the false rail f_w and the true rail t_w, and the three valid states are unknown $(0,0)$, false $(1,0)$, and true $(0,1)$. Input wires are initialized to their specified Boolean values and internal wires to unknown, while a constant-one wire, if the circuit has one, is initialized to $(0,1)$ and carries the identity update like an input. Writing $u \vee v = u + v - uv$ for the polynomial OR, the rules for a gate g with parent wires a, b are

Gate	f_g^+	t_g^+
AND	$f_a \vee f_b$	$t_a t_b$
XOR	$(f_a f_b) \vee (t_a t_b)$	$(f_a t_b) \vee (t_a f_b)$

Each rule has degree at most four and reads at most four parent coordinates, and repeated parent wires are permitted.

Definition A.2 (Finite delay)**.** A circuit has delay at most D on a specified input if these synchronous rules resolve every internal wire within D rounds. Here an AND gate may resolve to false after learning only one false input, whereas an XOR gate requires both input values, and the directed circuit may contain cycles.

The decrease identity will count rail changes, which the following lemma identifies with newly resolved gates.

Lemma A.3 (Monotone resolution)**.** *Along the trajectory of the synchronous rail rules from the initialization above, every rail is nondecreasing and each wire remains in one of the three valid states. In a complete finite-delay run, exactly one rail per gate changes from zero to one, and no rail changes more than once.*

Proof. Each rail rule is coordinatewise monotone, being composed of AND and OR. Since the initial internal rails are zero and the inputs stay fixed, $z^1 \geq z^0$ and induction gives $z^{t+1} \geq z^t$. Validity is also inductive. An AND output is true only when both parents are true, precluding its false rail. The two XOR rules require equal and unequal known parent values respectively, so cannot both fire on valid parents. Thus no wire asserts both rails. Finite delay resolves every gate, which therefore changes exactly one rail once. □

A.3 Exact simulation and decrease

Proof of Theorem 2.1. Applying Lemma A.1 with identity rules on the input rails and $V(z) = -\sum_{i \in I} z_i$ on the internal rails produces the objective (2), so at every reached Boolean state the unit PGD update is the synchronous rail update. Induction on the round then gives the claimed trajectory and its completion within D rounds. Once all gates have resolved, the state is stationary: monotonicity (Lemma A.3) gives $A(b) \geq b$ coordinatewise and validity forbids a second rail on a resolved gate, so $A(b) = b$ and $\nabla \Phi_C(b) = b - A(b) = 0$.

The degree and size count is direct. Since q and r have degree three and $\deg R_i \leq 4$, a correction term has degree at most $3 + 3 \cdot 4 = 15$ and involves its target together with at most four parents. Combining each internal rail's unary value term with its correction term yields one factor per internal rail. With $2S$ internal rails, $2B$ input rails, and at most two constant coordinates, the representation and its incidence lists have size $O(B + S)$.

Let E_t be the rails changing in round t. Each changes from zero to one, so the objective, which equals $-\sum_{i \in I} z_i$ at Boolean states, falls by $|E_t|$. The gradient is -1 on those rails and zero elsewhere, giving

$$\Phi_C(z^{t+1}) - \Phi_C(z^t) = -|E_t| = -\|z^{t+1} - z^t\|_2^2 = -\|\nabla \Phi_C(z^t)\|_2^2.$$

The cache argument below (Lemma A.4) proves the total work and storage bounds. Local intermediate arithmetic is bounded because the arguments are Boolean and the formulas have fixed degree and integer coefficients. The accumulated gradient entries lie in $\{-1, 0, 1\}$; indices and objective values in $[-S, 0]$ fit in the stated $O(\log(B + S))$-bit words. □

An Armijo line search trying the unit step first accepts it along this trajectory for every sufficient-decrease parameter in $(0, 1)$. This gives no step-size bound at fractional states.

A.4 Counting the cost of a full simultaneous gradient

Caching local derivatives avoids reevaluating factors whose arguments have not changed. The following accounting also covers the objective values.

Lemma A.4 (Cached factor evaluation)**.** *Let $\Phi(z) = \sum_{a=1}^{M} \phi_a(z_{I_a})$ be a sum of M factors on a product box in $\mathbb{R}^N$, where I_a is the set of coordinates read by factor a, and suppose that evaluating a factor and all its partial derivatives costs $c_a \geq \max\{1, |I_a|\}$ operations. If coordinate i changes κ_i times before the PGD trajectory becomes stationary, exact simultaneous PGD and its objective values can be evaluated in total work*

$$O\left(N + \sum_i \kappa_i + \sum_a c_a \left[1 + \sum_{i \in I_a} \kappa_i\right]\right).$$

When factor arity and cost are constant and each coordinate changes a constant number of times, the total work is $O(N + M)$.

Proof. Store the vector, factor values and partial derivatives, gradient sums $g_i = \sum_{a: i \in I_a} \partial_i \phi_a$, and reverse incidence lists. Maintain the active coordinates whose projected displacement $d_i = \Pi_{K_i}(z_i - g_i) - z_i$ is nonzero. Initialization costs $O(N + \sum_a c_a)$.

In each round, save the active displacements and commit them simultaneously. Reevaluate each factor incident to a changed coordinate once, replacing its contributions to the gradient sums and objective value. Refresh active membership for coordinates whose value or gradient contribution changed. Unchanged factors retain valid derivatives, so induction gives exactly full simultaneous PGD.

Factor a is refreshed at most $\sum_{i \in I_a} \kappa_i$ times, and $c_a \geq |I_a|$ pays for both its evaluation and incidence updates. Saving and committing changes costs $O(\sum_i \kappa_i)$. Every active round changes a coordinate, so the same quantity pays for round bookkeeping. Membership flags and array positions give constant-time active-set updates, completing the bound. □

For the circuit construction, $N = O(B + S)$, $M = O(S)$, every rail changes at most once, and each factor has constant arity and cost, so the lemma gives the $O(B+S)$ work bound of Theorem 2.1. The argument tolerates unbounded fanout because the total number of factor incidences remains $O(S)$. The evaluator stops when the active set is empty, and a caller who wants dense copies of every iterate pays their output cost in addition.

A.5 A strict local minimum at the terminal state

A modified value term makes the computed terminal state a strict local minimum without changing the execution.

Proposition A.5. *Clamp the input rails and replace the value term in (2) by $\widetilde{V}(q(z))$, where*

$$\widetilde{V}(f, t) = \sum_{w \text{ internal}} (-f_w - t_w + 2 f_w t_w).$$

The trajectory, work bound, and decrease identity are unchanged, and at the terminal state the Hessian on the internal coordinates is $6I$, so the terminal state is a strict local minimum.

Proof. On valid wire states $f_w t_w = 0$, so $\widetilde{V}$ still counts minus the resolved gates. Lemma A.1 gives the same trajectory and decrease, and the constant-arity modification preserves the work bound.

At the terminal state, each correction term has zero residual, zero first residual derivatives, and zero leading factor $r(z_i)$, so its Hessian vanishes by the product rule. In $\widetilde{V}(q(z))$, mixed derivatives vanish because $q' = 0$. At a resolved pair $(f, t) = (1, 0)$ the diagonal entries are $(-1+2q(t))q''(f) = 6$ and $(-1 + 2q(f))q''(t) = 6$, with the same values at $(0, 1)$. Thus the free-coordinate Hessian is $6I$ at a stationary point, proving strict local minimality. □

This local-minimum guarantee concerns the terminal state reached from the encoded initialization; it does not assert convergence from other starts or a quantitative perturbation radius.

A.6 Worked example with a feedback edge

A two-gate circuit with a feedback edge illustrates that finite delay, and not acyclicity, is what the construction needs. Consider the two gates

$$g = x \text{ AND } h, \qquad h = g \text{ XOR } 1,$$

with input $x = 0$. Although the graph has the cycle $g \to h \to g$, the known false input resolves g in the first round, after which the second round resolves h. In the coordinate order (f_g, t_g, f_h, t_h),

Round	Internal rails	Resolved gate values	Φ_C
0	$(0,0,0,0)$	none	0
1	$(1,0,0,0)$	$g = 0$	-1
2	$(1,0,0,1)$	$g = 0,\ h = 1$	-2

After clamping x and the constant wire, the update rules simplify to $R_{f_g} = 1$, $R_{t_g} = 0$, $R_{f_h} = t_g$, and $R_{t_h} = f_g$, and writing $z = (z_1, z_2, z_3, z_4)$ gives the explicit objective

$$\begin{aligned}\Phi_C(z) = & -\sum_{i=1}^{4} q(z_i) + r(z_1)[q(z_1) - 1] + r(z_2)q(z_2) \\ & + r(z_3)[q(z_3) - q(z_2)] + r(z_4)[q(z_4) - q(z_1)].\end{aligned}$$

Its first two gradients are $(-1, 0, 0, 0)$ and $(0, 0, 0, -1)$ and its terminal gradient is zero, so the two active steps each reduce the objective by one, as Theorem 2.1 asserts.

A.7 Message passing

The variable–factor graph computes a gradient step in two communication phases.

Proposition A.6 (Exact realization on reached states)**.** *The PGD circuit construction has a fixed-weight ReLU message-passing realization with $O(B + S)$ nodes and edges and constant state and message dimension, in which two communication phases implement each PGD step. Total work is $O(B + S)$ with incremental evaluation and $O((B + S)(D + 1))$ with full graph evaluation in every round.*

Proof. Make one node for each coordinate and one for each factor, with an edge for every incidence; a finite node type identifies the polynomial template and finite edge labels identify argument positions. In the first phase, variable nodes send their current values to adjacent factor nodes, and in the second phase each factor returns its partial derivative to each incident variable. Summing the returned values gives the full gradient, after which each variable applies its coordinate projection, so the states at phase boundaries are the PGD iterates.

It remains to realize each local derivative table by a ReLU network. For $d \le 5$ arguments and $a \in \{0,1\}^d$, let the bump

$$\beta_a(x) = \mathrm{ReLU}\left(1 - \sum_{j=1}^{d} |x_j - a_j|\right).$$

On the Boolean cube β_a equals one at a and zero at every other vertex, so any local table p is represented on the reached states by $\sum_{a\in\{0,1\}^d} p(a)\beta_a(x)$, with a constant number of terms and a constant number of derivative outputs. Projection is the ReLU formula $\Pi_{[0,1]}(s) = \mathrm{ReLU}(s) - \mathrm{ReLU}(s-1)$.

Caching messages and aggregate sums as in Lemma A.4 gives $O(B+S)$ work, with all changes committed simultaneously. Full graph evaluation in each phase costs $O((B+S)(D+1))$ instead. □

The ReLU tables agree with the polynomial derivatives on Boolean arguments, as needed for this trajectory. Communication on a prescribed input graph, rather than the compiled factor graph, is a separate requirement.

A.8 Computational models and resource bounds

Finite-delay circuits store wires that resolve once; recurrent constructions reuse changing coordinates. Their space and precision costs differ.

General programs through circuits. For a bounded-word PRAM in the model of Heath (2023, Theorem 1), let n be its input-size parameter, B its encoded input bit count, $W(n)$ a supplied polynomial worst-case work bound, and T the parallel time of the specified execution. Its words have $\Theta(\log n)$ bits, and its primitive operations satisfy the compiler's circuit-size and delay requirements, so the compiler produces $S = O(W(n)\log^4 n)$ gates with delay $D = O(T\log^3 n)$. If a gate description can be generated in time $u(n) = \mathrm{polylog}(n)$, then generating the objective and executing PGD takes $O(B + W(n)\log^4 n\, u(n))$ work, while the representation and the cached execution use $O(B + W(n)\log^4 n)$ space and work respectively. Because the supplied budget may exceed the work of an easy input, the bound keeps its dependence on $W(n)$.

A bounded clock for compact local programs. For completeness, the same lift gives a particularly simple reusable Boolean construction. Suppose A updates d Boolean state bits by constant-arity rules R_i and has an absorbing halting state, in the sense that $A(b) = b$ whenever the halt bit satisfies $b_h = 1$, and suppose the specified execution halts within L transitions. With the correction sum Λ and a clock coordinate c, let

$$\Lambda(z) = \sum_{i=1}^{d} r(z_i)[q(z_i) - R_i(q(z))], \qquad \Phi_{\mathrm{loc}}(z,c) = \Lambda(z) + dc(1-q(z_h)).$$

Use the fixed box $[0,1]^d \times [0, dL]$ and initialize $c = dL$. At a reached Boolean state b before halting,

$$\nabla_z \Phi_{\mathrm{loc}} = b - A(b), \qquad \partial_c \Phi_{\mathrm{loc}} = d,$$

so each unit PGD step performs one source transition and decreases c by d. Once $b_h = 1$, the clock term has zero gradient in both z and c because $q'(1) = 0$ and $1 - q(1) = 0$, and the correction gradient $b - A(b)$ vanishes by the absorbing hypothesis, so the entire state is stationary. Each active step decreases the objective by at least d^2, because at a Boolean state the objective equals $dc(1-b_h)$ and c falls by d, while its squared gradient norm is at most $d^2 + d \le 2d^2$. There are $O(d)$ constant-arity factors, so if bit i changes κ_i times and occurs in δ_i factors, Lemma A.4 bounds the work by $O(d + L + \sum_i \kappa_i \max\{1, \delta_i\})$, and for an execution of length $\ell \le L$ the additive L can be replaced by ℓ. The clock uses $O(\log(dL))$ bits.

Randomness. Fixing a finite random tape reduces a bounded randomized execution to a deterministic program, so the exact-output guarantee applies pathwise.

B Circuit evaluation at a strongly convex minimizer

We prove Proposition 2.2 by placing a circuit's answer at the unique minimizer of a strongly convex energy. Flat gate interpolants remove backward derivatives on Boolean states, and decreasing coordinate scales control the Hessian globally.

B.1 Circuit normalization and coordinate scales

Let V_0 be the set of clamped input sites and V_1 the set of gates whose states are optimized, and write $n = |V_1|$ and $S_{\mathrm{tot}} = |V_0| + n$ for the total circuit size, which therefore includes the input sites. The circuit is a directed acyclic graph over AND, OR, NOT, and COPY gates with at most two parents per gate, and constant inputs may be included among the clamped sites. Prune sites with no path to an output, and replace an AND or OR gate with identical parents by COPY, so each parent set P_v has distinct sites. Write $v \to w$ for a dependency edge. All counts refer to this normalized circuit; an arbitrary two-input truth table requires only constant size and depth overhead in this basis.

Inputs $b \in \{0,1\}^{V_0}$ are clamped at depth zero. Gate depth is one plus maximum parent depth, $D \geq 1$ is the largest depth, and $r \geq 1$ counts sinks, all of which are outputs after pruning. An output that is an input is read directly from b.

Let $z^*(b) \in \{0,1\}^{V_1}$ be the correct gate-value vector for the input b. Positive dyadic scales a_v, which depend only on the circuit and not on b, will place this vector at $x^*_v = a_v z^*_v(b)$, and an output bit at gate o is decoded by the test $x_o > a_o/2$.

The following scale budget controls arbitrary fanout without an excessive bit cost.

Lemma B.1 (Dyadic scales for arbitrary fanout)**.** *For every $0 < \theta \leq 1$, there are integer powers of two $a_v \in (0,1]$ such that*

$$\sum_{w : v \to w} (a_w/a_v)^2 \leq \theta \quad \text{for every site } v. \tag{11}$$

Their largest scale exponent $P = \log_2(1/\min_v a_v)$ satisfies

$$P \leq \tfrac{1}{2}\log_2 r + \tfrac{D}{2}\log_2(8/\theta). \tag{12}$$

Proof. Assign weights in reverse topological order: sinks have $W_v = 1$, and each nonsink receives the least power of four at least $\theta^{-1}\sum_{w:v\to w} W_w$. With $W_{\max} = \max_v W_v$, put $a_v = \sqrt{W_v/W_{\max}}$. These are powers of two and satisfy (11).

Rounding up to a power of four gives $W_v \leq (4/\theta)\sum_{w:v\to w} W_w$. Substitution bounds W_v by $\sum_\pi (4/\theta)^{|\pi|}$ over paths from v to sinks. Each path has length at most D; unfolding the dependencies of a fixed sink into a binary tree shows at most 2^D paths from v to that sink. Hence

$$W_v \leq r\, 2^D (4/\theta)^D = r(8/\theta)^D.$$

Since the sink weights are one, $\min_v a_v = W_{\max}^{-1/2}$, which proves (12). □

For the remaining construction fix $\theta = 1/4096$, so that $\log_2(8/\theta) = 15$ and (12) reads $P \leq 7.5D + \frac{1}{2}\log_2 r$, regardless of the circuit's maximum fanout.

B.2 The local energy

Define the clipped scalar interpolation p, which reads a parent value, and the bounded output extension H, which reads a gate's own output, by

$$p(t) = \begin{cases} 0, & t \le 0, \\ 3t^2 - 2t^3, & 0 \le t \le 1, \\ 1, & t \ge 1, \end{cases} \qquad H(t) = \begin{cases} -4, & t \le -8, \\ t + t^2/16, & -8 \le t \le 0, \\ t, & 0 \le t \le 1, \\ t - (t-1)^2/16, & 1 \le t \le 9, \\ 5, & t \ge 9. \end{cases} \tag{13}$$

Their values and first derivatives agree at every breakpoint, so both functions are globally $C^{1,1}$ with

$$|p'| \le 3/2, \quad |p''| \le 6, \quad |H| \le 5, \quad |H'| \le 1, \quad |H''| \le 1/8, \tag{14}$$

where second derivatives here and throughout this appendix are interpreted almost everywhere. At the Boolean points $p(z) = z$, $p'(z) = 0$, $H(z) = z$, and $H'(z) = 1$, which is what makes a unit gradient step act as an exact gate evaluation.

Write $y_v = x_v/a_v$ for the normalized value of each optimized coordinate. For gate v, let $G_v(y; b)$ be its gate polynomial with every optimized parent value y_i replaced by $p(y_i)$ and every clamped parent replaced by b_i. Here AND uses the product of its two parent values, OR their sum minus their product, and NOT and COPY the complement and the identity; since p takes values in $[0, 1]$, so does G_v. Differentiating these gate polynomials and applying (14) bounds each first partial by $3/2$, each pure second partial by 6, and each mixed second partial by $(3/2)^2 = 9/4$. A first partial vanishes whenever the differentiated parent coordinate is Boolean because p' does. Distinct parents ensure these bounds; repeated AND/OR inputs were therefore replaced by COPY before choosing the scales.

Given the clamped input b, define the energy

$$E_b(x) = \frac{1}{2} \sum_{v \in V_1} x_v^2 - \sum_{v \in V_1} a_v^2 H(x_v/a_v) G_v(y; b), \tag{15}$$

whose first sum is quadratic in each coordinate and whose second sum contains one interaction factor per optimized gate. Each factor involves one output and at most two optimized parents, the input enters only through its clamped slots, and there are $O(S_{\mathrm{tot}})$ factors, each with a constant number of polynomial pieces of degree at most eight.

Theorem B.2 (The ordinary-gradient circuit compiler)**.** *The energy (15) is globally $C^{1,1}$, $3/4$-strongly convex, and $5/4$-smooth. Its unit-GD map is globally $1/4$-contractive, and*

$$E_b(x - \nabla E_b(x)) \le E_b(x) - \tfrac{3}{8} \|\nabla E_b(x)\|_2^2. \tag{16}$$

From every initialization $x^0 = a \odot z^0$ with $z^0 \in \{0, 1\}^{V_1}$, it reaches the unique global minimizer $x^* = a \odot z^*(b)$ within D iterations. At this point,

$$\nabla E_b(x^*) = 0, \qquad E_b(x^*) = -\tfrac{1}{2} \sum_{v : z_v^*(b) = 1} a_v^2. \tag{17}$$

Each local numerical coefficient, breakpoint, and exact orbit coordinate has an $O(D + \log r)$-bit rational description. Dense evaluation of the D steps uses $O(S_{\mathrm{tot}} D)$ arithmetic operations.

B.3 Global conditioning and exact execution

Proof of Theorem B.2. For regularity, the scalar functions H and p and their first derivatives are bounded, and their first derivatives are globally Lipschitz. For fixed positive scales, finite products of their scaled versions therefore have globally Lipschitz first derivatives, so each interaction factor, and with it E_b, is globally $C^{1,1}$. The pieces are separated by finitely many coordinate hyperplanes, and the largest factor degree is $2+3+3=8$.

Off the break hyperplanes, decompose $\nabla^2 E_b - I$ into output–output, output–parent, and parent–parent derivatives. Differentiating a factor twice in its own output gives a diagonal matrix with entries $-H''(y_v)G_v$, whose operator norm is at most $1/8$ by (14) and $0 \le G_v \le 1$.

For the output–parent derivatives, define a matrix M by

$$M_{vi} = (a_v/a_i)H'(y_v)\partial_i G_v \quad (i \in P_v \cap V_1),$$

with all other entries zero, where $\partial_i G_v$ differentiates the normalized parent variable. Fan-in at most two, the bounds $|H'| \le 1$ and $|\partial_i G_v| \le 3/2$, and the scale budget (11) give

$$\begin{aligned}\|Mu\|_2^2 &\le 2(3/2)^2 \sum_v \sum_{i \in P_v \cap V_1} (a_v/a_i)^2 u_i^2 \\ &\le (9/2)\theta \|u\|_2^2,\end{aligned}$$

so the output–parent correction $-(M + M^\top)$ has norm at most $2\|M\|_{\mathrm{op}} \le 3\sqrt{2\theta}$.

For the parent–parent block, $|H| \le 5$ and the pure second-partial bound 6 show that the pure second derivatives contribute at most

$$5 \cdot 6 \sum_v \sum_{i \in P_v \cap V_1} (a_v/a_i)^2 u_i^2 \le 30\theta \|u\|_2^2$$

to the absolute value of its quadratic form. If gate v has two optimized parents i, j, the mixed bound $9/4$ together with $2|st| \le s^2 + t^2$ shows that its mixed term is at most

$$2 \cdot 5 \cdot \frac{9}{4} \left| \frac{a_v}{a_i} u_i \frac{a_v}{a_j} u_j \right| \le \frac{45}{4} \left((a_v/a_i)^2 u_i^2 + (a_v/a_j)^2 u_j^2 \right).$$

Summing over gates and using (11) bounds this part by $(45/4)\theta\|u\|_2^2$. Since all three matrices are symmetric, their quadratic-form bounds also bound their operator norms, and adding the three bounds gives

$$\|\nabla^2 E_b - I\|_{\mathrm{op}} \le \frac{1}{8} + 3\sqrt{2\theta} + \frac{165}{4}\theta < \frac{1}{4}, \qquad \theta = 1/4096, \tag{18}$$

where for the chosen θ the right side is smaller than $1/8 + 3/32 + 1/64 = 15/64$.

Integrate this bound along the finitely many smooth pieces of a generic line segment; continuity of the gradient joins them, and approximation covers segments in break hyperplanes. The spectral interval $[3/4, 5/4]$ gives strong convexity and smoothness, while integrating $I - \nabla^2 E_b$ gives

$$\|(y - \nabla E_b(y)) - (x - \nabla E_b(x))\|_2 \le \tfrac{1}{4}\|y - x\|_2.$$

Finally, the smoothness inequality yields

$$E_b(x - \nabla E_b(x)) \le E_b(x) - (1 - \tfrac{5}{8})\|\nabla E_b(x)\|_2^2,$$

proving (16).

Exact execution and the minimizer. Differentiating every factor in which x_v appears, including those of the downstream gates w with $v \to w$, gives

$$\partial_v E_b(x) = x_v - a_v H'(y_v) G_v - \sum_{w: v \to w} \frac{a_w^2}{a_v} H(y_w) \partial_v G_w. \tag{19}$$

When all normalized optimized coordinates are Boolean, $H'(y_v) = 1$ and every derivative $\partial_v G_w$ in the downstream sum vanishes because p' does, so unit GD satisfies

$$x_v - \partial_v E_b(x) = a_v \, \mathrm{gate}_v(y_{P_v}; b). \tag{20}$$

Thus the scaled Boolean set is invariant. Depth-one gates become correct in one update, and a gate becomes correct once its parents are correct. Induction on depth proves that all gates are correct by time D from every scaled Boolean start.

At that state, (20) is a fixed-point identity, so $\nabla E_b(x^*) = 0$, and strong convexity makes x^* the unique global minimizer. Substituting $H(z_v^*) = z_v^*$ and $G_v(z^*; b) = z_v^*$ into (15) gives (17).

Numerical descriptions and work. Every scale is a power of two with exponent at most P. The coefficients of a local degree-eight piece are fixed dyadic constants multiplied by a bounded number of scales or their reciprocals, the breakpoints are fixed dyadic constants times scales, and the orbit coordinates are zero or a_v. By Lemma B.1, each such numerical datum has $O(P+1) = O(D + \log r)$ bits. This is a statement about individual local numerical data; graph indices have their usual $O(\log S_{\mathrm{tot}})$-bit cost.

Each constant-size factor costs $O(1)$ arithmetic operations and comparisons to differentiate. Dense execution therefore costs $O(S_{\mathrm{tot}} D)$ operations, with finite-arithmetic bit costs charged separately. □

Global contraction holds from every real start; exact execution in D steps uses a scaled Boolean start. The output scale may be exponentially small in depth, so its bit cost and decoding margin $a_o/2$ are essential. The energy is $C^{1,1}$, with no higher regularity claimed at breakpoints.

C Recurrent computation in a fixed convex energy

To reuse coordinates across time, encode each Boolean register in the direction of a signal–clock pair while descent shrinks its scale. One fixed energy then simulates every finite prefix of a recurrent computation, although its unique minimizer contains no answer. Energy-dissipating simulation has precedents such as Šíma and Orponen (2001); the construction here specifies ordinary discrete GD, constant conditioning, local factors, and the arithmetic of this representation.

C.1 The recurrent model and the local-clock theorem

Let $m \geq 1$ denote the number of Boolean registers and $F : \{0,1\}^m \to \{0,1\}^m$ their synchronous transition map. For each register i, the parent list P_i is an ordered list of at most two parent registers and F_i is an arbitrary Boolean truth table on those input occurrences, with constants, self-inputs, repeated inputs, and directed cycles all permitted. Let $S_i := \{i\} \cup \{j : j \text{ occurs in } P_i\}$ be the scope of factor i, the set of registers it uses, and define the maximum factor incidence

$$\Delta := \max_{1 \leq j \leq m} |\{i : j \in S_i\}|,$$

so that $1 \le \Delta \le m$. This incidence counts a factor once for each distinct register it contains, although the formulas below still count repeated input occurrences when taking products and derivatives.

To represent a register at a changing physical scale, use a signal $x_i \in \mathbb{R}$ and a clock $c_i \in \mathbb{R}$, and write $u = (x, c) \in \mathbb{R}^{2m}$ for the full physical state. At a synchronized positive clock value c, the Boolean configuration z is represented by $x_i = cz_i$ and $c_i = c$ for every i; the clocks are coordinates of the autonomous descent algorithm rather than an externally supplied time schedule.

Theorem C.1 (Local clocks for recurrent Boolean computation)**.** *For the transition map F above, fix the universal constant $K_0 := 200000$ together with the scale exponent k and the scale ratio τ given by*

$$k := \lceil \log_2(4(1 + K_0\Delta)) \rceil, \qquad \tau := 2^{-k}.$$

There is an explicit energy $E_F : \mathbb{R}^{2m} \to \mathbb{R}$, independent of the initial Boolean configuration and the time horizon, with the following properties.

1. *The energy is a sum of $O(m)$ factors, each involving at most six real coordinates; it is globally $C^{1,1}$ and $3/4$-strongly convex, has $5/4$-Lipschitz gradient, and its unique minimizer is the origin.*

2. *Ordinary gradient descent with step size one, initialized at $x_i^0 = z_i^0$ and $c_i^0 = 1$ for any $z^0 \in \{0,1\}^m$, satisfies*

$$c_i^t = \tau^t, \qquad x_i^t = \tau^t (F^t(z^0))_i \quad (i = 1, \dots, m;\ t \ge 0). \tag{21}$$

   The Boolean register at any finite time T is decoded by $x_i^T - c_i^T/2 > 0$.

3. *The unit-step map $u \mapsto u - \nabla E_F(u)$ is globally $1/4$-Lipschitz, and every exact update $u^+ = u - \nabla E_F(u)$ obeys*

$$E_F(u^+) \le E_F(u) - \frac{3}{8}\|u^+ - u\|_2^2. \tag{22}$$

4. *Each update uses a gather stage and a return stage on the gate incidence graph, followed by local aggregation, and takes $O(m)$ arithmetic operations on explicit, constant-size piecewise rational factor formulas, whose bit cost is additional.*

The dimension is independent of the horizon, while the decoder margin is $\tau^T/2$. Computation is read from the transient; the minimizer is zero. We establish the orbit by cancelling backward gate and clock derivatives, and prove conditioning by a uniform bound on each interaction.

C.2 Gate interpolation and the clock factors

Use the clipped smoothstep p from (13), with $|p'| \le 3/2$, $|p''| \le 6$ almost everywhere, and $p'(0) = p'(1) = 0$. For a two-input truth table with values $b_{00}, b_{01}, b_{10}, b_{11} \in \{0,1\}$, let g be its multilinear extension

$$g(a,b) := b_{00} + (b_{10} - b_{00})a + (b_{01} - b_{00})b + (b_{11} - b_{10} - b_{01} + b_{00})ab,$$

with unary and constant gates using the corresponding affine and constant extensions, and define the interpolated gate $G_i(z) := g_i(p(z_j) : j \text{ occurs in } P_i)$. Because the multilinear extension is a convex combination of truth-table values on the unit square, $0 \le G_i \le 1$ on all of $\mathbb{R}^m$, while on the Boolean cube $G_i = F_i$ and every input derivative vanishes. A repeated parent contributes both chain-rule terms, each of which still contains p'.

The clock derivative needs a second cancellation in addition to flat gate inputs. Define the bounded interpolant $H : \mathbb{R} \to \mathbb{R}$ by

$$H(t) := \begin{cases} -1/2, & t \le -1, \\ t + t^2/2, & -1 \le t \le 0, \\ t - 3t^2/2 + t^3, & 0 \le t \le 1, \\ 1/2 + (t-1) - (t-1)^2/2, & 1 \le t \le 2, \\ 1, & t \ge 2, \end{cases}$$

whose pieces match in value and first derivative, so that H is $C^{1,1}$ with $|H| \le 1$, $|H'| \le 1$, and $|H''| \le 3$ almost everywhere, and whose derivatives are supported in $[-1, 2]$. For a Boolean argument $z \in \{0, 1\}$ it satisfies

$$H(z) = z/2, \qquad H'(z) = 1, \qquad 2H(z) - zH'(z) = 0, \tag{23}$$

which cancels the own-clock derivative on Boolean states.

Because incomparable local clocks would make derivatives of their ratios unbounded, interactions between them are suppressed by the clock bump $\chi : \mathbb{R} \to [0, 1]$, defined by

$$\chi(r) := \begin{cases} p(2r-1), & r \le 1, \\ p(2-r), & r \ge 1. \end{cases}$$

This $C^{1,1}$ function is supported in $[1/2, 2]$ and satisfies $\chi(1) = 1$, $\chi'(1) = 0$, $|\chi'| \le 3$, and $|\chi''| \le 24$ almost everywhere; its vanishing derivative at one will leave synchronized clocks unchanged by the interaction terms.

For nonzero involved clocks, write $z_j := x_j/c_j$ for the normalized signal and define the local interaction

$$V_i(x, c) := c_i^2 H(z_i) G_i(z) \prod_{j \text{ occurs in } P_i} \chi(c_i/c_j), \tag{24}$$

and set $V_i = 0$ if any clock indexed by S_i is zero. A self-input has a constant ratio factor $\chi(c_i/c_i) = 1$ when $c_i \neq 0$, so that factor can be omitted. The proposed energy is

$$E_F(x, c) := \frac{1}{2} \sum_{i=1}^{m} x_i^2 + \frac{1-\tau}{2} \sum_{i=1}^{m} c_i^2 - \tau \sum_{i=1}^{m} V_i(x, c). \tag{25}$$

These formulas are piecewise rational in physical coordinates; their values at zero clocks are part of the definition, and their differentiability there is the principal point of Lemma C.2.

C.3 Global regularity and the exact descent orbit

Uniformly bounded interaction Hessians allow a small coefficient to preserve the quadratic energy's conditioning, including at zero clocks.

Lemma C.2 (Uniform regularity of one local interaction)**.** *Each factor V_i in (24), extended by zero as specified, is globally $C^{1,1}$ with a K_0-Lipschitz gradient, where $K_0 = 200000$, and its gradient is zero whenever an involved clock is zero. The bound holds for arbitrary truth tables, self-inputs, and repeated inputs.*

Proof. Because χ, χ', and its almost-everywhere second derivative all vanish outside $[1/2, 2]$, each parent clock obeys

$$\frac{1}{2} \le \frac{c_i}{c_j} \le 2$$

whenever a nonzero value, first-derivative term, or second-derivative term occurs. Every active term therefore has clocks of the same sign and comparable magnitudes. Every unbounded normalized signal in a derivative is multiplied by a derivative of its own H or p, whose compact derivative supports bound that normalized signal by two in absolute value.

To obtain an explicit constant, expand the gate polynomial into monomials. The absolute sum of its coefficients is at most $1 + 1 + 1 + 2 = 5$, a bound attained by XNOR. A monomial of V_i, apart from its scalar coefficient, has the form $s^2 A$, where $s := c_i$ is the own clock and A is a product of at most five functions, each of absolute value at most one. These consist of one H, at most two p functions, and at most two clock bumps. For each physical coordinate pair a, b, every nonzero product derivative term satisfies the conservative individual-function bounds

$$|\partial_a f| \le \frac{12}{|s|}, \qquad |\partial_{ab} f| \le \frac{432}{s^2}. \tag{26}$$

These bounds concern factors within an active product term, since an isolated $p(x_j/c_j)$ need not satisfy them when its clock is incomparable with s.

For $H(x/s)$, first-derivative numerators are H' and $-zH'$, bounded by two, and second-derivative numerators are H'', $-(H' + zH'')$, and $2zH' + z^2 H''$, bounded by sixteen. The analogous bounds for $p(x/c_j)$ are $3/|s|$ and $36/s^2$. For $\chi(s/c_j)$ they are $12/|s|$ and $4(2 \cdot 2 \cdot 3 + 2^2 \cdot 24)/s^2 = 432/s^2$. Self-input ratio factors were omitted before differentiation.

Applying the product rule in the physical variables gives

$$|\partial_a A| \le \frac{60}{|s|}, \qquad |\partial_{ab} A| \le \frac{5 \cdot 432 + 20 \cdot 12^2}{s^2} = \frac{5040}{s^2}.$$

Since a Hessian entry of $s^2 A$ collects the terms $2A$, two copies of $2s\,\partial A$, and $s^2 \partial^2 A$, it has absolute value at most $2 + 120 + 120 + 5040 = 5282$. A factor involves at most six physical coordinates and the coefficient absolute sum is at most five, so the row-sum (Gershgorin) bound $\|W\|_2 \le \max_i \sum_j |W_{ij}|$ for a symmetric matrix W gives

$$\|\nabla^2 V_i\|_2 \le 6 \cdot 5 \cdot 5282 = 158460 < K_0 \tag{27}$$

almost everywhere away from zero clocks. This estimate already includes repeated inputs, because their repeated p and χ factors occur among the at most five product factors, and identifying a self-input adds no multiplicity factor either, because the product rule was applied in physical coordinates throughout.

For the zero extension, $|V_i| \le s^2$ and every gradient entry is at most $5(2 + 60)|s| = 310|s|$. If one involved clock is zero and another is nonzero, a ratio bump vanishes locally. If all are zero, these estimates prove differentiability with zero gradient and gradient continuity, uniformly in the signals.

On a generic line segment, clock zeros and polynomial-piece boundaries have finitely many crossings. Integrating (27) between them and using gradient continuity gives the global Lipschitz bound. Exceptional segments follow by approximation, and the gradient is identically zero on an involved zero-clock hyperplane. This proves the claimed $C^{1,1}$ regularity. □

Proof of Theorem C.1. For a vector $v \in \mathbb{R}^{2m}$, write v_{S_i} for its signal and clock entries belonging to registers in S_i. Lemma C.2 gives

$$\left| v^{\mathsf{T}} \nabla^2 \Big(\sum_i V_i \Big) v \right| \le K_0 \sum_i \|v_{S_i}\|_2^2 \le K_0 \Delta \|v\|_2^2$$

almost everywhere, and comparing (25) with $\|u\|_2^2/2$ yields

$$\|\nabla^2 E_F - I\|_2 \le \tau(1 + K_0\Delta) \le \frac{1}{4}.$$

The segment-integration argument in the proof of Theorem B.2 now gives strong convexity, smoothness, contraction, and (22).

For the exact orbit, suppose $c_i = c \neq 0$ and $x_i = cz_i$ with $z \in \{0,1\}^m$. All clock bumps then equal one with vanishing first derivatives, and all gate-input derivatives vanish as well, so the only surviving signal derivative is

$$\partial_{x_i} \sum_j V_j = cH'(z_i)F_i(z) = cF_i(z).$$

Every clock derivative is zero: the input and ratio terms vanish, and the own-clock term is $c[2H(z_i) - z_i H'(z_i)]F_i(z) = 0$ by (23). This calculation includes self-inputs, because their gate-input derivatives are still zero. The full energy gradient is therefore

$$\partial_{x_i} E_F = c(z_i - \tau F_i(z)), \qquad \partial_{c_i} E_F = (1-\tau)c, \tag{28}$$

and subtracting it gives $(x_i^+, c_i^+) = (\tau c F_i(z), \tau c)$, so induction from $c = 1$ proves (21). The origin has zero gradient by the zero-clock extension, and strong convexity makes it the unique minimizer.

Each factor gathers the signal–clock pairs in its scope, evaluates its constant-size formula, and returns derivative contributions. Summing those contributions gives the gradient in $O(m)$ operations because each factor meets at most three registers. Arithmetic bit costs remain additional. □

The proof supplies rational local modules, and implementing their arithmetic with finite-bit messages incurs an additional bit cost. Appendix D provides a rounded construction with an explicit precision guarantee.

D Local rounded gradient computation

We now give a local energy with explicit finite arithmetic for Theorem 3.1. Affine neighborhoods replace the pointwise clock cancellation of Appendix C, so each update restores the next Boolean state despite persistent data errors.

Fix a synchronous Boolean transition map $F : \{0,1\}^m \to \{0,1\}^m$ in which each coordinate has at most two distinct parents, repeated parent occurrences having been collapsed into a unary truth table. Write P_i for its parent set and $S_i := \{i\} \cup P_i$ for its scope, and assume that the maximum factor incidence satisfies $\Delta = \max_j |\{i : j \in S_i\}| \le 4$. The tape pipeline of Appendix E has $\Delta \le 3$, which discharges this hypothesis there. The dynamic coordinates are $X = (x, u) \in \mathbb{R}^{2m}$, and the fixed constants are the scale ratio $\tau = 2^{-25}$, the clock rescaling $\alpha = 64$, and the time-indexed scale $\gamma_t := \tau^t$.

Here u_i is the clock coordinate, $c_i = u_i/\alpha$ its signal scale, and the decoder is $x_i > u_i/(2\alpha)$.

Theorem D.1 (Rounded local computation in a convex energy)**.** *For every transition map F above, there is an explicit fixed energy $E_F : \mathbb{R}^{2m} \to \mathbb{R}$, independent of the initial bits and the horizon, with the following properties.*

1. *The energy is a sum of $O(m)$ factors of arity at most six; it is globally $C^{1,1}$, $3/4$-strongly convex, and $5/4$-smooth, with unique minimizer zero, and its step-one GD map is globally $1/4$-contractive.*

2. *For every finite horizon T, a specified implementation uses 37 significand bits, round to nearest, gradual underflow, and a finite exponent range of magnitude $O(T)$. Initialized by $x_i^0 = z_i^0 \in \{0,1\}$ and $u_i^0 = \alpha$, it satisfies*

$$u_i^t = \alpha\gamma_t, \qquad \left| x_i^t/\gamma_t - (F^t(z^0))_i \right| \le 1/16 \quad (0 \le t \le T).$$

   After each raw update, each stored data value may be replaced by another represented value within $\gamma_{t+1}/64$ of it; clocks receive no such additional errors. Each exponent uses $O(\log(T+2))$ storage bits.

3. *Each update uses two message stages on the original gate-incidence graph, two dynamic features per register, and constant-size local modules. A finite family of ReLU modules implements the gradient formulas on every covered state, and its exact-real extension is globally 64τ-contractive, although off those states it need not be a gradient map.*

4. *Every covered rounded and perturbed update obeys $E_F(X^{t+1}) < 2\tau^2 E_F(X^t) < E_F(X^t)$.*

The energy is independent of T; only the arithmetic's exponent range grows with the horizon. Rounding fixes the clock scale exactly and each update regenerates the data, preventing accumulated logical errors.

D.1 The local factors and their purpose

With the register scopes and coordinates fixed above, set

$$K := 25, \qquad \tau := 2^{-K}, \qquad p := K + 12 = 37, \qquad \alpha := 64, \qquad c_i := u_i/\alpha, \tag{29}$$

where p is the number of binary significand bits and c_i is the local signal scale. All scale changes in (29) are powers of two, and the rescaling α will make a clock feedback force smaller than a rounding interval without changing the number of local coordinates.

To suppress backward derivatives near a Boolean input, define the clipped quintic $P_0 : \mathbb{R} \to [0,1]$ by

$$P_0(t) := \begin{cases} 0, & t \le 0, \\ 10t^3 - 15t^4 + 6t^5, & 0 < t < 1, \\ 1, & t \ge 1, \end{cases}$$

whose first two derivatives agree at the endpoints, so that $P_0 \in C^2$. On $(0,1)$ they are $P_0'(t) = 30t^2(1-t)^2$ and $P_0''(t) = 60t(1-t)(1-2t)$, so $|P_0'| \le 15/8$ and $|P_0''| \le 6$. Define the bounded bump χ_0, the affine-neighborhood interpolant H, and the gate-input plateau R by

$$\begin{aligned} \chi_0(s) &:= P_0(2 - 8|s|), \\ H(q) &:= q\chi_0(q) + (q - \tfrac{1}{2})\chi_0(q-1), \\ R(q) &:= P_0((q - 1/8)/(3/4)), \end{aligned}$$

where the absolute value introduces no singularity because χ_0 is constant near zero, and where the two summands of H have disjoint supports. For $z \in \{0,1\}$ and $|q-z| \le 1/8$, these functions satisfy

$$H(q) = q - z/2, \qquad H'(q) = 1, \qquad H''(q) = 0, \qquad R(q) = z, \tag{30}$$

so that the affine part supplies a nonzero output derivative while the plateau makes every input derivative vanish.

Let g_i be the multilinear truth polynomial from Appendix C, with unary and constant gates using the corresponding affine and constant polynomials, and for a normalized data vector $q \in \mathbb{R}^m$ define the interpolated gate $G_i(q) := g_i(R(q_j) : j \in P_i)$. As before, $0 \le G_i \le 1$ and its coefficient absolute sum is at most five. On a Boolean $1/8$-neighborhood, $G_i = F_i(z)$ and all parent derivatives of G_i are zero.

Local clocks require a further cutoff to prevent unbounded derivatives when their scales are incomparable. Define the clock cutoff

$$\chi(r) := P_0(4r-2)P_0(4-2r),$$

which is supported in $[1/2, 2]$ and equals one on $[3/4, 3/2]$, so that its derivatives vanish near the synchronized ratio one. The two factors have derivative supports $(1/2, 3/4)$ and $(3/2, 2)$, which are disjoint, so at most one product-rule term is active at any r and

$$|\chi'| \le 4 \cdot \frac{15}{8} = \frac{15}{2} < 12, \qquad |\chi''| \le 16 \cdot 6 + 16(15/8)^2 + 4 \cdot 6 < 180.$$

The constants 12 and 180 are convenient upper bounds carried through the calculation below, the second retaining all three product-rule terms.

For nonzero clocks indexed by S_i, define the local interaction

$$V_i(X) := c_i^2 H(x_i/c_i) G_i((x_j/c_j)_{j=1}^m) \prod_{j \in P_i \setminus \{i\}} \chi(c_i/c_j), \tag{31}$$

and set $V_i(X) := 0$ if one of those clocks vanishes. A self-parent contributes no clock cutoff because its ratio would be identically one. The proposed energy is

$$E_F(X) := \frac{1}{2}\sum_i x_i^2 + \frac{1-\tau}{2}\sum_i u_i^2 - \tau \sum_i V_i(X). \tag{32}$$

Each factor involves at most six physical coordinates, and the formulas are piecewise rational, including the explicitly defined values at zero clocks.

D.2 Global conditioning, including zero clocks

As in Appendix C, clock cutoffs bound the interaction Hessians globally. The new interpolants require the following constants.

Lemma D.2 (Regularity of a rounded-computation factor)**.** *Each factor V_i in (31), with its specified zero extension, is globally $C^{1,1}$ with gradient Lipschitz constant at most $C_0 := 2{,}000{,}000$, and its gradient is zero whenever an involved clock is zero. As a result, E_F in (32) is globally $3/4$-strongly convex and $5/4$-smooth, has unique minimizer zero, and its step-one GD map is globally $1/4$-contractive.*

Proof. The support of χ and its derivatives implies that every nonzero product-derivative term has

$$\frac{1}{2} \le c_i/c_j \le 2 \qquad (j \in P_i), \tag{33}$$

a condition that a self-parent satisfies identically, so all involved clocks have the same sign and comparable magnitudes in an active term. An unbounded normalized data coordinate can occur in a derivative only multiplied by a derivative of its own H or R, whose compact derivative supports then bound that normalized coordinate.

Since $|\chi_0'| \le 15$ and $|\chi_0''| \le 384$, disjointness of the two bumps gives

$$|H| \le 1, \qquad |H'| \le 13, \qquad |H''| \le 320.$$

For example, on the second bump $|q-1/2| \le 3/4$, so the last two bounds follow from $1+(3/4)15 < 13$ and $2 \cdot 15 + (3/4)384 = 318 < 320$, while the first bump has smaller bounds. The data support of a nonzero derivative of H lies in $|q| \le 5/4$, and the plateau R has $|R'| \le 5/2$, $|R''| \le 11$, and derivative support in $[1/8, 7/8]$.

To translate these estimates into physical-coordinate derivatives, put $s := c_i$ within this proof. For $H(x_i/s)$, the first-derivative numerators are H' and $-qH'$, and the second-derivative numerators are H'', $-(H' + qH'')$, and $2qH' + q^2H''$. Since $|q| \le 5/4$ on the derivative support, $|H'| \le 13$, and $|H''| \le 320$, the first-derivative numerators are bounded by $(5/4) \cdot 13 < 17$ and the second-derivative numerators by $2(5/4)13 + (5/4)^2\, 320 < 533$. For $R(x_j/c_j)$, (33) bounds the rescaled first derivatives by $5/|s|$ and the second derivatives by $52/s^2$, the largest clock numerator being at most

$$4(2(7/8)(5/2) + (7/8)^2 \cdot 11) < 52.$$

For $\chi(s/c_j)$, the rescaled first derivatives are at most $48/|s|$, and the largest second derivative is the parent-clock derivative, whose bound is

$$\frac{4(2 \cdot 2 \cdot 12 + 2^2 \cdot 180)}{s^2} = \frac{3072}{s^2},$$

while the own-clock and mixed-clock bounds are smaller. These calculations show that each elementary factor f of an active product obeys

$$|f| \le 1, \qquad |\partial_a f| \le 48/|s|, \qquad |\partial_{ab} f| \le 3072/s^2 \tag{34}$$

for all pairs of physical coordinates a, b, including $a = b$.

The product count in Lemma C.2 now applies with first- and second-derivative constants 48 and 3072. Explicitly, a monomial is $s^2 A$ with at most five elementary factors, so

$$|\partial_a A| \le 240/|s|, \qquad |\partial_{ab} A| \le (5 \cdot 3072 + 20 \cdot 48^2)/s^2 = 61440/s^2.$$

A Hessian entry of $s^2 A$ is at most $2 + 4 \cdot 240 + 61440 = 62402$. The coefficient absolute sum is at most five and there are at most six coordinates, giving

$$\|\nabla^2 V_i\|_{\mathrm{op}} \le 6 \cdot 5 \cdot 62402 = 1{,}872{,}060 < C_0. \tag{35}$$

Self-input contributions are included because differentiation is in physical coordinates. Rescaling c to u multiplies the Hessian on both sides by a diagonal contraction and cannot increase this bound.

Moreover, $|V_i| \le c_i^2$ and $\nabla V_i = O(|c_i|)$ uniformly in the signals. Thus the zero-extension and segment-integration argument of Lemma C.2 applies unchanged: mixed zero and nonzero clocks

make a cutoff vanish locally, while all-zero clocks give zero value and gradient by these estimates. Consequently the factor is globally $C^{1,1}$ with gradient Lipschitz constant C_0.

Finally, let $h \in \mathbb{R}^{2m}$ be a test vector and write h_{S_i} for its entries at both coordinates of each register in S_i. Incidence counting gives

$$\left| h^T \nabla^2 \Big(\sum_i V_i \Big) h \right| \le C_0 \sum_i \|h_{S_i}\|^2 \le C_0 \Delta \|h\|^2,$$

and comparing (32) with $\|X\|^2/2$ gives

$$\|\nabla^2 E_F - I\|_{\mathrm{op}} \le \tau(1 + C_0\Delta) \le \frac{8{,}000{,}001}{33{,}554{,}432} < \frac{1}{4}.$$

Integrating these almost-everywhere Hessian bounds proves strong convexity, smoothness, and the contraction of $I - \nabla E_F$, and since every factor gradient vanishes at zero, zero is the unique minimizer. □

D.3 The gradient on Boolean tubes and the numerical invariant

Inside a Boolean neighborhood, flat input derivatives and an affine output factor give a simpler gradient that restores the next bit. For $z \in \{0,1\}^m$ and $c > 0$, define the synchronized tube

$$\mathcal{T}(z,c) := \{X : u_i = \alpha c \text{ for every } i,\ |x_i/c - z_i| \le \delta \text{ for every } i\}, \qquad \delta := 1/16, \tag{36}$$

The clock equality will be maintained by rounding (Lemma D.5).

Lemma D.3 (Gradient on a Boolean tube)**.** *For $X \in \mathcal{T}(z,c)$, let $b := F(z)$ be the successor Boolean state. The energy gradients satisfy*

$$\partial_{x_i} E_F = x_i - \tau c b_i, \qquad \partial_{u_i} E_F = (1-\tau)u_i - (\tau/\alpha) b_i (x_i - z_i c), \tag{37}$$

and on an open neighborhood of each tube state the same identities hold with the Boolean labels retained and the common c on the right-hand sides replaced by the individual $c_i = u_i/\alpha$. These identities include all backward derivatives.

Proof. Each normalized data argument lies strictly inside the $1/8$ neighborhood in (30), so $G_i = b_i$ and all its first derivatives vanish; every clock ratio is one, so its cutoff is one and its derivative is zero. The surviving part of the own factor is

$$c_i^2 H(x_i/c_i) b_i = b_i (c_i x_i - z_i c_i^2/2),$$

whose data derivative is $c_i b_i$ and whose c_i derivative is $b_i(x_i - z_i c_i)$, the latter acquiring the factor $1/\alpha$ when differentiated with respect to u_i. Self-inputs are included in this calculation, because their plateau derivatives vanish, and substitution into (32) gives (37). Strict containment in the flat regions also proves the neighborhood claim, with the varying individual c_i in the formulas. □

On an exactly encoded state the clock correction vanishes, which gives the exact orbit that the prediction theorem uses for its exact-arithmetic model.

Corollary D.4 (Exact execution by the same local energy)**.** *For the energy E_F in (32), ordinary gradient descent with step size one and exact arithmetic, initialized at $x_i^0 = z_i^0 \in \{0,1\}$ and $u_i^0 = \alpha$, satisfies*

$$x_i^t = \tau^t (F^t(z^0))_i, \qquad u_i^t = \alpha \tau^t \qquad (t \ge 0), \tag{38}$$

so the comparison $x_i^T > u_i^T/(2\alpha)$ returns $(F^T(z^0))_i$, with absolute margin $\tau^T/2$.

Proof. At $x_i = cz_i$, $u_i = \alpha c$, the clock correction in (37) vanishes. Subtracting the gradient gives

$$x_i^+ = \tau c(F(z))_i, \qquad u_i^+ = \alpha\tau c.$$

Induction from $c = 1$ proves the orbit. Its displacement from the decoder threshold is $\tau^T((F^T(z^0))_i - 1/2)$, giving the margin. □

Before analyzing the arithmetic, fix its model. Let RN round a real elementary result to a nearest radix-two number with p significand bits, with ties to even, and let $v := 2^{-p} = \tau/4096$ denote the unit roundoff. Until Lemma D.8, the exponent range is taken large enough for the specified horizon that underflow and overflow are excluded in the operations used. A nonzero rounded result then has relative error at most v, multiplication by a power of two is exact, and comparisons and ReLU preserve an already represented argument exactly.

The only other arithmetic fact needed is the exact subtraction of nearby floating-point numbers, commonly called Sterbenz's lemma (Higham, 2002b, Theorem 2.5): if represented positive numbers a, b satisfy $a/2 \le b \le 2a$, then $a - b$ is represented exactly, provided the exponent range contains it. To verify this, assume $a \ge b$ and choose an integer e with $2^e \le b < 2^{e+1}$. Both numbers are integer multiples of 2^{e-p+1} because $a \le 2b < 2^{e+2}$, so their difference is a multiple of the same spacing and satisfies $0 \le a - b \le b < 2^{e+1}$, which means it has at most p significant bits.

Lemma D.5 (Local clocks remain exactly synchronized)**.** *Suppose $X \in \mathcal{T}(z, c)$ is represented, $c = \tau^t$ for a nonnegative integer t, and $b = F(z)$. Compute, with every operation rounded,*

$$\begin{aligned} B_i &:= \mathrm{RN}((1-\tau)u_i), & r_i &:= \mathrm{RN}(x_i - z_i c), & & \\ h_i &:= (\tau/\alpha)b_i r_i, & g_{u_i} &:= \mathrm{RN}(B_i - h_i), & u_i^+ &:= \mathrm{RN}(u_i - g_{u_i}). \end{aligned} \tag{39}$$

Then $u_i^+ = \alpha\tau^{t+1}$ for every i.

Proof. Since the clock $u_i = \alpha\tau^t$ is a positive power of two and $1 - \tau$ has at most K significand bits, $B_i = (1-\tau)u_i$ without rounding error. This number is an interior point of the binade $[u_i/2, u_i)$, whose spacing is vu_i, so its distance to either nearest rounding boundary is $vu_i/2$.

The residual r_i is also exact: for $z_i = 0$ it subtracts zero, and for $z_i = 1$ the tube gives $x_i/c \in [15/16, 17/16]$, so Sterbenz's lemma applies. Bit selection and power-of-two scaling make h_i exact, and its magnitude obeys

$$|h_i| \le \frac{\tau}{\alpha} c\delta = \frac{\tau u_i \delta}{\alpha^2} = vu_i\delta < vu_i/2,$$

because $\alpha^2 = 4096$. Hence $g_{u_i} = B_i$ regardless of the nearest tie rule, and the final difference is the exactly represented number $u_i - B_i = \tau u_i$, which proves the clock identity without any communication between clocks. □

The next lemma controls the cancellation in the data update and restores the successor bit at the new clock scale.

Lemma D.6 (Data regeneration with persistent errors)**.** *Under the assumptions of Lemma D.5, compute*

$$g_i := \mathrm{RN}(x_i - \tau c b_i), \qquad x_i^{\mathrm{raw}} := \mathrm{RN}(x_i - g_i), \tag{40}$$

and after this computation allow any representable stored value x_i^+ with $|x_i^+ - x_i^{\mathrm{raw}}| \le \tau c/64$. Together with the clocks of Lemma D.5, the next state belongs to $\mathcal{T}(F(z), \tau c)$.

Proof. If $b_i = 0$, the gradient is x_i and its subtraction gives zero, both without rounding error. If $b_i = 1$, put $a := x_i - \tau c$ and write $g_i = a + \eta$ with $|\eta| \le v|a|$. Rounding the final subtraction gives

$$\begin{aligned} |x_i^{\mathrm{raw}} - \tau c| &\le v|a| + v(\tau c + v|a|) \\ &\le v\big((1+v)(1+\delta+\tau)+\tau\big)c < 3vc, \end{aligned}$$

since $|x_i| \le (1+\delta)c$, and the permitted stored perturbation then gives

$$\left| \frac{x_i^+}{\tau c} - b_i \right| \le \frac{3v}{\tau} + \frac{1}{64} = \frac{3}{4096} + \frac{1}{64} < \frac{1}{16} = \delta.$$

The next tube property therefore holds for every coordinate and for every history-dependent choice of the permitted data errors. □

D.4 ReLU messages and finite exponent storage

Local ReLU modules recover scaled bits on the tube without division or a Boolean oracle. Their complete recurrent block is contractive even away from the tube.

Lemma D.7 (Local ReLU realization and contraction)**.** *There is a fixed finite family of constant-size ReLU modules, selected by static gate and port types, that executes (37) on every represented synchronized tube state, with every operation rounded as specified in the preceding lemmas. One complete exact-real recurrent block extends to a globally Lipschitz map $\mathcal{M}_F : \mathbb{R}^{2m} \to \mathbb{R}^{2m}$ with*

$$\mathrm{Lip}(\mathcal{M}_F) \le 32\tau\sqrt{\Delta} \le 64\tau,$$

which equals $I - \nabla E_F$ on the synchronized tubes, although its behavior elsewhere need not be conservative. Each complete block uses a gather stage and a return stage on the factor graph and constant dynamic width.

Proof. Let $\sigma(a) := \max(a, 0)$ denote ReLU, and for a positive local signal scale c define the restoration map

$$\mathcal{R}_c(x) := \sigma(2x - c/2) - \sigma(2x - 3c/2), \tag{41}$$

whose value is cz whenever $|x/c - z| \le 1/16$. To execute it in floating arithmetic, first compute $s := \mathrm{RN}(2x - c/2)$, then compute $\mathrm{RN}(s - c)$, apply the two ReLUs, and subtract their outputs. When $z = 0$, the first preactivation is in $[-5c/8, -3c/8]$ before rounding, so both ReLU outputs are zero. When $z = 1$, it is in $[11c/8, 13c/8]$ before rounding and its rounded value remains in $(c, 2c)$, so $s - c$ is exact by Sterbenz's lemma and the final difference is the represented number $s - (s - c) = c$. The arithmetic thus recovers 0 or c without approximation.

Use these restored amplitudes in a disjunctive normal form for the supplied truth table, where complement is $c - A$ for an amplitude A and conjunction and disjunction are minimum and maximum, both ReLU operations, for example $\min(A, B) = A - \sigma(A - B)$. A two-input truth table uses at most four terms of two literals, so the expression has fixed size, and all Boolean intermediates belong to $\{0, c\}$, which makes every operation exact. Let $J_i := cb_i$ denote the resulting gate-output amplitude and $Z_i := cz_i$ the receiver's restored old amplitude.

The clock gradient also uses the signed data residual. Compute $r_i := x_i - Z_i$ as in Lemma D.5, and select it using

$$Q_i := \sigma(r_i + (J_i - c)) - \sigma(-r_i + (J_i - c)), \tag{42}$$

where the parentheses prescribe the arithmetic order. If $J_i = c$, the inner difference is zero and the expression returns r_i, both without rounding error. If $J_i = 0$, both preactivations are strictly

negative because $|r_i| \le \delta c$, so it returns zero. Hence $Q_i = b_i r_i$ exactly, and using τJ_i in the data gradient and $(\tau/\alpha)Q_i$ in the clock gradient reproduces the formulas (37) with the operation order already analyzed.

The same ReLU expressions define a map off the tube: receiver i uses $c_i = u_i/\alpha$ to restore its own and parent data, evaluates the same minimum/maximum truth expression, and forms Q_i by (42). Constant gates produce the output zero or c_i directly. In exact arithmetic, the full update is

$$(\mathcal{M}_F(X))_i = \tau(J_i,\ u_i + Q_i/\alpha), \tag{43}$$

a definition that makes sense for all real clock signs, although only its agreement with the energy gradient on the tube is needed for the computational claim.

To bound the extension globally, compare two physical states X and $\widetilde{X}$, let d_j be the Euclidean distance between their pairs at register j, and let $D_i := \max_{j \in S_i} d_j$ be the largest local input difference to gate i. ReLU's unit Lipschitz constant bounds the restoration difference by $4|\Delta x_j| + 2|\Delta c_i|$, and a complemented literal has bound $4|\Delta x_j| + 3|\Delta c_i|$. Minimum and maximum are nonexpansive in the sup norm of their arguments, so $|\Delta J_i| \le 7D_i$, and the residual has $|\Delta r_i| \le 7D_i$ as well. The selected residual therefore satisfies

$$|\Delta Q_i| \le 2|\Delta r_i| + 2|\Delta J_i - \Delta c_i| \le 30 D_i.$$

Since $\alpha \ge 1$, one output pair before the factor τ has squared difference at most $(7^2+31^2)D_i^2 < 32^2 D_i^2$. Summing over scopes gives $\sum_i D_i^2 \le \Delta \sum_j d_j^2$, which proves the displayed Lipschitz constant for the completed block.

Each factor gathers its scope and returns derivative contributions. On a tube all parent derivatives are zero, leaving the own-coordinate formulas above. Static types select from finitely many constant-size ReLU modules, with constant widths because arity and incidence are bounded. Their agreement with the rational energy gradient and the contraction assertion concern the covered states and the completed block respectively. □

A finite format must also accommodate arbitrarily small represented data errors. Gradual underflow supplies the required absolute-error bound.

Lemma D.8 (A finite exponent range suffices)**.** *Fix an integer horizon $T \ge 0$, and use $p = 37$ significand bits, round to nearest, gradual underflow, and normal binary exponents ranging from $e_{\min} := -K(T+2) - 11$ to $e_{\max} := 10$. Then all conclusions of Lemmas D.5–D.7 remain valid through time T for every permitted data perturbation represented in this format, and each exponent can be stored in $O(\log(T+2))$ bits.*

Proof. With $\gamma_T = \tau^T$ the final signal scale, the smallest positive subnormal is

$$q_{\min} := 2^{e_{\min}-p+1} = 2^{-K(T+2)-p-10} = 2^{-2K}\,\frac{\upsilon\gamma_T}{1024} \le \frac{\upsilon\gamma_T}{1024},$$

Every elementary result a in range satisfies

$$|\,\mathrm{RN}(a) - a| \le \upsilon|a| + q_{\min}/2, \tag{44}$$

because for normal results the relative bound suffices, while for subnormal results rounding to the grid of spacing $q_{\min}$ gives the absolute term. Sterbenz subtraction remains exact with gradual underflow: if the difference is normal, the earlier significand argument applies, and if it is subnormal, both operands are integer multiples of $q_{\min}$, so the difference lies exactly on the subnormal grid.

The clock and restoration values are far above the smallest normal value, because every covered signal scale is at least $\gamma_T = 2^{-KT}$, whereas the smallest normal value is $2^{-K(T+2)-11}$. The baseline clock gradient $B_i = (1-\tau)u_i$ and its final subtraction therefore remain exactly represented, and only the very small feedback product h_i can underflow. Because that product is a power-of-two scaling of a represented residual, it is exact unless it enters the subnormal range, where its error is at most $q_{\min}/2$. The computed value $\widehat{h}_i$ therefore obeys

$$|\widehat{h}_i| \le v u_i \delta + q_{\min}/2 < v u_i/2,$$

and the same nearest rounding interval as before yields $\mathrm{RN}(B_i - \widehat{h}_i) = B_i$, so the clocks still update exactly.

For the data update with $b_i = 1$, retain $a := x_i - \tau c$ and write $g_i = a + \eta$, where now (44) gives $|\eta| \le v|a| + q_{\min}/2$. Applying it once more to the final subtraction bounds the raw data error by

$$\begin{aligned} |x_i^{\mathrm{raw}} - \tau c| &\le v\big((1+v)(1+\delta+\tau)+\tau\big)c + (1+v/2)q_{\min} \\ &< 3vc. \end{aligned}$$

The last inequality follows from $q_{\min} \le v\gamma_T/1024 \le vc/1024$, valid because $c = \tau^t \ge \gamma_T$ for $t \le T$, together with the fixed values of δ, τ, v, the first coefficient being less than $1.064v$. The zero-bit case still subtracts a represented number from itself exactly, so the same normalized data-error estimate closes.

Restoration preactivations, Boolean amplitudes, and clock scales are normal, so their sign and exact-subtraction arguments remain valid. Selected residuals may be subnormal, but ReLU, sign reversal, and subtraction of zero preserve them exactly; only the feedback scaling can underflow, as already bounded. Non-clock intermediates have magnitude at most $4c \le 4$, and clock-gradient intermediates at most $65 < 2^7$, excluding overflow with $e_{\max} = 10$. The allowed exponent interval has length $O(K(T+2)+p)$ with K, p fixed constants, which proves the bit cost. □

D.5 Descent and completion of the local theorem

The exact clocks dominate the state norm, allowing global conditioning to turn the tube invariant into strict energy decrease.

Lemma D.9 (Energy decrease under the permitted data errors)**.** *If $X \in \mathcal{T}(z,c)$ and X^+ is any update covered by Lemmas D.6 and D.8, then*

$$E_F(X^+) < 2\tau^2 E_F(X) < E_F(X).$$

Proof. The old clock coordinates imply $\|X\|^2 \ge m\alpha^2 c^2$, while the next clock scale is τc and the next tube bounds every data coordinate by $(1+\delta)\tau c$, so

$$\|X^+\|^2 \le \tau^2 \left(1 + \frac{(1+\delta)^2}{\alpha^2}\right) \|X\|^2.$$

Lemma D.2, together with $E_F(0) = 0$ and $\nabla E_F(0) = 0$, gives $3\|X\|^2/8 \le E_F(X) \le 5\|X\|^2/8$, and combining these estimates shows

$$E_F(X^+) \le \frac{5}{3}\tau^2 \left(1 + \frac{(17/16)^2}{4096}\right) E_F(X) < 2\tau^2 E_F(X).$$

Since the positive clock makes $X \ne 0$, so that $E_F(X) > 0$, and since $2\tau^2 < 1$, both strict inequalities follow. □

Proof of Theorem D.1. Use the energy (32). Lemma D.2 proves the first item. The initial state lies in $\mathcal{T}(z^0, 1)$; Lemmas D.5, D.6, and D.8 preserve the tube through the chosen horizon for every permitted error history, proving the second item and the readout because $\delta < 1/2$. Lemma D.7 gives the third item, and Lemma D.9 the fourth. □

The exponent range is a computational resource that grows with the horizon, and the construction uses ordinary round-to-nearest arithmetic with the specified order of operations. The allowed additional errors affect completed data registers only: independent clock faults could destroy the exactly represented baseline used by Lemma D.5, so no such stronger robustness statement follows from the theorem.

E The prediction problem and its complexity

The exact orbit of Corollary D.4 and the rounded implementation of Theorem D.1 use the same local energy. We place that energy on a fixed path architecture and prove hardness and membership for the resulting prediction language.

E.1 Fixed architecture and arithmetic

Fix a deterministic Turing machine deciding the language TQBF of true fully quantified Boolean formulas, which is PSPACE-complete (Stockmeyer and Meyer, 1973), and choose a polynomial bound on its tape usage. Modify the machine so that after deciding it returns its head to the left boundary and enters an absorbing accepting or rejecting state, a modification that preserves polynomial space. An input path contains a finite-alphabet encoding of the input word, a canonical initial head, blank padding, and endpoint markers; its syntax is checkable in polynomial time, and invalid encodings are rejected.

A cell records its tape symbol and the machine state when the head is present, so its next symbol depends only on itself and its two neighbors. Extend this fixed finite-alphabet rule arbitrarily to invalid local states.

Encode a cell symbol by a fixed number of Boolean bits, including a dedicated bit for the absorbing accepting state, and implement each next-state bit by a fixed circuit with fan-in at most two. Binary COPY trees reduce fan-out to at most two, including the broadcasts from a cell to the adjacent receiving circuits. Level the circuits, insert COPY registers until every wire joins consecutive levels, and include the feedback from the output level to the configuration registers. The resulting pipeline has a constant number D of levels, because the machine and its local rule are fixed.

Treat all circuit gates and configuration bits as persistent registers updated synchronously, with the following level convention. The configuration registers sit at level 0 and the remaining registers occupy levels $1, \dots, D-1$; every wire joins consecutive levels, and the feedback wires from level $D-1$ are the only inputs of the configuration registers. A register at level ℓ at time $t+\ell$ is then a fixed function of the level-0 block at time t. Writing $q(t)$ for the configuration-register block and $\mathcal{C}$ for the tape transition, substitution through the D levels gives

$$q(t+D) = \mathcal{C}(q(t)) \qquad (t \geq 0),$$

whatever values the other registers hold initially. Initialize $q(0)$ by the encoded tape and all other registers by zero; the subsequence $q(kD)$ then follows the machine, and the intermediate pipeline phases do not affect it. Each register is the output of one factor and an input to at most two factors, so the factor incidence satisfies $\Delta \leq 3$, which discharges the hypothesis $\Delta \leq 4$ of Appendix D.

For each tape length M, these fixed interior and boundary circuits define a Boolean register map F_M in which only the initial register values depend on the input word. Apply the local energy construction of Appendix D with $\tau = 2^{-25}$ and $\alpha = 64$, grouping all register pairs belonging to a tape cell into one vertex of the communication path $1 - 2 - \cdots - M$. The resulting energy, denoted by E_M, is independent of the input word and the horizon.

Each cell has constantly many registers, and each factor meets constantly many neighboring cells. Hence constantly many nearest-neighbor stages gather its inputs and return derivatives, using fixed left/right ports and endpoint markers. Lemma D.7 supplies the finite family of constant-size ReLU modules, independent of M, the input, and the horizon. A complete recurrent block implements one register update (one exact GD step), and D blocks implement a machine transition. The horizon below counts complete blocks.

E.2 Exact and rounded evaluation

An input specifies the initial finite-alphabet tape and an integer $T \geq 0$ in binary; each data register is initialized by its encoded Boolean value and each clock by α. The only queried output is the fixed accepting-bit register at vertex 1, whose register index we denote by i_* and whose choice belongs to the architecture rather than to the input. In either arithmetic model, the input is accepted if and only if

$$x_{i_*}^T > u_{i_*}^T/(2\alpha). \tag{45}$$

For exact evaluation, every recurrent block is ordinary GD with step size one. Writing $z^t = F_M^t(z^0)$ for the Boolean orbit, Corollary D.4 proves

$$x_i^t = \tau^t z_i^t, \qquad u_i^t = \alpha\tau^t,$$

so that (45) returns $z_{i_*}^T$ for every input tape and every nonnegative horizon, including horizons between the simulated machine transitions.

For rounded evaluation, fix the operation order from Appendix D and use 37 significand bits, normal exponents from $-25(T+2) - 11$ through 10, gradual underflow, and round to nearest with ties to even. No stored perturbation is applied in this decision language. From an exact Boolean encoding, every intermediate in the prescribed order is a dyadic multiple of the current scale with at most $25 < 37$ significant bits. Thus every operation is exact and the rounded and exact orbits coincide. More generally, Theorem D.1 guarantees

$$u_i^t = \alpha\tau^t, \qquad |x_i^t/\tau^t - z_i^t| \leq 1/16 \qquad (0 \leq t \leq T),$$

which also suffices for the decoder. Consequently the two decision languages coincide on every valid input. The exponent bounds have $O(\log(T+2))$-bit descriptions; constant dynamic width counts coordinates, each rounded coordinate requires $O(\log(T+2))$ bits of storage, and one block uses $O(M)$ local arithmetic operations over the whole path.

E.3 Hardness and membership

Proof of Theorem 3.1. For hardness, let a TQBF instance have length n and choose a tape length M, polynomial in n, that accommodates the fixed decider. Its number of configurations on this tape is at most 2^{CM} for a fixed integer C, after increasing C to cover the head position, the control state, and the return to the left boundary. Since the decider halts, it has reached its absorbing decision by that time. Construct the canonical path encoding and set

$$T = D\,2^{CM}.$$

The path has polynomial length and T has $O(M)$ binary digits, so both are produced in polynomial time. The pipeline identity shows that the fixed output bit at time T is the TQBF decision, and the exact orbit identity and the rounded invariant both equate this bit with (45). This proves PSPACE-hardness for both arithmetic models under polynomial-time many-one reductions.

For membership, reject malformed inputs and simulate the Boolean register network synchronously. Since there are $O(M)$ registers, two Boolean buffers and a binary iteration counter use $O(M + \log(T + 2))$ space, and both arithmetic models return the final Boolean output by the identities above. Because this argument applies to every valid input and horizon, it proves membership for each complete language, and it never materializes a fixed-point denominator of length proportional to T.

Lemma D.2 gives regularity, conditioning, the zero minimizer, and exact-GD contraction; grouping coordinates into path vertices preserves the Euclidean norm. Lemma D.7 identifies the fixed ReLU modules with GD on the encoded orbit and with the gradient formulas on covered rounded states, without requiring their off-tube extension to be conservative. Lemma D.9 gives strict rounded energy decrease. □

Corollary E.1 (Consequences for prediction and certificates)**.** *A uniform algorithm deciding the language of Theorem 3.1 in time polynomial in the path length and* $\log(T + 2)$ *implies* $\mathsf{P} = \mathsf{PSPACE}$*. If all accepting instances have polynomial-length certificates accepted by a single polynomial-time verifier that is sound on all rejecting instances, then* $\mathsf{NP} = \mathsf{PSPACE}$*.*

Proof. The reduction places TQBF in P or NP, respectively. PSPACE-completeness and the reverse containments give the equalities. □

These statements concern a prescribed finite time given in binary, and polynomial or unary horizons admit direct polynomial-time simulation. Since the energy minimizer is the known origin, the classification supplies no hardness for approximating that minimizer to a fixed absolute accuracy. At time T the exact decoder margin is $\tau^T/2$ and the rounded margin is at least $7\tau^T/16$. These shrinking margins, together with an exponent range that must grow with T, are the reason the theorem gives no robustness guarantee for fixed absolute noise.

F Why correlation coordinates matter

A mean-only network represents independent spins, whereas a correlation matrix can describe their dependence. We compare the improvement directions available at the unbiased state. This is a local geometric statement, with no complexity lower bound.

A spin assignment $\sigma \in \{-1,1\}^n$ specifies a cut. Let $A \in \mathbb{R}^{n\times n}$ be a symmetric, nonzero, entrywise nonnegative weight matrix with zero diagonal, and let $W = \sum_{i<j} A_{ij} > 0$ be its total weight. The *weighted cut score* of an assignment counts the weight of the edges whose endpoints carry opposite signs,

$$c_A(\sigma) = \frac{1}{2}\sum_{i<j} A_{ij}(1-\sigma_i\sigma_j).$$

For $m \in [-1,1]^n$, let μ_m be the independent spin law with means m_i, and set $C_A(m) = \mathbb{E}_{\mu_m} c_A(\sigma)$. General spin laws have second-moment matrices in the elliptope, and their expected cut score is the linear functional

$$\mathcal{E}_n = \{X \in \mathbb{R}^{n\times n} \mid X \succeq 0, \operatorname{diag} X = \mathbf{1}\}, \qquad L_A(X) = \frac{W}{2} - \frac{1}{4}\langle A, X\rangle,$$

where the Frobenius product counts both orientations of each edge. Write $X_{\mathrm{prod}}(m) = \mathbb{E}_{\mu_m}\sigma\sigma^\top$ for the moment map of the product law.

Proposition F.1 (First-order access to correlations)**.** *The product extension and its moment map satisfy*

$$C_A(m) = \frac{W}{2} - \frac{1}{4}m^\top A m, \qquad \nabla C_A(m) = -\frac{1}{2}Am, \tag{46}$$

$$X_{\mathrm{prod}}(m) = mm^\top + \operatorname{diag}(1-m_i^2) \in \mathcal{E}_n, \qquad C_A = L_A \circ X_{\mathrm{prod}}. \tag{47}$$

At the unbiased center, $DX_{\mathrm{prod}}(0) = 0$. Hence every differentiable parameterization $\theta \mapsto m_\theta \in [-1,1]^n$ with $m_{\theta_0} = 0$ obeys

$$\nabla_\theta[-C_A(m_\theta)]\big|_{\theta=\theta_0} = 0,$$

regardless of its parameter count. In contrast, the matrices $X_t = I - tA$ satisfy

$$X_t \in \mathcal{E}_n \quad (0 \le t \le \|A\|_{\mathrm{op}}^{-1}), \qquad L_A(X_t) = \frac{W}{2} + \frac{t}{4}\|A\|_F^2. \tag{48}$$

For $0 \le t \le 1/W$, X_t is the second-moment matrix of a zero-mean spin distribution. The product center is a saddle.

Proof. Independence gives $\mathbb{E}_{\mu_m}\sigma_i\sigma_j = m_i m_j$ for $i \neq j$, proving (46), and since the diagonal moments equal one, the moment matrix is $mm^\top + \operatorname{diag}(1-m_i^2)$. Both summands are positive semidefinite on the cube, which establishes (47). The diagonal of this matrix is constant and every off-diagonal entry is quadratic in m, so its differential vanishes at zero, and the chain rule then gives the parameter-gradient assertion.

For the stated spectral bound on t, every eigenvalue of $I - tA$ is nonnegative and its diagonal is one, and since $\langle A, I\rangle = 0$, substitution gives (48). To show that X_t is realizable by a spin law for sufficiently small t, set

$$h(\sigma) = \sum_{i<j} A_{ij}\sigma_i\sigma_j, \qquad \mu_t(\sigma) = 2^{-n}(1 - t h(\sigma)).$$

The inequality $|h(\sigma)| \le W$ makes μ_t nonnegative when $t \le 1/W$, its total mass is one because $\mathbb{E}_{\mathrm{unif}} h = 0$, and orthogonality of spin monomials under the uniform law gives

$$\mathbb{E}_{\mu_t} \sigma_i = 0, \qquad \mathbb{E}_{\mu_t} \sigma_i \sigma_j = -tA_{ij} \quad (i \neq j),$$

so that its moment matrix is $I - tA$.

Finally, a nonzero symmetric matrix of trace zero has eigenvalues of both signs, so the Hessian $-A/2$ of C_A is indefinite at the interior point $m = 0$, which is the saddle assertion. □

At zero means, any differentiable mean-only network has zero gradient, while a correlation direction improves the score. Accessible mean perturbations or second-order methods can still escape the saddle. The product extension does retain the discrete optimum,

$$\max_{m \in [-1,1]^n} C_A(m) = \max_{\sigma \in \{-1,1\}^n} c_A(\sigma),$$

because every product expectation is an average of discrete scores and the product family includes every deterministic assignment; what correlation coordinates change is the set of available directions and rounding possibilities.

The following examples separate correlation information from exact representability by a spin law.

Example F.2 (Zero means with different cut values)**.** Suppose the positive-weight graph is connected and bipartite, and let $s \in \{-1, 1\}^n$ record its bipartition. The correlated law $(\delta_s + \delta_{-s})/2$ has zero means and cut score W, whereas the product law with the same zero means has score $W/2$.

In this case the SDP $\max_{X \in \mathcal{E}_n} L_A(X)$ has the unique optimum $ss^\top$. To see this, factor $X = VV^\top$ into unit rows and note that every edge contributes at most its weight, so attaining W forces $\langle v_i, v_j \rangle = -1$ and hence $v_i = -v_j$ across every positive edge. Connectivity then gives $v_i = s_i u$ for one unit vector u, so that $X = ss^\top$. Shared-Gaussian hyperplane rounding returns s or $-s$ almost surely, and both describe the same optimal cut. Since the graph itself is also solved by bipartite traversal, this example illustrates a representation difference without a computational separation.

Example F.3 (The boundary between a moment law and its relaxation)**.** For the unit-weight triangle, the matrix with diagonal one and every off-diagonal entry $-1/2$ is positive semidefinite, with eigenvalues $0, 3/2, 3/2$ and cut SDP score $9/4$, whereas every discrete cut has score at most 2. This matrix cannot be the second moment of any spin distribution, because

$$\sigma_1\sigma_2 + \sigma_1\sigma_3 + \sigma_2\sigma_3 \ge -1$$

holds for every spin assignment while the proposed matrix makes the expectation of the left side $-3/2$. Arbitrary elliptope points therefore require a rounding analysis of their own, which for cuts the Goemans–Williamson guarantee supplies (Goemans and Williamson, 1995), with the optimization error accounted for separately.

G Ordinary gradient descent with a polynomial spectral regularizer

We prove Theorem 4.1 and Corollary 4.2 by replacing the log-determinant barrier with a finite polynomial. Its gradient requires only matrix products. Although the polynomial is finite at singular matrices, the prescribed initialization and step size keep every normalized iterate above an explicit eigenvalue floor. On that region, a weighted log-determinant inequality gives geometric convergence and a certificate from the gradient. We then specialize to affine feasibility and explain the obstructions at deficient initial span or smaller search rank.

G.1 The polynomial loss and the general theorem

The general convex objective below covers both affine residuals and the matrix losses in Appendix H. Write $\mathbb{S}^d$ for the real symmetric $d \times d$ matrices and let

$$\mathcal{D}_d = \{X \in \mathbb{S}^d \mid X \succeq 0,\ \operatorname{tr} X = 1\}, \qquad d \geq 2,$$

be the set of density matrices, that is, of positive semidefinite matrices of unit trace; all matrix inner products are Frobenius inner products $\langle A, H\rangle = \operatorname{tr}(AH)$. Suppose the objective ϕ is convex and twice continuously differentiable on an open convex neighborhood of $\mathcal{D}_d$, with a gradient bound $L > 0$ and a matrix smoothness constant $M \geq 0$ satisfying

$$\|\nabla\phi(X)\|_{\mathrm{op}} \leq L, \qquad 0 \leq \nabla^2\phi(X)[H,H] \leq M\|H\|_F^2 \quad (X \in \mathcal{D}_d,\ H \in \mathbb{S}^d). \tag{49}$$

Fix a barrier weight $\mu > 0$ and define the scale T, the eigenvalue level r, and the polynomial degree p by

$$T = L + \mu d, \qquad r = \frac{\mu}{4T}, \qquad p = \left\lceil \frac{1}{r} \log \frac{64T}{\mu} \right\rceil. \tag{50}$$

The polynomial regularizer, the geometric series h_p that appears in its gradient, and that gradient are

$$R_p(X) = \mu \sum_{j=1}^{p} \frac{\operatorname{tr}[(I-X)^j]}{j}, \qquad h_p(X) = \sum_{j=0}^{p-1} (I-X)^j, \qquad \nabla R_p(X) = -\mu h_p(X); \tag{51}$$

since $-\mu \log\det X = \mu \sum_{j\geq 1} \operatorname{tr}[(I-X)^j]/j$ whenever $0 \prec X \preceq I$, the regularizer is the degree-p truncation of the log-determinant barrier and $h_p(X)$ the corresponding truncation of X^{-1}. For every nonzero square factor $V \in \mathbb{R}^{d\times d}$, set

$$X(V) = \frac{VV^\top}{\|V\|_F^2}, \qquad \psi_p(X) = \phi(X) + R_p(X), \qquad \mathcal{L}_p(V) = \psi_p(X(V)). \tag{52}$$

Because the normalization is defined for every nonzero V and R_p is a polynomial, the loss $\mathcal{L}_p$ is smooth on all nonzero factors, singular ones included, and the work of the analysis is to show that the iterates nevertheless stay well conditioned.

Theorem G.1 (Ordinary GD with a finite polynomial regularizer; Theorem 4.1 restated)**.** *Under (49), start at the scaled-identity initialization $V_0 = I_d/\sqrt{d}$ and apply*

$$V_{t+1} = V_t - \eta \nabla \mathcal{L}_p(V_t), \qquad 0 < \eta \leq \frac{1}{64T + 4M}. \tag{53}$$

Write $s_t = \|V_t\|_F^2$ for the squared radius, $W_t = V_t/\sqrt{s_t}$ for the normalized factor, $G_t = \sqrt{s_t}\nabla\mathcal{L}_p(V_t)$ for the sphere gradient, and $h_{\min} = \eta/(1+4\eta L)$ for the smallest effective step. Every iterate satisfies

$$X(V_t) \succeq r I_d, \qquad 1 \leq s_t \leq 1 + 4\eta L. \tag{54}$$

The function ψ_p has a unique minimizer X_* on $\mathcal{D}_d$, with $X_* \succeq r I_d$. Moreover,

$$\mathcal{L}_p(V_t) - \psi_p(X_*) \leq 2L(1 - 4\mu h_{\min})^t, \tag{55}$$

$$\phi(X(V_t)) - \min_{X\in\mathcal{D}_d} \phi(X) \leq \mu d + \frac{\|G_t\|_F^2}{8\mu}. \tag{56}$$

Apart from evaluating $\nabla\phi$, the update and the certificate use only matrix products and scalar operations, and the factor stores d^2 real coordinates.

The eigenvalue floor makes a constant step possible. Gradient domination then turns descent into geometric contraction, and the same gradient bounds the original objective gap up to the regularization cost μd.

For a unit factor W, put $X = WW^\top$ and center the matrix gradient:

$$A = \nabla\phi(X), \quad B = A - \mu h_p(X), \quad \lambda_B = \langle B, X\rangle, \quad S = B - \lambda_B I_d. \tag{57}$$

Differentiating the normalization gives

$$G(W) = 2SW, \qquad \langle W, G(W)\rangle = 0, \qquad \nabla\mathcal{L}_p(V) = \frac{2}{\|V\|_F^2}(B - \langle B, X(V)\rangle I_d)V, \tag{58}$$

Here $G(W)$ is the sphere gradient and the raw gradient is its rescaling. The algorithm evaluates h_p by repeated squaring, as described below. Inverse matrices appear only in the analysis.

We first locate the matrix minimizer inside the invariant region. This allows the later curvature bound to be integrated from an iterate to it.

Lemma G.2 (An interior matrix minimizer)**.** *The minimizer X_* in Theorem G.1 exists, is unique, and satisfies $X_* \succeq rI_d$.*

Proof. For $0 \le x \le 1$, the geometric-series identity and Bernoulli's inequality $(1-x)^p \le (1+px)^{-1}$ give

$$xh_p(x) = 1 - (1-x)^p \in [0,1], \qquad h_p(x) \ge \frac{1}{x + 1/p}, \tag{59}$$

with the value at zero defined by continuity. Consequently $c(X) = \mathrm{tr}[Xh_p(X)] \le d$. Each $\mathrm{tr}[(I-X)^j]$ is convex on $\mathcal{D}_d$ and the quadratic term is strictly convex ($p \ge 1/r \ge 8$). Compactness therefore gives a unique minimizer.

If a minimizing X had a unit null vector u, the feasible direction $uu^\top - X$ would have objective derivative at most $2L + \mu d - \mu p$, which is negative because $p \ge 1/r = 4T/\mu$, so the minimizer is positive definite, and stationarity on the trace-one affine space makes its matrix gradient a scalar multiple of the identity. Testing this stationarity condition against an eigenvector of X_* for its smallest eigenvalue x_* gives

$$\mu h_p(x_*) \le 2L + \mu c(X_*) \le 2T,$$

and the second relation in (59) turns this into $x_* \ge \mu/(2T) - 1/p \ge r$, as claimed. □

G.2 A lower bound on the smallest eigenvalue along every trial step

To justify a constant step, derivative bounds must hold along the whole normalized trial path. We prove that the eigenvalue floor survives there.

Lemma G.3 (Invariant normalized trial path)**.** *If $\|W\|_F = 1$ and $WW^\top \succeq rI_d$, then*

$$W(h) = \frac{W - hG(W)}{\|W - hG(W)\|_F} \quad \textit{satisfies} \quad W(h)W(h)^\top \succeq rI_d \quad \left(0 \le h \le \frac{1}{64T}\right). \tag{60}$$

Proof. We compare with the exact barrier: its trial point gains an eigenvalue margin larger than the polynomial truncation error. The exact sphere gradient is

$$G_\infty(W) = 2[(A - \langle A, X\rangle I_d + \mu d I_d)W - \mu W^{-\top}].$$

Set $Q = (I - X)^p$ and $\delta = e^{-pr} \le \mu/(64T)$, so that $0 \preceq Q \preceq \delta I_d$ because $X \succeq rI_d$. Since $h_p(X) = X^{-1}(I - Q)$ and $\langle h_p(X), X\rangle = d - \operatorname{tr} Q$, the two gradients differ by

$$E = G(W) - G_\infty(W) = 2\mu[X^{-1}Q - \operatorname{tr}(Q)I_d]W.$$

Because Q commutes with X, $\|X^{-1}QW\|_F^2 = \operatorname{tr}(Q^2X^{-1}) \le \delta^2 \operatorname{tr} X^{-1} \le \delta^2 d/r$, while $\|\operatorname{tr}(Q)W\|_F = \operatorname{tr} Q \le d\delta$. Bounding the two terms of G_∞ separately by $4L + 2\mu d \le 4T$ and by $2\mu\|W^{-\top}\|_F \le 2\mu\sqrt{d/r} \le 4T$, where $\mu\sqrt{d/r} = 2\sqrt{\mu dT} \le 2T$ because $\mu d \le T$, we obtain

$$\|G_\infty(W)\|_F \le 8T, \qquad \|E\|_F \le 2\mu\delta(\sqrt{d/r} + d) \le 6T\delta, \qquad \|G(W)\|_F \le 9T, \tag{61}$$

the last bound because $\delta \le \mu/(64T) \le 1/128$.

Let $Y_\infty = W - hG_\infty(W)$ and $Y = W - hG(W)$ be the unnormalized trial points, with squared norms $q_\infty = \|Y_\infty\|_F^2$ and $q = \|Y\|_F^2$. With

$$D = I_d - 2h(A - \langle A, X\rangle I_d + \mu d I_d), \qquad a = 1 - 4hT,$$

we have $D \succeq aI_d$, and since $W^{-\top} = X^{-1}W$,

$$Y_\infty Y_\infty^\top = DXD + 4h\mu D + 4h^2\mu^2 X^{-1}.$$

Here $DXD \succeq rD^2 \succeq ra^2 I_d$ without any commutation between D and X, tangency $\langle W, G_\infty(W)\rangle = 0$ gives $q_\infty = 1 + h^2\|G_\infty(W)\|_F^2 \le 1 + 64h^2T^2$, and substituting $r = \mu/(4T)$ yields

$$Y_\infty Y_\infty^\top - rq_\infty I_d \succeq (2h\mu - 28h^2\mu T)I_d. \tag{62}$$

For $h \le 1/(64T)$, both $\|Y\|_F$ and $\|Y_\infty\|_F$ are below two, so (61) and tangency give

$$\begin{aligned}\|YY^\top - Y_\infty Y_\infty^\top\|_{\mathrm{op}} &\le 4h\|E\|_F \le 24hT\delta,\\ |q - q_\infty| &\le h^2\|E\|_F(\|G(W)\|_F + \|G_\infty(W)\|_F) \le 102h^2T^2\delta.\end{aligned}$$

Passing from Y_∞ to Y therefore lowers the smallest eigenvalue of the left side of (62) by at most $24hT\delta + r \cdot 102h^2T^2\delta \le (24 + 102/64)hT\delta < 26hT\delta \le 26h\mu/64$, where we used $r \le 1/(4d) < 1$, $h \le 1/(64T)$, and $T\delta \le \mu/64$; for the same range of h the right side of (62) is at least $2h\mu - 28h\mu/64 = 100h\mu/64$. Hence $YY^\top \succeq rqI_d$, and dividing by $q \ge 1$ gives $W(h)W(h)^\top \succeq rI_d$, which is the claim. □

G.3 Descent and a weighted gradient inequality

We next compare the polynomial curvature with the exact barrier. The upper bound controls the step size, and the lower bound gives gradient domination and the certificate.

Lemma G.4 (Spectral curvature comparison)**.** *For $X \in \mathcal{D}_d$ with $X \succeq rI_d$ and every symmetric H,*

$$\frac{\mu}{2}\operatorname{tr}(X^{-1}HX^{-1}H) \le \nabla^2 R_p(X)[H, H] \le \mu \operatorname{tr}(X^{-1}HX^{-1}H). \tag{63}$$

The upper bound also holds at every positive definite $X \in \mathcal{D}_d$.

Proof. We first compare scalar second derivatives, then use spectral divided differences. For $r_p(x) = \mu\sum_{j=1}^p (1 - x)^j/j$, differentiation gives

$$r_p''(x) = \frac{\mu}{x^2}[1 - (1 - x)^p - px(1 - x)^{p-1}],$$

which lies between zero and μ/x^2. If $x \geq r$, then $z = px \geq pr \geq \log(64T/\mu) \geq \log 128 > 4$, because $T/\mu \geq d \geq 2$, and since $(1-x)^p \leq e^{-z}$ and $px(1-x)^{p-1} \leq ez\,e^{-z}$,

$$(1-x)^p + px(1-x)^{p-1} \leq (1+ez)e^{-z} \leq \frac{1}{2},$$

since $(1+ez)e^{-z}$ decreases for $z \geq 1$ and is below $1/2$ at $z = 4$. Thus $r_p''(x) \geq \mu/(2x^2)$ on $[r, 1]$. In an eigenbasis of X with eigenvalues x_i, the Hessian of $\operatorname{tr} r_p(X)$ has diagonal coefficients $r_p''(x_i)$ and off-diagonal coefficients $(r_p'(x_i) - r_p'(x_j))/(x_i - x_j)$, and integrating the scalar bounds on r_p'' between x_i and x_j compares each coefficient with the corresponding coefficient $\mu/(x_i x_j)$ of $-\mu \log\det X$. This proves (63), coincident eigenvalues being covered by continuity; the upper bound uses only $0 \leq r_p'' \leq \mu/x^2$, which holds on all of $(0, 1]$. □

The decrease estimate holds for every admissible trial step, allowing its later use at the smallest effective step $h_{\min}$.

Lemma G.5 (Uniform trial decrease)**.** *Under the hypotheses of Lemma G.3,*

$$\psi_p(W(h)W(h)^\top) \leq \psi_p(WW^\top) - \frac{h}{2}\|G(W)\|_F^2 \quad \left(0 \leq h \leq \frac{1}{64T + 4M}\right). \tag{64}$$

Proof. Lemma G.3 keeps the trial path in the region where we bound the ambient derivatives. Consider $F(U) = \phi(UU^\top) + R_p(UU^\top)$ on ambient factors, which agrees with $\psi_p(UU^\top)$ at unit factors. At a unit factor with $UU^\top \succeq rI_d$,

$$\|\nabla F(U)\|_F \leq 2L + 2\mu\sqrt{d/r} \leq 6T.$$

The Hessian contribution from ϕ has norm at most $2L + 4M$, and for the regularizer the chain rule gives

$$\begin{aligned}\nabla^2(R_p \circ \mathrm{Gram})(U)[H, H] = {} & 2\langle -\mu h_p(X), HH^\top\rangle \\ & + \nabla^2 R_p(X)[UH^\top + HU^\top, UH^\top + HU^\top].\end{aligned}$$

The first term lies between $-2\mu\|H\|_F^2/r$ and zero because $0 \preceq h_p(X) \preceq X^{-1} \preceq I_d/r$, while the second is nonnegative and, by Lemma G.4 together with $\|X^{-1/2}U\|_{\mathrm{op}} = 1$ and $\|X^{-1/2}H\|_F \leq \|H\|_F/\sqrt{r}$, at most $4\mu\|H\|_F^2/r$. The regularizer therefore contributes a quadratic form between $-8T\|H\|_F^2$ and $16T\|H\|_F^2$, and altogether

$$\|\nabla^2 F(U)\|_{\mathrm{op}} \leq 18T + 4M.$$

Put $g = \|G(W)\|_F$. Since $G(W)$ is tangent to the sphere at W, the trial path is the unit-circle parametrization $W(h) = \cos\theta(h)\, W - \sin\theta(h)\, G(W)/g$ with $\theta(h) = \arctan(hg)$ when $g > 0$ and is constant when $g = 0$, so that $\|W'(h)\|_F \leq g$ and $\|W''(h)\|_F \leq 3g^2$ in either case. Because $h \leq 1/(64T + 4M) \leq 1/(64T)$, Lemma G.3 places every point of the path in the region where the derivative bounds above hold, and the second derivative of $F(W(h))$ is therefore at most $\|\nabla^2 F\|_{\mathrm{op}}\|W'\|_F^2 + \|\nabla F\|_F\|W''\|_F \leq (36T + 4M)g^2$, while its first derivative at zero is $-g^2$. Taylor's theorem gives $F(W(h)) \leq F(W) - hg^2 + \frac{h^2}{2}(36T + 4M)g^2 \leq F(W) - \frac{h}{2}g^2$, which is (64). □

A factor-Hessian variant. Only the bound $2L + 4M$ on the Hessian of $\phi \circ \mathrm{Gram}$ used M. If that Hessian instead has norm at most $2L + M_F$ on unit factors with $UU^\top \succeq rI_d$, the same argument gives a path-curvature bound $(36T + M_F)\|G\|_F^2$. Thus Lemma G.5 and Theorem G.1 hold with step bound $(64T + M_F)^{-1}$. For log-sum-exp this saves a factor d over the generic matrix-Hessian estimate.

The next inequality weights the displacement on one side by $X^{-1/2}$. That weighting pairs with the factor $X^{1/2}$ in the sphere gradient and will convert convexity into gradient domination.

Lemma G.6 (A weighted log-determinant inequality)**.** *For positive definite $X, Y \in \mathcal{D}_d$, define*

$$D_{-\log\det}(Y, X) = \operatorname{tr}(X^{-1}Y) - d - \log\det(X^{-1}Y).$$

Then

$$D_{-\log\det}(Y, X) \geq \|(Y - X)X^{-1/2}\|_F^2. \tag{65}$$

The constant one is sharp.

Proof. The proof reduces to a scalar inequality for the eigenvalues of the relative matrix $Q = X^{-1/2}YX^{-1/2}$, the two trace constraints entering as weights. Diagonalize Q with eigenvalues $q_i > 0$ and orthonormal eigenvectors u_i, and set $x_i = u_i^\top X u_i$, so that $x_i > 0$ and $\sum_i x_i = \sum_i x_i q_i = 1$ by the two trace constraints. In this notation the divergence is $\sum_i f(q_i)$ with $f(q) = q - 1 - \log q \geq 0$, while the squared norm in (65) is $\sum_i x_i(q_i - 1)^2$. If all $q_i = 1$, both sides vanish. Otherwise let $a = \min_i q_i < 1 < b = \max_i q_i$; averaging $(q_i - a)(b - q_i) \geq 0$ with the weights x_i gives

$$\sum_i x_i(q_i - 1)^2 \leq (b - 1)(1 - a),$$

whereas

$$\sum_i f(q_i) \geq f(a) + f(b) \geq (b - 1)(1 - a),$$

because $f \geq 0$ and $f(a) + f(b) - (b - 1)(1 - a) = ab - 1 - \log(ab) \geq 0$. Comparing the two displays proves the inequality, with no commutation assumption on X and Y. Equality holds for two distinct diagonal density matrices of size two whose diagonal entries are interchanged, so the constant one cannot be improved. □

The weighted inequality now bounds both the regularized gap and the Frank–Wolfe gap of the original objective by the same gradient norm.

Lemma G.7 (Gradient domination and an objective certificate)**.** *At every unit factor with $X = WW^\top \succeq rI_d$,*

$$\|G(W)\|_F^2 \geq 8\mu[\psi_p(X) - \psi_p(X_*)], \tag{66}$$

$$\langle A, X\rangle - \lambda_{\min}(A) \leq \mu d + \frac{\|G(W)\|_F^2}{8\mu}. \tag{67}$$

Proof. For the gradient inequality we integrate the lower Hessian bound along the segment from X to X_* and then apply the weighted inequality; for the certificate we complete a matrix square. The segment from X to X_* lies in $\{Y \succeq rI_d\}$ by Lemma G.2, so integrating the lower bound in (63) along it and then invoking Lemma G.6 gives

$$\begin{aligned}\psi_p(X) - \psi_p(X_*) &\leq \langle S, X - X_*\rangle - \frac{\mu}{2}\|(X_* - X)X^{-1/2}\|_F^2 \\ &\leq \frac{\|SX^{1/2}\|_F^2}{2\mu} = \frac{\|G(W)\|_F^2}{8\mu},\end{aligned}$$

where the second line is the weighted Cauchy–Schwarz inequality $\langle S, X - X_*\rangle \leq \|SX^{1/2}\|_F\|(X_* - X)X^{-1/2}\|_F$ followed by completing the scalar square, and $\|SX^{1/2}\|_F^2 = \operatorname{tr}(SXS) = \|G(W)\|_F^2/4$ because $G(W) = 2SW$; this proves (66).

For the certificate, set $\lambda_A = \langle A, X\rangle$ and $c = \operatorname{tr}[Xh_p(X)] \le d$. Because $(I-X)^p \preceq e^{-pr} I_d \preceq I_d/2$, the series identity $h_p(X) = X^{-1}(I-(I-X)^p)$ gives $h_p(X) \succeq X^{-1}/2$. Substituting the decomposition

$$\lambda_A I_d - A = -S - \mu h_p(X) + \mu c I_d$$

into the matrix square

$$S + \frac{\mu}{2} X^{-1} + \frac{1}{2\mu} SXS \succeq 0$$

yields

$$A \succeq \left(\lambda_A - \mu d - \frac{\operatorname{tr}(SXS)}{2\mu}\right) I_d,$$

where the positive semidefinite matrix SXS was bounded by its trace times the identity. Since $\operatorname{tr}(SXS) = \|G(W)\|_F^2/4$, the smallest eigenvalue of A is at least $\lambda_A - \mu d - \|G(W)\|_F^2/(8\mu)$, which is (67). □

Proof of Theorem G.1. The eigenvalue floor propagates, descent bounds the radius, and gradient domination supplies the rate. Homogeneity in (58) gives the identities

$$W_{t+1} = \frac{W_t - h_t G_t}{\|W_t - h_t G_t\|_F}, \qquad s_{t+1} = s_t + \frac{\eta^2}{s_t}\|G_t\|_F^2, \qquad h_t = \frac{\eta}{s_t}.$$

Initially $s_0 = 1$ and $X_0 = I_d/d \succeq r I_d$ because $r \le 1/(4d)$; since s_t is nondecreasing, $h_t \le \eta \le 1/(64T)$, and the lower eigenvalue bound follows inductively from Lemma G.3. Lemma G.5 applied with $h = h_t$ gives $\psi_p(X_{t+1}) \le \psi_p(X_t) - \frac{h_t}{2}\|G_t\|_F^2$, and substituting $\|G_t\|_F^2 = (s_t/\eta^2)(s_{t+1} - s_t)$ turns this into

$$s_{t+1} - s_t \le 2\eta[\psi_p(X_t) - \psi_p(X_{t+1})].$$

The regularizer is minimized over $\mathcal{D}_d$ at I_d/d by convexity and unitary invariance, and the operator-norm bound on $\nabla\phi$ gives $|\phi(X) - \phi(Y)| \le L\|X - Y\|_* \le 2L$ for density matrices, where $\|\cdot\|_*$ is the nuclear norm; hence $\psi_p(X_0) - \psi_p(X_*) \le 2L$, and telescoping the radius inequality proves $s_t \le 1 + 4\eta L$, which is $h_t \ge h_{\min}$. Combining the descent estimate at step $h_t \ge h_{\min}$ with (66) yields

$$\psi_p(X_{t+1}) - \psi_p(X_*) \le (1 - 4\mu h_{\min})[\psi_p(X_t) - \psi_p(X_*)],$$

and iterating from $\psi_p(X_0) - \psi_p(X_*) \le 2L$ gives (55). Finally, convexity of ϕ bounds the unregularized gap by the Frank–Wolfe gap, $\phi(X) - \min_{\mathcal{D}_d}\phi \le \max_{Y\in\mathcal{D}_d}\langle A, X - Y\rangle = \langle A, X\rangle - \lambda_{\min}(A)$, and (67) turns this into (56). □

G.4 Affine feasibility and a finite stopping time

For affine feasibility we sharpen the factor-Hessian estimate and read the gradient certificate as a lower bound on the best possible violation. Consider signed affine residuals

$$a_i(X) = b_i + \langle B_i, X\rangle \quad (1 \le i \le m), \qquad a_0(X) = 0,$$

with symmetric coefficient matrices B_i, where the trivial residual has $B_0 = 0$ and $b_0 = 0$; let $q = m + 1 \ge 2$ count the residuals, and fix a positive bound $L \ge \max_i \|B_i\|_{\mathrm{op}}$ (if every B_i vanishes, the residuals are constant and feasibility is decided by inspection). Define the maximum constraint

violation v, its minimum value γ over density matrices, the log-sum-exp smoothing ϕ_τ at temperature $\tau > 0$, and the softmax weights ω_i by

$$\begin{aligned} v(X) &= \max_{0\le i\le m} a_i(X), \qquad \gamma = \min_{X\in\mathcal{D}_d} v(X), \\ \phi_\tau(X) &= \tau \log \sum_{i=0}^{m} e^{a_i(X)/\tau}, \qquad \omega_i(X) = \frac{e^{a_i(X)/\tau}}{\sum_{j=0}^{m} e^{a_j(X)/\tau}}. \end{aligned} \tag{68}$$

The zero residual ensures $\gamma \ge 0$, and compactness of $\mathcal{D}_d$ shows that the affine system is feasible if and only if $\gamma = 0$.

Theorem G.8 (Ordinary GD for bounded affine systems)**.** *Fix $0 < \varepsilon \le L$ and set*

$$\mu = \frac{\varepsilon}{4d}, \qquad \tau = \frac{\varepsilon}{4\log q}, \qquad \eta = \frac{1}{64(L+\mu d) + 4L^2/\tau}. \tag{69}$$

Use T, r, p from (50) and apply ordinary GD to $\mathcal{L}_p$ with $\phi = \phi_\tau$ from the scaled-identity initialization $V_0 = I_d/\sqrt{d}$. The invariants and the geometric convergence of Theorem G.1 hold, and at every iterate the quantity

$$\underline{\gamma} = \sum_i \omega_i(X) a_i(X) - \mu d - \frac{\|G\|_F^2}{8\mu} \le \gamma \le v(X) \tag{70}$$

brackets the smallest achievable maximum constraint violation, with

$$v(X) - \underline{\gamma} \le \tau \log q + \mu d + \frac{\|G\|_F^2}{8\mu}. \tag{71}$$

The stopping condition

$$\|V_t\|_F^2 \|\nabla \mathcal{L}_p(V_t)\|_F^2 = \|G_t\|_F^2 \le 4\mu\varepsilon \tag{72}$$

holds for every $t \ge N$, where

$$N = \left\lceil \frac{1}{4\mu h_{\min}} \max\left\{0, \log \frac{L}{\mu\varepsilon h_{\min}}\right\} \right\rceil, \qquad h_{\min} = \frac{\eta}{1+4\eta L}. \tag{73}$$

At termination, either $v(X) \le \varepsilon$ or $\underline{\gamma} > 0$ certifies infeasibility of the exact affine system, and the iteration bound is $\widetilde{O}(d(L/\varepsilon)^2 \log q)$ in exact real arithmetic.

Proof. The gradient $A = \sum_i \omega_i B_i$ is a convex combination of the coefficient matrices, so $\|A\|_{\mathrm{op}} \le L$; a valid matrix smoothness constant is $M = dL^2/\tau$, but the factor curvature that the step size needs is smaller by a factor d. At every unit factor U,

$$\nabla^2(\phi_\tau \circ \mathrm{Gram})(U)[H,H] = 2\langle A, HH^\top\rangle + \frac{4}{\tau}\,\mathrm{Var}_{i\sim\omega}\langle B_i U, H\rangle.$$

The covariance operator is positive semidefinite with trace at most $4\sum_i \omega_i \|B_i U\|_F^2/\tau \le 4L^2/\tau$, so the factor Hessian satisfies $|\nabla^2(\phi_\tau \circ \mathrm{Gram})(U)[H,H]| \le (2L + 4L^2/\tau)\|H\|_F^2$. The factor-Hessian variant of Lemma G.5 with $M_F = 4L^2/\tau$ therefore gives Theorem G.1 for $0 < \eta \le 1/(64T + 4L^2/\tau)$, which is the step size in (69), together with the invariants and the geometric convergence.

For the certificate, (67) states that

$$A \succeq \alpha I_d, \qquad \alpha = \langle A, X\rangle - \mu d - \frac{\|G\|_F^2}{8\mu},$$

so for every density matrix Y,

$$v(Y) \geq \sum_i \omega_i(X) a_i(Y) \geq \sum_i \omega_i(X) b_i + \alpha = \underline{\gamma}.$$

Minimizing over Y gives $\underline{\gamma} \leq \gamma$, and $\gamma \leq v(X)$ holds by definition, which proves (70). For the gap, the entropy identity $\phi_\tau(X) = \sum_i \omega_i a_i(X) - \tau \sum_i \omega_i \log \omega_i$, the inequality $v(X) \leq \phi_\tau(X)$, and the entropy bound $-\sum_i \omega_i \log \omega_i \leq \log q$ combine into (71).

For the stopping time, let $\Delta_t = \psi_p(X_t) - \psi_p(X_*)$. Lemma G.5 applied at W_t with the trial step $h_{\min} \leq \eta$, whether or not this is the step the algorithm takes, produces a point of value at least $\psi_p(X_*)$, so by (55) and $1 - x \leq e^{-x}$,

$$\|G_t\|_F^2 \leq \frac{2\Delta_t}{h_{\min}} \leq \frac{4L}{h_{\min}} e^{-4\mu h_{\min} t}.$$

The right-hand side is at most $4\mu\varepsilon$ for every $t \geq N$ in (73). At the stated parameters, $64T = 64L + 16\varepsilon \leq 80L$ and $4L^2/\tau = 16L^2 \log q/\varepsilon$, so

$$h_{\min}^{-1} = \eta^{-1} + 4L \leq 84L + \frac{16L^2 \log q}{\varepsilon},$$

and $N = \widetilde{O}(d(L/\varepsilon)^2 \log q)$ follows. At stopping, the right side of (71) is at most $\tau \log q + \mu d + \varepsilon/2 = \varepsilon$; if $v(X) > \varepsilon$, the lower certificate $\underline{\gamma} \geq v(X) - \varepsilon$ is positive, and otherwise every affine residual is at most ε. □

The parameters allot one quarter of the tolerance to smoothing, one quarter to regularization, and one half to the gradient term. The stopping condition therefore returns either approximate feasibility or an exact infeasibility certificate.

It remains to add the slack coordinate of Section 4.1. For a positive semidefinite variable $Z \in \mathbb{S}^n$ with a known bound $\operatorname{tr} Z \leq R$, set $d = n + 1$ and read out $Z = RX_{1:n,1:n}$ from $X \in \mathcal{D}_d$; this slack-augmented trace-one embedding represents the entire trace-bounded cone, because

$$X = \operatorname{diag}(Z/R, 1 - \operatorname{tr} Z/R)$$

is a density matrix for every such Z. A residual $b_i + \langle C_i, Z \rangle$ becomes $a_i(X) = b_i + \langle B_i, X \rangle$ with $B_i = R \operatorname{diag}(C_i, 0)$, an equality contributes both of its signs, and positive rescalings of the residuals fix the units in which the final tolerance is measured. Applying Theorem G.8 to the embedded system proves Corollary 4.2: the output is positive semidefinite with trace at most R by construction, satisfies the remaining affine constraints to tolerance ε, and comes with the certificate $\underline{\gamma}$. Recovering a planted solution, rounding, or repairing the affine constraints exactly requires an argument specific to the application.

G.5 Arithmetic implementation

Since the theorem is stated for exact arithmetic, we record what one gradient step costs. The gradient (58) uses the geometric series $h_p(X)$ rather than the $1/j$-weighted value of $R_p(X)$, and the series is computed by repeated squaring: with $Q = I - X$, $P_j = Q^j$, and $S_j = \sum_{i=0}^{j-1} Q^i$, starting from $P_1 = Q$ and $S_1 = I_d$, the binary recursion

$$P_{2j} = P_j^2, \qquad S_{2j} = S_j + P_j S_j, \qquad P_{2j+1} = P_{2j} Q, \qquad S_{2j+1} = S_{2j} + P_{2j}$$

yields $h_p(X) = S_p$ after $O(\log p)$ dense matrix multiplications. The update and the stopping test use only this gradient and its norm, so the loss value is never evaluated, and for q dense affine residuals one step costs $O(d^3 \log p + qd^2)$ scalar arithmetic operations plus q exponentials. At the accuracy parameters of Theorem G.8,

$$p = O\Big(\frac{dL}{\varepsilon} \log \frac{dL}{\varepsilon}\Big),$$

and the factor occupies d^2 entries in addition to the storage for the input. The guarantee concerns exact real arithmetic, so a finite-precision implementation would need its own error analysis, although the recursion is at least well scaled: along the exact trajectory the spectrum of Q lies in $[0, 1 - r]$, so every power that appears has operator norm at most one.

G.6 Initialization and search-rank obstructions

The scaled identity supplies full span and a uniform eigenvalue floor. The obstructions below concern deficient initial span and search rank $k < d$. They do not show that isotropy is necessary: the invariant-path lemma already applies to any unit factor with $WW^\top \succeq rI_d$.

Proposition G.9 (A missed linear direction)**.** *Let a nonzero initial factor have a proper column space $\mathcal{S}$. There is a linear convex objective with a rank-one optimum such that ordinary GD on its normalized Gram loss remains suboptimal by one. The conclusion persists after adding any differentiable unitarily invariant spectral regularizer, whenever the iterates remain nonzero.*

Proof. Choose a unit vector $u \perp \mathcal{S}$ and set $\phi(X) = -u^\top X u$, whose minimum on $\mathcal{D}_d$ is -1; on factors with columns in $\mathcal{S}$ its value is zero and its factor gradient vanishes. Because the matrix gradient of a unitarily invariant spectral regularizer commutes with X and preserves its support, each factor gradient has columns in $\mathcal{S}$, so induction keeps every iterate in that subspace and the original objective stays at zero. If the initial Gram matrix is $P_{\mathcal{S}}/\dim \mathcal{S}$, its nonzero eigenvalues coincide and the regularizer's factor gradient vanishes as well, so the trajectory is stationary. □

This proposition concerns a fixed proper initial subspace and permits an adversarial objective after the subspace has been chosen, so it leaves open guarantees for an informative, instance-dependent initialization.

Proposition G.10 (A strict local minimum for smoothed affine residuals)**.** *For $d = 2$ and rank-one factors on the unit sphere, a feasible system with three affine residuals can have a strict suboptimal local minimum of the factor loss built from its log-sum-exp smoothing. Every finite unitarily invariant spectral regularizer preserves this local minimum.*

Proof. Take a feasible minimum at $e_2 e_2^\top$ and a competing rank-one factor e_1. Take $a > 0$, $\tau > 0$, and $c^2 > 2a\tau$, and set

$$a_0(X) = 0, \qquad a_\pm(X) = aX_{11} \pm cX_{12}.$$

Their log-sum-exp smoothing is

$$\phi_\tau(X) = \tau \log\big[1 + 2e^{aX_{11}/\tau} \cosh(cX_{12}/\tau)\big],$$

whose minimum is $\tau \log 3$ at $e_2 e_2^\top$ because $X_{11} \geq 0$ and $\cosh(cX_{12}/\tau) \geq 1$; since the maximum constraint violation is $v(X) = aX_{11} + |cX_{12}|$, the original system is feasible there. For $w(\theta) =$

$(\cos\theta, \sin\theta)$, expansion at $\theta = 0$ gives

$$\phi_\tau(w(\theta)w(\theta)^\top) = \tau \log(1 + 2e^{a/\tau}) + \omega \left(\frac{c^2}{2\tau} - a \right) \theta^2 + O(\theta^4),$$

$$\omega = \frac{2e^{a/\tau}}{1 + 2e^{a/\tau}} > 0,$$

so $w = e_1$ is a strict local minimum with violation a and positive objective gap $\tau \log[(1 + 2e^{a/\tau})/3]$. All rank-one density matrices have the same spectrum, so every finite spectral regularizer is constant on the factor domain and cannot remove this minimum. The coefficient matrices have operator norm $(a + \sqrt{a^2 + c^2})/2$, and the choice $a = c = 1$ gives a fixed coefficient bound for every temperature $0 < \tau < 1/2$. □

Proposition G.11 (Suboptimal local minima at every proper search rank)**.** *For every $1 \le k < d$, there is a convex smooth matrix objective with $\|\nabla\phi\|_{\mathrm{op}} \le 5$ and $\nabla^2\phi[H, H] \le 4\|H\|_F^2$ on $\mathcal{D}_d$ that has a rank-one global optimum and a rank-k factor local minimum on the unit Frobenius sphere with objective gap $2/k$. The local minimum is strict modulo right orthogonal transformations. Every finite convex unitarily invariant regularizer preserves a nonglobal local minimum at the same factor.*

Proof. Mass can leave a uniformly occupied subspace only by first creating penalized cross terms. Split $\mathbb{R}^d = \mathcal{S} \oplus \mathcal{S}^\perp$ with $\dim \mathcal{S} = k$ and write

$$X = \begin{pmatrix} B & Z \\ Z^\top & C \end{pmatrix}, \qquad \phi(X) = \frac{1}{k} \operatorname{tr} B + \|B\|_F^2 + 4\|Z\|_F^2.$$

This objective is nonnegative, equals zero at a rank-one projector in $\mathcal{S}^\perp$, and takes the value $2/k$ at $W_0 = [I_k \;\; 0]^\top/\sqrt{k}$. Writing a nearby unit factor as $W = [A^\top \;\; V^\top]^\top$ with $t = \|V\|_F^2$, we have $B = AA^\top$, $Z = AV^\top$, and $\operatorname{tr} B = 1 - t$, and in a sufficiently small neighborhood $\sigma_{\min}(A)^2 > 3/(4k)$, whence

$$\phi(WW^\top) - \frac{2}{k} \ge -\frac{t}{k} + \frac{(1-t)^2 - 1}{k} + 4\sigma_{\min}(A)^2 t > 0 \qquad (t > 0).$$

When $t = 0$, equality requires $AA^\top = I_k/k$, which is the orbit of W_0 under right orthogonal transformations, so the local minimum is strict in the stated sense. The matrix Hessian is $2\|H_B\|_F^2 + 8\|H_Z\|_F^2 \le 4\|H\|_F^2$, and since $\|B\|_{\mathrm{op}} \le 1$ and $\|Z\|_{\mathrm{op}} \le 1/2$ on $\mathcal{D}_d$, the block gradient has norm at most five.

A convex unitarily invariant regularizer is minimized over density matrices of rank at most k at the uniform rank-k spectrum, as one sees by averaging over all rotations within a k-dimensional space containing the support, so it preserves the local minimum at W_0. Replacing one column of W_0 by a vector of norm $1/\sqrt{k}$ in $\mathcal{S}^\perp$ produces a Gram matrix with the same spectrum and regularizer value but original objective $2(k-1)/k^2$, so the regularized local minimum has gap at least $2/k^2$, while the unregularized gap remains $2/k$. □

Shifted log-determinants and the finite polynomial regularizer above are among the regularizers covered by these statements, and two natural modifications fare no better. A log-determinant applied only to the rectangular right Gram matrix $W^\top W$ preserves the column space in Proposition G.9, and at search rank one its normalized right Gram matrix is the scalar one, so both obstructions persist.

H Matrix factorization and completion

We apply the polynomial-GD theorem to three convex matrix problems: nuclear-norm-bounded factorization, positive semidefinite sensing, and nuclear-norm completion. The classical block-PSD representation (Vandenberghe and Boyd, 1996, Eq. (37)) turns a nuclear-norm budget into a trace bound; certified factorized optimization over this ball is already known (Jaggi and Sulovský, 2010, Theorem 3). Our contribution here is the prescribed ordinary-GD realization and its explicit accuracy certificate. For PSD sensing a single slack coordinate suffices, while completion adds an entry-repair step and bisection over the budget.

H.1 A common accuracy prescription

For each application, let ϕ be convex and C^2 on an open convex neighborhood of the density-matrix domain $\mathcal{D}_d$, $d \geq 2$, with known $L > 0$ and $M \geq 0$ such that

$$\|\nabla\phi(X)\|_{\mathrm{op}} \leq L, \qquad 0 \leq \nabla^2\phi(X)[K,K] \leq M\|K\|_F^2 \quad (X \in \mathcal{D}_d, K \in \mathbb{S}^d). \tag{74}$$

For target accuracy $0 < \varepsilon \leq L$, set

$$\begin{aligned} \mu &= \frac{\varepsilon}{2d}, & T &= L + \mu d, & r &= \frac{\mu}{4T}, \\ p &= \left\lceil \frac{1}{r} \log \frac{64T}{\mu} \right\rceil, & \eta &= \frac{1}{64T + 4M}. \end{aligned} \tag{75}$$

For a nonzero raw factor $V \in \mathbb{R}^{d\times d}$, the normalized Gram matrix and the regularized loss are

$$X(V) = \frac{VV^\top}{\|V\|_F^2}, \qquad \mathcal{F}_p(V) = \phi(X(V)) + \mu \sum_{j=1}^{p} \frac{\mathrm{tr}[(I_d - X(V))^j]}{j}. \tag{76}$$

The following common prescription converts the regularized guarantee to accuracy for the original objective.

Lemma H.1 (Original-objective accuracy)**.** *Under (74), start from the scaled-identity initialization $V_0 = I_d/\sqrt{d}$ and perform*

$$V_{t+1} = V_t - \eta\nabla\mathcal{F}_p(V_t), \tag{77}$$

checking the stopping condition below at every iterate, including iteration zero. At the first iterate satisfying

$$\|V_t\|_F^2\|\nabla\mathcal{F}_p(V_t)\|_F^2 \leq 4\mu\varepsilon, \tag{78}$$

the matrix $X_t = X(V_t)$ obeys

$$\phi(X_t) \leq \min_{X\in\mathcal{D}_d} \phi(X) + \varepsilon. \tag{79}$$

Every iterate has $X_t \succeq rI_d$ and $1 \leq \|V_t\|_F^2 \leq 1 + 4\eta L$, and with $h_{\min} = \eta/(1 + 4\eta L)$ a sufficient iteration count is

$$N = \left\lceil \frac{1}{4\mu h_{\min}} \max\left\{0, \log \frac{L}{\mu\varepsilon h_{\min}}\right\} \right\rceil = \widetilde{O}\left(\frac{d(L+M)}{\varepsilon}\right). \tag{80}$$

Proof. Theorem G.1 supplies the invariants and, with $G_t = \|V_t\|_F\nabla\mathcal{F}_p(V_t)$, the certificate

$$\phi(X_t) - \min_{\mathcal{D}_d} \phi \leq \mu d + \frac{\|G_t\|_F^2}{8\mu} \leq \varepsilon$$

at stopping. Its regularized gap Δ_t satisfies

$$\Delta_t \le 2Le^{-4\mu h_{\min} t}, \qquad \|G_t\|_F^2 \le 2\Delta_t / h_{\min},$$

where the second inequality follows by applying Lemma G.5 at trial step $h_{\min}$ and comparing with the regularized minimum. These bounds give (80), with $h_{\min}^{-1} = 64T + 4M + 4L \le 100L + 4M$ since $\varepsilon \le L$. □

Two boundary cases are handled separately: a zero gradient bound makes ϕ constant on $\mathcal{D}_d$, so there is nothing to optimize, whereas a requested accuracy larger than a positive bound L is simply replaced by the accuracy L. All iteration counts below are exact-real counts, and each step is additionally charged the operations needed to evaluate the matrix loss and its gradient.

H.2 Rectangular factorization over a nuclear-norm ball

The nuclear norm $\|Y\|_*$ is the sum of the singular values. The next lemma expresses its budget through the trace-one Gram readout.

Lemma H.2 (A trace-one representation of a nuclear-norm ball)**.** *Let $a, b \ge 1$, $R > 0$, and $d = a + b$, and partition $X \in \mathcal{D}_d$ into blocks of sizes a and b. The readout*

$$Y = 2RX_{12}$$

maps $\mathcal{D}_d$ onto the nuclear-norm ball $\{Y \in \mathbb{R}^{a\times b} \mid \|Y\|_* \le R\}$. If $X = WW^\top$ with $\|W\|_F = 1$, and the rows of W are partitioned as $W_1 \in \mathbb{R}^{a\times d}$ and $W_2 \in \mathbb{R}^{b\times d}$, then

$$U = \sqrt{2R}\, W_1, \qquad H = \sqrt{2R}\, W_2, \qquad Y = UH^\top, \qquad \|U\|_F^2 + \|H\|_F^2 = 2R. \tag{81}$$

Proof. The factor readout gives the short bound

$$\|UH^\top\|_* \le \|U\|_F \|H\|_F \le \frac{\|U\|_F^2 + \|H\|_F^2}{2} = R.$$

For surjectivity, use the classical equivalence (Vandenberghe and Boyd, 1996, Eq. (37)): a rectangular block B belongs to a PSD matrix of trace one exactly when $\|B\|_* \le 1/2$. Apply it to $B = Y/(2R)$. The factor identities follow directly from $X = WW^\top$. □

It remains to compute the derivative bounds for the composed rectangular loss and apply Lemma H.1.

Corollary H.3 (Matrix factorization over a nuclear-norm ball)**.** *Let f be convex and C^2 on an open convex neighborhood of the nuclear-norm ball of radius $R > 0$ in $\mathbb{R}^{a\times b}$. Suppose that known constants $L_f > 0$ and $M_f \ge 0$ satisfy*

$$\|\nabla f(Y)\|_{\mathrm{op}} \le L_f, \qquad 0 \le \nabla^2 f(Y)[K, K] \le M_f \|K\|_F^2$$

for every Y in that ball and every rectangular perturbation K. Set

$$d = a + b, \qquad L = RL_f, \qquad M = 2R^2 M_f, \qquad \phi(X) = f(2RX_{12}). \tag{82}$$

For $0 < \varepsilon \le L$, apply (75)–(77) to this ϕ. At the stopping condition (78), the factors in (81), formed from $W = V/\|V\|_F$, satisfy

$$\|UH^\top\|_* \le R, \qquad f(UH^\top) \le \min_{\|Y\|_* \le R} f(Y) + \varepsilon. \tag{83}$$

The raw variable has d^2 entries, both output factors have d columns, and the iteration bound is (80).

Proof. The linear readout preserves convexity and C^2 regularity. Its gradient is

$$\nabla\phi(X) = \begin{pmatrix} 0 & R\nabla f(2RX_{12}) \\ R\nabla f(2RX_{12})^\top & 0 \end{pmatrix}, \qquad \|\nabla\phi(X)\|_{\mathrm{op}} \le RL_f,$$

and, for symmetric K,

$$\nabla^2\phi(X)[K,K] \le 4R^2M_f\|K_{12}\|_F^2 \le 2R^2M_f\|K\|_F^2.$$

Lemma H.1 therefore applies with the displayed constants, and Lemma H.2 identifies both the comparison class and the factor readout. □

GD acts on the raw square factor V; U, H are output factors. The comparison covers the nuclear-norm ball, with no prescribed output rank or guarantee for GD applied directly to $f(UH^\top)$. The scalar case makes the representation intuitive.

Example H.4 (A scalar product with an exact norm bound)**.** Take $a = b = 1$, so that the matrix variable is a scalar y whose nuclear-norm ball is the interval $[-R, R]$. The density matrix

$$X = \frac{1}{2}\begin{pmatrix} 1 & y/R \\ y/R & 1 \end{pmatrix}$$

is positive semidefinite if and only if $|y| \le R$, and its off-diagonal readout is $2RX_{12} = y$. One pair of output row factors is

$$U = \sqrt{R/2}(\sqrt{1+y/R}, \sqrt{1-y/R}), \qquad H = \sqrt{R/2}(\sqrt{1+y/R}, -\sqrt{1-y/R}),$$

which satisfy $UH^\top = y$ and $\|U\|^2 + \|H\|^2 = 2R$, so the factor directions encode every scalar in the interval while the combined factor norm stays fixed. Because the density matrix is positive definite in the open interval and singular at the endpoints, the GD theorem returns an objective-accurate interior output even when the comparison optimum is an endpoint.

H.3 Positive semidefinite matrix sensing

For PSD sensing, one slack coordinate embeds the trace ball. The sensing map can be arbitrary because the result compares fitting objectives over that ball.

Corollary H.5 (Positive semidefinite sensing)**.** *Let $n \ge 1$, let $\mathcal{A}\colon \mathbb{S}^n \to \mathbb{R}^m$ be a linear sensing map, let $y \in \mathbb{R}^m$ be the measurement vector, let $R > 0$ be the trace bound, and let $\kappa > 0$ be a known constant with*

$$\|\mathcal{A}(K)\|_2 \le \kappa\|K\|_F \quad (K \in \mathbb{S}^n).$$

Define

$$\begin{aligned} \mathrm{OPT}_R = \min_{Z\succeq 0,\ \mathrm{tr}\, Z\le R} \frac{1}{2}\|\mathcal{A}(Z) - y\|_2^2, \qquad & d = n+1, \\ L = R\kappa(R\kappa + \|y\|_2), \qquad & M = R^2\kappa^2. \end{aligned} \tag{84}$$

For $0 < \varepsilon \le L$, run the prescription of Lemma H.1 with

$$\phi(X) = \frac{1}{2}\|\mathcal{A}(RX_{1:n,1:n}) - y\|_2^2$$

and, at its stopping condition, read out the factor

$$U = \sqrt{R}\,\frac{V_{1:n,:}}{\|V\|_F} \in \mathbb{R}^{n\times(n+1)}. \tag{85}$$

Then

$$\|U\|_F^2 \le R, \qquad \frac{1}{2}\|\mathcal{A}(UU^\top) - y\|_2^2 \le \mathrm{OPT}_R + \varepsilon, \tag{86}$$

and the iteration count is (80) *with the constants in* (84).

Proof. The slack lift $X = \mathrm{diag}(Z/R, 1 - \mathrm{tr}\, Z/R)$ covers exactly the PSD trace ball, and the output has $UU^\top = RX(V_t)_{1:n,1:n}$. The nonzero block of $\nabla\phi(X)$ is

$$R\mathcal{A}^*(\mathcal{A}(RX_{1:n,1:n}) - y).$$

The adjoint has Euclidean-to-Frobenius norm at most κ, and a PSD matrix has Frobenius norm at most its trace. Thus the gradient has operator norm at most $R\kappa(R\kappa + \|y\|_2) = L$. Also,

$$\nabla^2\phi(X)[K,K] = R^2\|\mathcal{A}(K_{1:n,1:n})\|_2^2 \le R^2\kappa^2\|K\|_F^2.$$

Lemma H.1 gives the claim. □

Each iteration evaluates $\mathcal{A}$ and $\mathcal{A}^*$ once and spends $O(d^3 \log p)$ additional dense arithmetic operations on the polynomial gradient and the factor update, where $p = O((dL/\varepsilon)\log(dL/\varepsilon))$ for $0 < \varepsilon \le L$; no matrix inverse, eigendecomposition, or projection is required. The zero sensing map, for which the fitting objective is constant, needs no iteration at all.

No restricted-isometry or sampling assumption is needed for this optimization guarantee. Recovery of a particular low-rank matrix requires identifiability, which fails, for example, for the zero sensing map. Prescribed-rank and unregularized-factor results require their own hypotheses (Bhojanapalli et al., 2016; Zhang et al., 2023; Ge et al., 2017).

H.4 Exact observed entries and approximate minimum nuclear norm

For nuclear-norm completion (Candès and Recht, 2009), correcting one observed entry costs at most its correction magnitude in nuclear norm. We combine this elementary repair with bounded-affine feasibility and an outer bisection over the norm budget. Each feasibility call is one fixed-loss GD run.

Let $a, b \ge 1$, let $\Omega \subseteq \{1,\dots,a\}\times\{1,\dots,b\}$ be a set of s distinct observed positions with specified values $b_{ij} \in \mathbb{R}$, and define the optimal value $\mathrm{OPT}_{\mathrm{comp}}$ and the total observed mass B_0 by

$$\begin{aligned} \mathrm{OPT}_{\mathrm{comp}} &= \min\{\|Y\|_* \mid Y \in \mathbb{R}^{a\times b},\ Y_{ij} = b_{ij} \text{ for } (i,j)\in\Omega\}, \\ B_0 &= \sum_{(i,j)\in\Omega} |b_{ij}|. \end{aligned} \tag{87}$$

The zero-filled observation matrix $Y^{(0)}$ is feasible with $\|Y^{(0)}\|_* \le B_0$, by decomposition into single-entry matrices, and because a bounded nuclear-norm sublevel set is compact the minimum is attained.

Corollary H.6 (Completion with exact observed entries)**.** *For any accuracy $\varepsilon > 0$, there is a finite procedure that returns a matrix $\widehat{Y}$ satisfying*

$$\widehat{Y}_{ij} = b_{ij} \quad ((i,j)\in\Omega), \qquad \|\widehat{Y}\|_* \le \mathrm{OPT}_{\mathrm{comp}} + \varepsilon. \tag{88}$$

If $B_0 = 0$ or $\varepsilon \geq B_0$, the zero-filled matrix suffices. Otherwise the procedure uses at most

$$J = \left\lceil \log_2 \frac{2B_0}{\varepsilon} \right\rceil \tag{89}$$

feasibility tests. Each test applies Theorem G.8 with the known inputs

$$d = a + b + 1, \qquad q = 2s + 2, \qquad L = 2B_0, \qquad \delta = \frac{\varepsilon}{4(s + 1/2)} \tag{90}$$

and target residual accuracy δ. Each test uses $\widetilde{O}(d(L/\delta)^2 \log q)$ ordinary GD steps on a square factor with d^2 entries, while entry repair and the outer interval updates use scalar arithmetic and require no singular-value decomposition.

Proof. Assume $0 < \varepsilon < B_0$; since $B_0 > 0$ forces at least one nonzero observed entry, $s \geq 1$. For a proposed budget $t \in [0, B_0]$, consider the system

$$Z = \begin{pmatrix} P & Y \\ Y^\top & Q \end{pmatrix} \succeq 0, \qquad \operatorname{tr} Z \leq 2t, \qquad Y_{ij} = b_{ij} \quad ((i,j) \in \Omega),$$

in a symmetric block matrix Z with blocks of sizes a and b. By Lemma H.2, this system is feasible exactly when a completion of nuclear norm at most t exists (the case $t = 0$ is immediate).

To pose this as a bounded affine system, use the fixed trace bound $R_0 = 2B_0$ for every test, with density dimension $d = a + b + 1$ and readout $Z = R_0 X_{1:a+b,1:a+b}$; because $t \leq B_0$, the bound excludes no feasible point of the proposed system. The affine residuals are both signs of $Y_{ij} - b_{ij}$ for each observed position, the single residual $\operatorname{tr} Z - 2t$, and the zero residual, which makes $q = 2s + 2$ residuals in all. An entry residual has a symmetric coefficient matrix with two off-diagonal entries $R_0/2$ and hence operator norm $R_0/2$, while the trace residual has coefficient $R_0 \operatorname{diag}(I_{a+b}, 0)$ of norm R_0. The parameters (90) are therefore valid coefficient bounds with the correct residual count, and $0 < \delta < L$ holds because $\varepsilon < B_0$.

At termination, Theorem G.8 either certifies that this exact system is infeasible or returns a positive semidefinite Z with

$$|Y_{ij} - b_{ij}| \leq \delta \quad ((i,j) \in \Omega), \qquad \operatorname{tr} Z \leq 2t + \delta, \tag{91}$$

and the fixed trace bound $\operatorname{tr} Z \leq 2B_0$ holds exactly by construction. If both conclusions hold, use the certificate. In the output case, overwrite only the observed entries of Y to obtain a matrix Y'; the correction is a sum of s single-entry matrices, each of nuclear norm at most δ, so

$$\|Y' - Y\|_* \leq s\delta, \qquad \|Y'\|_* \leq t + (s + 1/2)\delta = t + \varepsilon/4. \tag{92}$$

This bound is computed from t and ε alone, so the nuclear norm of Y' never needs to be evaluated.

For bisection, initialize $[\ell, u] = [0, B_0]$ and retain $Y^{(0)}$, whose norm is at most u. At $t = (\ell + u)/2$, a positive certificate gives $t < \mathrm{OPT}_{\mathrm{comp}}$ and we set $\ell = t$. Otherwise replace the upper bound and candidate by $t + \varepsilon/4$ and the repaired matrix whenever this improves u. Every outcome keeps the candidate feasible with norm at most u and preserves $\ell \leq \mathrm{OPT}_{\mathrm{comp}} \leq u$. The width $w = u - \ell$ obeys

$$w_{\mathrm{new}} \leq w/2 + \varepsilon/4, \qquad w_J \leq 2^{-J} B_0 + \frac{\varepsilon}{2}(1 - 2^{-J}) \leq \varepsilon.$$

Indeed, if the upper bound is retained, it already satisfies $u \leq t + \varepsilon/4$. The retained candidate then satisfies (88), which establishes both the accuracy and the stated test count. □

The result optimizes the convex nuclear-norm problem without a sampling assumption. Recovering an underlying low-rank matrix requires additional hypotheses, and entry repair can increase rank. The procedure comprises several fixed-loss GD runs plus explicit repairs, with exact-real counts.

I The exact log-determinant comparison

The exact barrier $-\mu \log\det X$ gives the same invariant-region argument as Appendix G, with larger admissible steps and a gradient certificate sharper by a factor two. We record only the changes to that analysis. Its gradient requires matrix inversion or linear solves, which motivates the polynomial loss used in the main theorem.

I.1 A general convex objective

Use the density-matrix domain $\mathcal{D}_d$ and inner product of Appendix G. Let ϕ be convex and C^2 on an open convex neighborhood of $\mathcal{D}_d$, with $L > 0$, $M \geq 0$ satisfying

$$\|\nabla\phi(X)\|_{\mathrm{op}} \leq L, \qquad 0 \leq \nabla^2\phi(X)[H,H] \leq M\|H\|_F^2 \quad (X \in \mathcal{D}_d,\ H \in \mathbb{S}^d). \tag{93}$$

Fix a barrier weight $\mu > 0$ and, on the invertible square factors, define

$$X(V) = \frac{VV^\top}{\|V\|_F^2}, \qquad \mathcal{L}_\mu(V) = \phi(X(V)) - \mu \log\det X(V). \tag{94}$$

The normalized Gram readout enforces trace one. Additional affine constraints, if present, enter through ϕ.

Theorem I.1 (Euclidean gradient descent on a square factor)**.** *Under* (93)*, set*

$$T = L + \mu d, \qquad r = \frac{\mu}{4T}, \qquad 0 < \eta \leq \frac{1}{32T + 4M}, \qquad h_{\min} = \frac{\eta}{1 + 4\eta L}. \tag{95}$$

Starting from the scaled-identity initialization $V_0 = I_d/\sqrt{d}$, perform ordinary Euclidean gradient descent with the constant step size η,

$$V_{t+1} = V_t - \eta\nabla\mathcal{L}_\mu(V_t). \tag{96}$$

Every iterate is invertible, and for all $t \geq 0$

$$X(V_t) \succeq rI_d, \qquad 1 \leq \|V_t\|_F^2 \leq 1 + 4\eta L. \tag{97}$$

The matrix objective $\psi(X) = \phi(X) - \mu\log\det X$ has a unique minimizer on the positive definite part of $\mathcal{D}_d$, and if its value is $\mathcal{L}_\mu^*$, then

$$\mathcal{L}_\mu(V_t) - \mathcal{L}_\mu^* \leq 2L(1 - 8h_{\min}\mu)^t. \tag{98}$$

Moreover, with $W_t = V_t/\|V_t\|_F$ and $G_t = \|V_t\|_F\nabla\mathcal{L}_\mu(V_t)$,

$$\phi(X(V_t)) - \min_{X\in\mathcal{D}_d}\phi(X) \leq \mu(d-1) + \frac{\|G_t\|_F^2}{16\mu}. \tag{99}$$

The factor stores d^2 real coordinates, and the update uses the gradient of one fixed loss throughout the trajectory.

The loss is gradient dominated along this trajectory despite its scaling and orthogonal symmetries. Normalized factors below serve only to analyze the Euclidean update.

For an invertible V, put $s = \|V\|_F^2$ for the squared radius, $W = V/\sqrt{s}$ for the normalized factor, $X = WW^\top$, $A = \nabla\phi(X)$, and $\lambda = \langle A, X\rangle$. Direct differentiation gives

$$\nabla\mathcal{L}_\mu(V) = \frac{2}{s}(A - \lambda I_d + \mu d I_d)V - 2\mu V^{-\top}, \tag{100}$$

whose evaluation requires a matrix inverse or linear solves. On the unit Frobenius sphere define

$$G(W) = 2\big[(A - \lambda I_d + \mu d I_d)W - \mu W^{-\top}\big], \tag{101}$$

the gradient of $W \mapsto \psi(WW^\top)$ along that sphere; it is tangent to the sphere and determines the raw gradient by homogeneity,

$$\langle W, G(W)\rangle = 0, \qquad \nabla\mathcal{L}_\mu(V) = G(W)/\sqrt{s}. \tag{102}$$

The trial path retains an eigenvalue floor, now for steps four times as large as in Lemma G.3.

Lemma I.2 (An invariant lower bound on the smallest eigenvalue)**.** *Suppose $\|W\|_F = 1$ and $WW^\top \succeq rI_d$, with T, r as in (95). For every $0 \le h \le 1/(16T)$, the normalized point*

$$W(h) = \frac{W - hG(W)}{\|W - hG(W)\|_F}$$

is well defined and satisfies $W(h)W(h)^\top \succeq rI_d$.

Proof. Here $G = G_\infty$ in the proof of Lemma G.3. Its exact-barrier calculation, through (62), uses only $h \le 1/(16T)$: with $D = I_d - 2h(A - \lambda I_d + \mu d I_d)$, one has $D \succeq (1 - 4hT)I_d \succeq 3I_d/4$ and

$$\|G(W)\|_F \le 8T. \tag{103}$$

Consequently, $Y = W - hG(W)$ satisfies

$$YY^\top - r\|Y\|_F^2 I_d \succeq (2h\mu - 28h^2\mu T)I_d \succeq 0.$$

Tangency gives $\|Y\|_F^2 = 1 + h^2\|G(W)\|_F^2 \ge 1$, so normalization proves the claim. No polynomial perturbation estimate is needed. □

The exact barrier also lowers the ambient Hessian bound, improving the sufficient-decrease step size.

Lemma I.3 (Uniform decrease along normalized trial steps)**.** *Under the hypotheses of Lemma I.2, every $0 \le h \le (32T + 4M)^{-1}$ satisfies*

$$\psi(W(h)W(h)^\top) \le \psi(WW^\top) - \frac{h}{2}\|G(W)\|_F^2. \tag{104}$$

Proof. At a unit factor with $UU^\top \succeq rI_d$, the ambient function $F(U) = \phi(UU^\top) - 2\mu\log|\det U|$ satisfies

$$\|\nabla F(U)\|_F \le 2L + 2\mu\sqrt{d/r} \le 6T, \qquad \|\nabla^2 F(U)\|_{\mathrm{op}} \le 2L + 4M + 2\mu/r \le 10T + 4M.$$

The ϕ bound is the same chain-rule estimate as in Lemma G.5, and the barrier Hessian has norm at most $2\mu\|U^{-1}\|_{\mathrm{op}}^2$. For $g = \|G(W)\|_F$, the normalized-circle calculation in that lemma gives $\|W'(h)\|_F \le g$ and $\|W''(h)\|_F \le 3g^2$. Lemma I.2 keeps the whole path in this region, hence

$$\frac{d^2}{dh^2}F(W(h)) \le (28T + 4M)g^2, \qquad \frac{d}{dh}F(W(h))\Big|_{h=0} = -g^2.$$

Taylor's theorem proves (104) for $h \le (32T + 4M)^{-1}$. □

The gradient inequalities below hold at every invertible factor, so their certificates also apply away from the prescribed trajectory.

Lemma I.4 (Gradient domination and an unregularized certificate)**.** *The objective ψ has a unique minimizer X_* on the positive definite part of $\mathcal{D}_d$. At every invertible unit factor with $X = WW^\top$,*

$$\|G(W)\|_F^2 \geq 16\mu\,[\psi(X) - \psi(X_*)]. \tag{105}$$

At the same point,

$$\lambda - \lambda_{\min}(A) \leq \mu(d-1) + \frac{\|G(W)\|_F^2}{16\mu}. \tag{106}$$

Proof. The barrier diverges at the boundary and ϕ is continuous on the compact domain, so ψ attains its minimum in the interior. There it is unique since

$$\nabla^2(-\mu \log\det X)[H,H] = \mu\,\mathrm{tr}(X^{-1}HX^{-1}H) \geq \mu\|H\|_F^2$$

for density matrices $X \preceq I_d$.

Remove the scalar part of the matrix gradient by setting

$$S = \nabla\psi(X) - \langle \nabla\psi(X), X\rangle I_d.$$

For every positive definite $Y \in \mathcal{D}_d$, convexity of ϕ and Lemma G.6 give

$$\begin{aligned}
\psi(X) - \psi(Y) &\leq \langle S, X - Y\rangle - \mu D_{-\log\det}(Y, X) \\
&\leq \|SX^{1/2}\|_F \|(X-Y)X^{-1/2}\|_F - \mu\|(X-Y)X^{-1/2}\|_F^2 \\
&\leq \frac{\|SX^{1/2}\|_F^2}{4\mu} = \frac{\|G(W)\|_F^2}{16\mu},
\end{aligned}$$

where the second line is the weighted Cauchy–Schwarz inequality and the third maximizes a scalar quadratic; taking $Y = X_*$ proves (105).

For the certificate, the same matrix S satisfies

$$A = (\lambda - \mu d)I_d + \mu X^{-1} + S, \qquad \mathrm{tr}(SX) = 0, \qquad \mathrm{tr}(SXS) = \frac{\|G(W)\|_F^2}{4}. \tag{107}$$

For a unit vector u, put $a = u^\top X^{-1}u \geq 1$. Because $\mathrm{tr}(SX) = 0$ and $\|(uu^\top - X)X^{-1/2}\|_F^2 = a - 1$ by expansion, the weighted Cauchy–Schwarz inequality gives

$$|u^\top S u| = |\langle S, uu^\top - X\rangle| \leq \frac{\|G(W)\|_F}{2}\sqrt{a-1},$$

so that, minimizing over $a \geq 1$,

$$u^\top(\mu X^{-1} + S)u \geq \mu a - \frac{\|G(W)\|_F}{2}\sqrt{a-1} \geq \mu - \frac{\|G(W)\|_F^2}{16\mu}.$$

Substituting into (107) shows that every eigenvalue of A is at least $\lambda - \mu(d-1) - \|G(W)\|_F^2/(16\mu)$, which is (106). Neither estimate used a lower bound on the smallest eigenvalue of X; that invariant is needed only to justify the constant step size. □

Proof of Theorem I.1. The radius argument from Theorem G.1 applies unchanged. In detail, with $s_t = \|V_t\|_F^2$ and $h_t = \eta/s_t$, homogeneity gives

$$W_{t+1} = \frac{W_t - h_t G(W_t)}{\|W_t - h_t G(W_t)\|_F}, \qquad s_{t+1} = s_t + \frac{\eta^2}{s_t}\|G(W_t)\|_F^2. \tag{108}$$

Thus $s_t \geq 1$, and Lemma I.2 propagates the eigenvalue floor from $X_0 = I_d/d$. Lemma I.3 gives

$$s_{t+1} - s_t \leq 2\eta[\psi(X_t) - \psi(X_{t+1})]. \tag{109}$$

The barrier is minimized at I_d/d and the variation of ϕ is at most $2L$, so $\psi(X_0) - \psi(X_*) \leq 2L$. Telescoping yields $s_t \leq 1 + 4\eta L$, hence $h_t \geq h_{\min}$. Combining sufficient decrease with (105) now contracts the gap by $1 - 8\mu h_{\min}$. Finally, convexity bounds the unregularized gap by $\lambda - \lambda_{\min}(A)$, and (106) gives its stated certificate. □

I.2 Affine constraints and a gradient-based certificate

Use the slack-coordinate embedding from Appendix G. For $Z \in \mathbb{S}^n$, $Z \succeq 0$, a justified bound $\operatorname{tr} Z \leq R$ with $R > 0$, and residuals $b_i + \langle C_i, Z\rangle \leq 0$, set $d = n + 1$ and $Z = RX_{1:n,1:n}$. Equalities contribute both signs, and any positive residual rescalings fix the units of accuracy. The embedded residuals are

$$B_i = R\operatorname{diag}(C_i, 0), \qquad a_i(X) = b_i + \langle B_i, X\rangle, \qquad a_0(X) = 0, \tag{110}$$

with $B_0 = 0$ and $b_0 = 0$. Thus

$$v(X) = \max_{0\leq i\leq m} a_i(X), \qquad \gamma = \min_{X\in\mathcal{D}_d} v(X), \tag{111}$$

satisfy $\gamma \geq 0$, with equality exactly when the trace-bounded system is feasible.

Let $q = m + 1 \geq 2$ count the residuals and take any positive bound $L \geq \max_i \|B_i\|_{\mathrm{op}}$; the main statement uses the maximum itself when it is positive, and if all B_i vanish the residuals are constant and feasibility can be checked directly. For a temperature $\tau > 0$ define the log-sum-exp smoothing and its softmax weights,

$$\phi_\tau(X) = \tau \log \sum_{i=0}^{m} \exp(a_i(X)/\tau), \qquad p_i(X) = \frac{\exp(a_i(X)/\tau)}{\sum_{j=0}^{m}\exp(a_j(X)/\tau)}. \tag{112}$$

Its gradient $A = \sum_i p_i B_i$ has operator norm at most L, and

$$\nabla^2\phi_\tau(X)[H,H] = \frac{1}{\tau}\operatorname{Var}_{i\sim p}\langle B_i, H\rangle \leq \frac{dL^2}{\tau}\|H\|_F^2. \tag{113}$$

As in Theorem G.8, a direct factor-Hessian bound avoids the factor d in the step size.

Lemma I.5 (Step size for the log-sum-exp smoothing)**.** *For $\phi = \phi_\tau$, every conclusion of Theorem I.1 remains valid under the step-size bound*

$$0 < \eta \leq \frac{1}{32(L + \mu d) + 4L^2/\tau}. \tag{114}$$

Proof. The covariance calculation in Theorem G.8 gives $\|\nabla^2(\phi_\tau \circ \mathrm{Gram})(U)\|_{\mathrm{op}} \leq 2L + 4L^2/\tau$ at every unit factor. Thus $4M$ is replaced by $4L^2/\tau$ throughout Lemma I.3, the only place where M enters the proof of Theorem I.1. □

Softmax weights and the gradient norm also give a lower bound on the best achievable violation, valid at every invertible factor.

Proposition I.6 (A gradient certificate for bounded affine systems)**.** *Let V be any invertible square factor, for instance an iterate covered by Lemma I.5, and let $X = X(V)$, $W = V/\|V\|_F$, $G = G(W)$, and $A = \sum_i p_i(X)B_i$. Define*

$$\alpha = \langle A, X\rangle - \mu(d-1) - \frac{\|G\|_F^2}{16\mu}, \qquad \underline{d} = \sum_i p_i(X)b_i + \alpha. \tag{115}$$

Then $A - \alpha I_d \succeq 0$ and

$$\underline{d} \le \gamma \le v(X), \qquad v(X) - \underline{d} \le \tau \log q + \mu(d-1) + \frac{\|G\|_F^2}{16\mu}. \tag{116}$$

The norm in this formula can be computed in the raw variables as $\|G\|_F = \|V\|_F\|\nabla\mathcal{L}_\mu(V)\|_F$.

Proof. Lemma I.4 gives $A \succeq \alpha I_d$ at every invertible unit factor, so for every $Y \in \mathcal{D}_d$,

$$v(Y) \ge \sum_i p_i(X)a_i(Y) = \sum_i p_i(X)b_i + \langle A, Y\rangle \ge \underline{d},$$

and minimizing over Y proves the lower certificate. If $H(p) = -\sum_i p_i \log p_i$ denotes the entropy of the weights, the softmax identity gives

$$v(X) \le \phi_\tau(X) = \sum_i p_i(X)b_i + \langle A, X\rangle + \tau H(p), \qquad H(p) \le \log q,$$

and subtracting (115) proves the gap bound; the raw-gradient formula is (102). □

I.3 Finite-time accuracy and arithmetic scope

We allocate one quarter of the tolerance to smoothing, one quarter to the barrier, and one half to the gradient term. Fix $0 < \varepsilon \le L$ and set

$$\mu = \frac{\varepsilon}{4d}, \qquad \tau = \frac{\varepsilon}{4\log q}, \qquad T = L + \mu d, \qquad r = \frac{\mu}{4T}, \qquad h_{\min} = \frac{\eta}{1 + 4\eta L}, \tag{117}$$

with η satisfying (114). The stopping condition

$$\|V_t\|_F^2\|\nabla\mathcal{L}_\mu(V_t)\|_F^2 \le 8\mu\varepsilon \tag{118}$$

makes the right side of (116) at most $\tau\log q + \mu d + \varepsilon/2 = \varepsilon$.

Finite-time accuracy for the exact log-determinant loss.

With $\Delta_t = \psi(X_t) - \psi(X_*)$, applying the trial-decrease bound at step $h_{\min}$ and using (98) gives

$$\|G(W_t)\|_F^2 \le \frac{2\Delta_t}{h_{\min}} \le \frac{4L}{h_{\min}} e^{-8h_{\min}\mu t}.$$

This is at most $8\mu\varepsilon$ for every $t \ge N$, where

$$N = \left\lceil \frac{1}{8h_{\min}\mu} \max\left\{0, \log\frac{L}{2h_{\min}\mu\varepsilon}\right\} \right\rceil. \tag{119}$$

At the largest allowed step, $h_{\min}^{-1} = \eta^{-1} + 4L \le 44L + 16L^2 \log q/\varepsilon$, so

$$N = \widetilde{O}\Big(d(L/\varepsilon)^2 \log q\Big). \tag{120}$$

The suppressed factors are logarithmic in the displayed parameters and d. At stopping, $v(X) - \underline{d} \le \varepsilon$. Hence either the readout is ε-feasible in the chosen residual units, or $v(X) > \varepsilon$ gives $\underline{d} > 0$ and certifies infeasibility. A request with $\varepsilon > L$ can use accuracy L. □

The certificate also has a direct reading in the original matrix variable. Put $\overline{C} = \sum_i p_i C_i$ with $C_0 = 0$. The embedded matrix $A = R\,\mathrm{diag}(\overline{C}, 0)$ has a zero bottom diagonal entry, so $A \succeq \alpha I_d$ forces $\alpha \le 0$, and with $y = -\alpha/R \ge 0$ the certificate becomes

$$\overline{C} + yI_n \succeq 0, \qquad \underline{d} = \sum_i p_i b_i - Ry.$$

Every feasible Z would then satisfy

$$0 \ge \sum_i p_i(b_i + \langle C_i, Z\rangle) \ge \sum_i p_i b_i - y\,\mathrm{tr}\, Z \ge \underline{d},$$

which is why a positive lower certificate proves infeasibility.

Each exact-real step uses $O(md^2 + d^3)$ dense scalar operations and q exponentials, including inversion or linear solves, and stores the d^2 factor entries in addition to the input. The invariant region controls derivatives that diverge near singular factors. Finite-precision stability and transcendental-function costs require separate analysis.

The certificate concerns the system restricted by the supplied trace bound. It requires no strict feasibility: singular feasible sets are approached through approximate affine feasibility. It does not give exact constraint repair, a bit-complexity bound, or bounds on sum-of-squares coefficients.

I.4 A local conditioning limit

The improved rate retains a factor of $d(L/\varepsilon)^2$, and the example below shows that this scale appears in the local conditioning of the exact log-determinant loss. Its conclusion concerns arbitrary nearby initializations at a minimizer, whereas the scaled-identity trajectory of Theorem I.1 has additional symmetry in this example.

Proposition I.7 (Two local curvature scales)**.** *Let $d \ge 4$, $0 < \varepsilon \le L$, $\mu = \varepsilon/(4d)$, and $\tau = \varepsilon/4$. There is a convex log-sum-exp smoothing of two affine functions with coefficient operator norms at most $L/2$ such that the normalized exact log-determinant loss has two positive tangent Hessian eigenvalues whose ratio is*

$$\kappa = 1 + 8a^2 d(L/\varepsilon)^2, \qquad \frac{1}{8} \le a \le \frac{1}{4}.$$

For the linearized ordinary-GD map, the best scalar step has worst contraction factor at least $(\kappa - 1)/(\kappa + 1)$ on these two modes.

Proof. The construction places the minimizer at a diagonal matrix with four large and $d - 4$ small eigenvalues, and then aligns the smoothed pair of affine functions with two of the large eigenvalues so that its covariance stiffens one tangent direction and leaves a neighboring one untouched. Set $\delta = 2\mu/L$, $a = [1 - (d-4)\delta]/4$, and

$$X_* = \mathrm{diag}(a, a, a, a, \delta, \dots, \delta), \qquad C = \mu X_*^{-1} - \frac{\mu}{a} I_d, \qquad B = \frac{L}{2}\,\mathrm{diag}(1, -1, 0, \dots, 0).$$

The assumptions give $1/8 \le a \le 1/4$, and since the first four diagonal entries of C vanish while the remaining ones lie in $[0, L/2]$, both $\|C + B\|_{\mathrm{op}}$ and $\|C - B\|_{\mathrm{op}}$ are at most $L/2$. Define

$$\phi(X) = \langle C, X\rangle + \tau \log(2\cosh(\langle B, X\rangle/\tau)).$$

At X_* the two softmax weights are equal and $\nabla\phi(X_*) = C$, so $\nabla(\phi - \mu \log\det)(X_*) = -(\mu/a)I_d$, and strict convexity makes X_* the unique minimizer over density matrices.

Let $W_* = X_*^{1/2}$. The unit tangent directions

$$H_{\mathrm{slow}} = \frac{1}{\sqrt{2}}\,\mathrm{diag}(0, 0, 1, -1, 0, \dots, 0), \qquad H_{\mathrm{stiff}} = \frac{1}{\sqrt{2}}\,\mathrm{diag}(1, -1, 0, 0, \dots, 0)$$

are Hessian eigenvectors, because the constrained matrix gradient at X_* is scalar and the sphere multiplier therefore cancels the first-order factor curvature. In either direction the matrix perturbation is $D = W_* H^\top + H W_*^\top = 2\sqrt{a}\,H$, to which the log-determinant Hessian contributes $\mu\,\mathrm{tr}(X_*^{-1} D X_*^{-1} D) = 4\mu/a$, while the softmax covariance, which is rank one on the diagonal tangent space, contributes zero in the slow direction and $2aL^2/\tau$ in the stiff one. Hence

$$\lambda_{\mathrm{slow}} = \frac{4\mu}{a}, \qquad \lambda_{\mathrm{stiff}} = \frac{4\mu}{a} + \frac{2aL^2}{\tau}, \qquad \frac{\lambda_{\mathrm{stiff}}}{\lambda_{\mathrm{slow}}} = 1 + \frac{a^2 L^2}{2\mu\tau},$$

and substituting μ and τ gives the stated ratio. A scalar-step linearized update has factors $1 - h\lambda_{\mathrm{slow}}$ and $1 - h\lambda_{\mathrm{stiff}}$ on the two modes, and minimizing their maximum absolute value over h gives $(\kappa - 1)/(\kappa + 1)$; changing the Frobenius radius divides both Hessian eigenvalues by the same scalar and so preserves their ratio. □

The scaled-identity initialization preserves equality of the third and fourth factor coordinates in this example and therefore never excites the slow mode. Proposition I.7 is thus a local conditioning limit for the exact loss rather than a matching iteration lower bound for the scaled-identity trajectory or for the polynomial loss of Corollary 4.2.

I.5 Applications with an explicit trace bound

For a linear objective $h(Z) = \langle C, Z\rangle$, encode a threshold c by $(h(Z) - c)/s_{\mathrm{obj}}$ with a specified $s_{\mathrm{obj}} > 0$. Approximate feasibility then gives $h(Z) \le c + s_{\mathrm{obj}}\varepsilon$. A positive certificate rules out the exact threshold system, whereas an approximately feasible objective value need not bound the exactly constrained optimum.

For the continuous example, consider the Peng–Wei relaxation from (150) with n data points and $1 \le K \le n$ clusters, whose constraints

$$Z \succeq 0, \qquad Z\mathbf{1} = \mathbf{1}, \qquad \mathrm{tr}\, Z = K, \qquad Z_{ij} \ge 0$$

supply the known trace bound $R = K$. Adding the slack coordinate gives dimension $d = n + 1$ and a square factor with $(n + 1)^2$ entries, so this application treats the relaxation of Appendix L.9 through a direct trace representation.

To state one explicit residual normalization, take squared distances $0 \le d_{ij} \le 1$ and put $h(Z) = (2n)^{-1}\sum_{i,j} d_{ij} Z_{ij}$. For an objective threshold c, use both signs of each equality residual and the inequalities

$$\frac{(Z\mathbf{1})_i - 1}{K\sqrt{n}}, \qquad \frac{\mathrm{tr}\, Z - K}{K}, \qquad -\frac{Z_{ij}}{K}\ (i < j), \qquad \frac{h(Z) - c}{K}, \tag{121}$$

where the displayed equality residuals stand for their positive and negative versions and positive semidefiniteness already enforces nonnegativity of the diagonal. After the embedding, every affine coefficient in (121) has operator norm at most one: the row-equation coefficient $(e_i\mathbf{1}^\top + \mathbf{1}e_i^\top)/(2\sqrt{n})$ has norm at most one, the trace coefficient has norm one, an off-diagonal coefficient has norm $1/2$, and the objective coefficient $(d_{ij})/(2n)$ has norm at most $1/2$. One may therefore take $L = 1$ and $q = 2n + 4 + \binom{n}{2}$, the zero residual included.

For $0 < \varepsilon \le 1$, the resulting Euclidean GD trajectory reaches its gradient stopping condition within $\widetilde{O}((n+1)\varepsilon^{-2}\log q)$ steps, and an approximate output obeys

$$|(Z\mathbf{1})_i - 1| \le K\sqrt{n}\,\varepsilon, \quad |\operatorname{tr} Z - K| \le K\varepsilon, \quad Z_{ij} \ge -K\varepsilon, \quad h(Z) \le c + K\varepsilon,$$

while being positive semidefinite with trace at most K by construction; alternatively, a positive lower certificate proves that the exact Peng–Wei threshold system is infeasible. This is a finite gradient-descent guarantee for a semidefinite relaxation in explicit residual units, although producing a feasible clustering from the output still requires a separate rounding argument.

Landscape bounds and applications

J Certificates and negative curvature at fixed search rank

This appendix proves Theorem 4.3(a) and Corollary 4.4, then studies the log-sum-exp loss without the polynomial regularizer of Appendix G, on factors whose search rank k may be smaller than the matrix dimension d. All factor Hessians use the metric induced by the Frobenius inner product. Write

$$\mathcal{D}_d = \{X \in \mathbb{S}^d \mid X \succeq 0,\ \operatorname{tr} X = 1\}, \qquad \mathcal{S}_{d,k} = \{W \in \mathbb{R}^{d\times k} \mid \|W\|_F = 1\}$$

for the set of density matrices, that is, positive semidefinite matrices of unit trace, and for the unit sphere of factors. The tangent space of the sphere at W consists of the matrices U with $\langle W, U\rangle = 0$. For a trace bound $R > 0$, take $d = n + 1$ and read out $Z = R(WW^\top)_{1:n,1:n}$, which is positive semidefinite with trace at most R. Conversely, adding one slack coordinate embeds any $Z \succeq 0$ with $\operatorname{tr} Z \le R$ as the density matrix $\operatorname{diag}(Z/R, 1 - \operatorname{tr} Z/R)$, so the unrestricted Gram domain represents the entire trace ball. At search rank k, the output has rank at most k and the factor stores dk coordinates.

J.1 A Hessian certificate for the convex duality gap

A factor-Hessian lower bound controls the convex duality gap even without stationarity. Let f be convex and C^2 on a neighborhood of $\mathcal{D}_d$, and for the factor objective $g(W) = f(WW^\top)$ put

$$X = WW^\top, \qquad A = \nabla f(X), \qquad \lambda = \langle A, X\rangle, \qquad \Delta(X) = \lambda - \lambda_{\min}(A).$$

The quantity $\Delta(X)$ is the Frank–Wolfe gap of f over $\mathcal{D}_d$ (Jaggi, 2013), and convexity gives

$$0 \le f(X) - \min_{S\in\mathcal{D}_d} f(S) \le \Delta(X). \tag{122}$$

Theorem J.1 (Trace certificate at fixed search rank; Theorem 4.3(a) restated)**.** *Let $k \ge 2$ and $\eta \ge 0$, and suppose that the sphere Hessian satisfies* $\operatorname{Hess} g(W) \succeq -\eta I$ *at a factor $W \in \mathcal{S}_{d,k}$. If, at the current matrix $X = WW^\top$,*

$$\nabla^2 f(X)[H, H] \le M\|H\|_F^2 \quad \textit{for every symmetric } H,$$

then

$$\Delta(X) \le \frac{M}{k-1} + \frac{\eta}{2}, \tag{123}$$

with no first-order condition required. If also $\operatorname{grad} g(W) = 0$*, the denominator $k - 1$ can be replaced by k.*

Proof. Probe the Hessian in rank-one tangent directions built from a bottom eigenvector of A. The sphere gradient and Hessian satisfy

$$\operatorname{grad} g(W) = 2(A - \lambda I)W, \tag{124}$$

$$\operatorname{Hess} g(W)[U, U] = 2\langle A - \lambda I, UU^\top\rangle + \nabla^2 f(X)[WU^\top + UW^\top, WU^\top + UW^\top]; \tag{125}$$

for the second identity, the sphere geodesic with initial velocity U has acceleration $-\|U\|_F^2 W$ at W, and this term produces the shift $-\lambda I$.

Choose a unit bottom eigenvector u of A and a unit vector $z \perp W^\top u$. Then $U = uz^\top$ is a unit tangent vector, since $\langle W, uz^\top \rangle = u^\top W z = 0$, and the same identity gives $u \perp Wz$, so that

$$H = WU^\top + UW^\top = (Wz)u^\top + u(Wz)^\top, \qquad \|H\|_F^2 = 2\|Wz\|^2.$$

Equation (125) becomes

$$\operatorname{Hess} g(W)[U,U] = -2\Delta(X) + \nabla^2 f(X)[H,H].$$

The space $(W^\top u)^\perp \subseteq \mathbb{R}^k$ has dimension $k-1$ if $W^\top u \neq 0$ and k if $W^\top u = 0$, so averaging $\|Wz\|^2$ over an orthonormal basis $\{z_j\}$ of this space gives at most $\|W\|_F^2/(k-1) = 1/(k-1)$ in both cases, because $\sum_j \|Wz_j\|^2 \le \|W\|_F^2$. Some unit z in the space therefore satisfies $\nabla^2 f(X)[H,H] \le 2M/(k-1)$, and the Hessian lower bound at this U reads $-\eta \le -2\Delta + 2M/(k-1)$, which rearranges to (123).

Under stationarity, $(A - \lambda I)W = 0$, and if $\Delta > 0$ then u is an eigenvector of $A - \lambda I$ with the nonzero eigenvalue $-\Delta$, so multiplying by $u^\top$ gives $u^\top W = 0$ and all k right directions are available for the average. The case $\Delta = 0$ needs no argument, since the left side of (123) then vanishes. Finally, convexity gives $f(X) - f(S) \le \langle A, X - S \rangle \le \Delta(X)$ for every $S \in \mathcal{D}_d$, which proves (122). □

The stationary constant M/k is sharp for the Frank–Wolfe gap.

Proposition J.2 (Sharp stationary constant)**.** *For $d > k$ and $M > 0$, let*

$$f(X) = \frac{M}{2}\|X - I_d/d\|_F^2, \qquad W = \frac{1}{\sqrt{k}}\begin{bmatrix} I_k \\ 0 \end{bmatrix}.$$

The factor W is a second-order stationary point of the factor problem, and

$$\Delta(WW^\top) = \frac{M}{k}, \qquad f(WW^\top) - \min_{X \in \mathcal{D}_d} f(X) = \frac{M}{2}\left(\frac{1}{k} - \frac{1}{d}\right).$$

Proof. A rank-at-most-k density matrix has squared Frobenius norm at least $1/k$ by Cauchy–Schwarz on its eigenvalues. The displayed factor attains this bound, hence globally minimizes the smooth factor problem and is second-order stationary. Since $A = M(WW^\top - I_d/d)$, $\lambda = M(1/k - 1/d)$, and $\lambda_{\min}(A) = -M/d$, the claimed gap is M/k; substitution in f gives the value gap. □

J.2 Affine feasibility and an explicit dual witness

We now specialize the certificate to the log-sum-exp smoothing of an affine feasibility system, where the dual witness becomes explicit. Let $m \ge 1$ and $q = m + 1$, consider signed affine residuals $a_i(X) = b_i + \langle B_i, X \rangle$ for $1 \le i \le m$ together with the zero residual $a_0 = 0$ (so $b_0 = 0$ and $B_0 = 0$), and suppose that the coefficient matrices satisfy $\|B_i\|_{\mathrm{op}} \le L$ with $L > 0$. For the trace-bounded formulation above one may take $B_i = R\,\mathrm{diag}(C_i, 0)$ when the original residual is $b_i + \langle C_i, Z \rangle$. With a smoothing temperature $\tau > 0$, define the maximum constraint violation, its minimum value, the smoothed objective, and the softmax weights by

$$\begin{aligned} v(X) &= \max_{0 \le i \le m} a_i(X), & \gamma &= \min_{X \in \mathcal{D}_d} v(X), \\ \Phi_\tau(X) &= \tau \log \sum_{i=0}^{m} e^{a_i(X)/\tau}, & p_i &= \frac{e^{a_i(X)/\tau}}{\sum_{j=0}^{m} e^{a_j(X)/\tau}}. \end{aligned} \tag{126}$$

In this subsection $g(W) = \Phi_\tau(WW^\top)$, its matrix gradient is $A = \sum_i p_i B_i$, and the dual value of a probability vector p is

$$\mathsf{d}(p) = \sum_{i=0}^{m} p_i b_i + \lambda_{\min}(A).$$

For every probability vector p, minimizing the weighted residual $\sum_i p_i a_i$ over $\mathcal{D}_d$ shows that $\mathsf{d}(p) \le \gamma$, and compact convex minimax gives $\gamma = \max_p \mathsf{d}(p)$ without any strict feasibility assumption. Applied to Φ_τ, whose curvature is controlled by the covariance of the coefficient matrices, the trace certificate yields a computable gap between the primal and dual values.

Corollary J.3 (Softmax certificate)**.** *If $k \ge 2$ and the sphere Hessian satisfies* $\operatorname{Hess} g(W) \succeq -\eta I$ *at W, then*

$$\mathsf{d}(p) \le \gamma \le v(X), \qquad v(X) - \mathsf{d}(p) \le \tau \log q + \frac{2L^2}{\tau(k-1)} + \frac{\eta}{2}, \tag{127}$$

and at exact first-order stationarity $k-1$ may be replaced by k.

Proof. Reuse the tangent directions of Theorem J.1, but bound the softmax curvature directly. With the unit bottom eigenvector u of A and the tangent direction $uz^\top$ from that proof, the nonlinear curvature term is

$$\nabla^2 \Phi_\tau(X)[H,H] = \frac{4}{\tau} z^\top Q z, \qquad Q = \sum_i p_i W^\top (B_i - A) u u^\top (B_i - A) W,$$

where Q is positive semidefinite because it is a weighted sum of squares. Its trace satisfies

$$\begin{aligned} \operatorname{tr} Q &= \sum_i p_i \|W^\top B_i u\|^2 - \|W^\top A u\|^2 \\ &\le \sum_i p_i \|B_i\|_{\mathrm{op}}^2 \|W\|_F^2 \le L^2. \end{aligned}$$

Averaging $z^\top Q z$ over an orthonormal basis of $(W^\top u)^\perp$, a space of dimension $k-1$ or k, therefore gives at most $L^2/(k-1)$, and the Hessian lower bound at the best z yields $\Delta \le 2L^2/(\tau(k-1)) + \eta/2$. The stationary improvement to k follows as in the preceding proof. The softmax identity

$$\Phi_\tau(X) = \sum_i p_i a_i(X) + \tau H(p), \qquad H(p) = -\sum_i p_i \log p_i \le \log q,$$

together with $v(X) \le \Phi_\tau(X)$ and $\mathsf{d}(p) = \sum_i p_i b_i + \lambda_{\min}(A)$, gives $v(X) - \mathsf{d}(p) \le \Delta + \tau \log q$, and combining this with the bound on Δ proves (127). □

J.3 Many negative-curvature directions

The same argument counts negative-curvature directions by bounding the trace of the smoothing's positive curvature. Write this trace as

$$\mathcal{B}(W) = \frac{4}{\tau}\left(\sum_i p_i \|B_i W\|_F^2 - \|AW\|_F^2\right) \le \frac{4L^2}{\tau}, \tag{128}$$

and for $a > 0$ let $r(a)$ count the eigenvalues of A that are at most $\lambda - a$, while $\nu_{-\eta}$ counts the eigenvalues of the factor Hessian strictly below $-\eta$.

Theorem J.4 (Spectral multiplicity of negative-curvature directions)**.** *For $0 \le \eta < 2a$, every factor point $W \in \mathcal{S}_{d,k}$ satisfies*

$$\nu_{-\eta} \ge \max\left\{0,\ r(a)k - 1 - \left\lfloor \frac{\mathcal{B}(W)}{2a-\eta} \right\rfloor \right\}, \tag{129}$$

and at an exact first-order stationary point $r(a)k-1$ may be replaced by $r(a)k$.

Proof. A bounded positive-curvature trace can stabilize only finitely many negative directions. On the tangent space, the Hessian quadratic form is

$$\operatorname{Hess} g(W)[U,U] = 2\langle A - \lambda I, UU^\top\rangle + \frac{4}{\tau}\operatorname{Var}_{i\sim p}\langle B_i W, U\rangle, \tag{130}$$

where the variance term defines a positive semidefinite operator whose trace on the ambient factor space equals $\mathcal{B}(W)$. Orthogonal compression to any subspace cannot increase that trace.

Let E be the spectral subspace of A for eigenvalues at most $\lambda - a$. The matrices whose columns lie in E form a space of dimension $r(a)k$, and intersecting it with the tangent space, which is defined by the single linear condition $\langle W, U\rangle = 0$, loses at most one dimension. On this intersection, $\langle A - \lambda I, UU^\top\rangle \le -a\|U\|_F^2$ because each column of U lies in E, so the first term in (130) is at most $-2a\|U\|_F^2$.

A positive semidefinite operator with trace at most $\mathcal{B}(W)$ has at most $\lfloor \mathcal{B}(W)/(2a-\eta)\rfloor$ eigenvalues that are at least $2a-\eta$. The compressed variance operator is therefore strictly below $2a-\eta$ on a subspace of the intersection whose codimension is at most that number, and on that subspace the complete Hessian is strictly below $-\eta\|U\|_F^2$. The min-max principle turns this subspace into the lower bound (129). Under stationarity, $(A-\lambda I)W = 0$ forces $P_E W = 0$, since $A - \lambda I$ is invertible on E with eigenvalues at most $-a < 0$; the entire $r(a)k$-dimensional space is then tangent, which gives the improved count. □

Remark J.5 (General convex objectives). Under the assumptions of Theorem J.1, the same argument on the space $\{uz^\top \mid z \perp W^\top u\}$ gives

$$\nu_{-\eta} \ge \max\left\{0,\ k - 1 - \left\lfloor \frac{2M}{2\Delta - \eta} \right\rfloor \right\}, \qquad 0 \le \eta < 2\Delta.$$

The nonlinear curvature operator restricted to this space is positive semidefinite by convexity, and its trace is at most $2M$ because $\|H\|_F^2 = 2\|Wz\|^2$ and $\sum_j \|Wz_j\|^2 \le 1$ over an orthonormal basis $\{z_j\}$. Stationarity again permits k in place of $k-1$.

Choosing the temperature and the search rank as in Corollary 4.4 turns the count into the alternative stated in the main text.

Corollary J.6 (Negative curvature on half the search rank)**.** *Let $\varepsilon > 0$ and choose*

$$\tau = \frac{\varepsilon}{4\log q}, \qquad k - 1 \ge \frac{32L^2\log q}{\varepsilon^2}. \tag{131}$$

At every factor point, either $v(X) - \mathsf{d}(p) \le \varepsilon$, or at least $\lceil (k-1)/2\rceil$ eigenvalues of its Hessian are strictly below $-\varepsilon/2$.

Proof. Suppose $v(X) - \mathsf{d}(p) > \varepsilon$. Since $\tau \log q = \varepsilon/4$, the entropy inequality $v(X) - \mathsf{d}(p) \le \Delta + \tau\log q$ from the proof of Corollary J.3 gives $\Delta > 3\varepsilon/4$. Apply Theorem J.4 with $a = \Delta$, for which $r(a) \ge 1$ because the bottom eigenvalue of A equals $\lambda - \Delta$, and with $\eta = \varepsilon/2$, so that $2a - \eta > \varepsilon$ and, by (128) and (131),

$$\frac{\mathcal{B}(W)}{2a-\eta} < \frac{16L^2\log q}{\varepsilon^2} \le \frac{k-1}{2}.$$

The count in (129) is then at least $k - 1 - \lfloor (k-1)/2\rfloor = \lceil (k-1)/2\rceil$, which proves the claim. □

Proof of Corollary 4.4. The main theorem is the preceding corollary with the alternative phrased through the Frank–Wolfe gap. The counting argument above uses only $\Delta > 3\varepsilon/4$, so it proves the negative-eigenvalue alternative whenever $\Delta > 3\varepsilon/4$. If instead $\Delta \le 3\varepsilon/4$, weak duality $\mathsf{d}(p) \le \gamma$ and the softmax entropy bound give

$$v(X) - \gamma_* \le v(X) - \mathsf{d}(p) \le \tau \log q + \Delta \le \varepsilon,$$

where γ_* in the main text denotes the same optimal maximum constraint violation as γ in this appendix; this is the first alternative of the theorem. □

J.4 Isotropic perturbations and local gradient descent

Many negative directions improve the alignment of an isotropic perturbation with the unstable subspace. The next two results quantify this advantage for a local gradient-only escape phase.

Lemma J.7 (Perturbation alignment)**.** *Let E_- be an s-dimensional subspace of $\mathbb{R}^D$ with $1 \le s \le D$, and let ξ be uniform on the sphere of radius $r > 0$. Then*

$$\Pr\left\{\|P_{E_-}\xi\| \ge \frac{r}{2}\sqrt{\frac{s}{D}}\right\} \ge 1 - e^{-s/16} - e^{-D/8}. \tag{132}$$

Proof. Both tails are chi-squared tails obtained from the moment-generating function. Write $\xi = rG/\|G\|$ for a standard Gaussian vector G, so that $\|P_{E_-}G\|^2$ and $\|G\|^2$ are chi-squared variables with s and D degrees of freedom. Recall that a chi-squared variable with j degrees of freedom has moment-generating function $(1-2t)^{-j/2}$ for $t < 1/2$. Markov's inequality at $t = -1/2$ gives $\Pr\{\|P_{E_-}G\|^2 < s/2\} \le e^{-s/16}$, and at $t = 1/4$ it gives $\Pr\{\|G\|^2 > 2D\} \le e^{-D/8}$. On the intersection of the two complementary events, $\|P_{E_-}\xi\|^2/r^2 = \|P_{E_-}G\|^2/\|G\|^2 \ge (s/2)/(2D) = s/(4D)$, which is (132). □

The alignment fraction then determines the local exit time.

Theorem J.8 (Local gradient-descent escape)**.** *Let F be C^2 on a neighborhood of the closed ball of radius b in $\mathbb{R}^D$, with an ℓ-Lipschitz gradient and a ρ-Lipschitz Hessian there, where $\ell > 0$ and $\rho \ge 0$. Assume that $\nabla^2 F(0)$ has at least $s \ge 1$ eigenvalues at most $-\gamma$ for some $\gamma > 0$, put $\chi = \sqrt{s/D}$ for the alignment fraction, and suppose*

$$b \le \frac{\gamma\chi}{16\rho}, \qquad \|\nabla F(0)\| \le \frac{\gamma b \chi}{32}, \tag{133}$$

where the first restriction is omitted when $\rho = 0$. Draw x_0 uniformly from the sphere of radius $b/4$ and apply

$$x_{t+1} = x_t - \alpha \nabla F(x_t), \qquad 0 < \alpha \le 1/\ell,$$

until the iterate leaves the open ball of radius b. With probability at least $1 - e^{-s/16} - e^{-D/8}$, the exit occurs after at most

$$1 + \left\lceil \frac{\log(16/\chi)}{\log(1+\alpha\gamma)} \right\rceil \tag{134}$$

gradient steps.

Proof. Track the unstable component, bounding its accumulated linearization error by a geometric series. Let P be the orthogonal projection onto an s-dimensional invariant subspace of $\nabla^2 F(0)$ spanned by eigenvectors with eigenvalues at most $-\gamma$. Put $J = I - \alpha\nabla^2 F(0)$ and denote by J_- its

restriction to this subspace, all of whose eigenvalues are at least $1+\alpha\gamma$. While the previous iterates remain in the ball, Taylor's theorem around the origin gives

$$x_{t+1} = Jx_t + e_t, \qquad \|e_t\| \le \alpha\left(\|\nabla F(0)\| + \frac{\rho}{2}b^2\right),$$

where the error collects the gradient at the origin and the quadratic remainder. Backward iteration on the invariant subspace yields

$$J_-^{-t}Px_t = Px_0 + \sum_{j=0}^{t-1} J_-^{-j-1}Pe_j,$$

and since $\|J_-^{-j-1}\| \le (1+\alpha\gamma)^{-j-1}$, the geometric series and (133) bound the norm of the sum by

$$\frac{\|\nabla F(0)\|}{\gamma} + \frac{\rho b^2}{2\gamma} \le \frac{b\chi}{16}.$$

Lemma J.7, applied with radius $b/4$, gives $\|Px_0\| \ge b\chi/8$ with the asserted probability, and on this event the triangle inequality leaves $\|J_-^{-t}Px_t\| \ge b\chi/16$, so that

$$\|Px_t\| \ge (1+\alpha\gamma)^t \frac{b\chi}{16}.$$

Because $\|Px_t\| \le \|x_t\| < b$ inside the open ball, remaining there until the time in (134) would contradict this growth, so the iterate has exited by that time. □

The step restriction $\alpha \le 1/\ell$ is useful for surrounding descent phases but is not needed for this exit argument. To apply the theorem on the factor sphere, use the normalized tangent retraction

$$\mathcal{R}_W(x) = \frac{W+x}{\sqrt{1+\|x\|_F^2}}, \qquad \langle W, x\rangle = 0, \qquad F_W(x) = g(\mathcal{R}_W(x)),$$

which is a second-order retraction, so that the pullback gradient and Hessian of F_W at zero agree with the intrinsic derivatives of g. Ordinary GD on F_W is then a gradient-only tangent-space phase, and the lemma below supplies the Lipschitz constants that the theorem requires.

Lemma J.9 (Explicit pullback regularity)**.** *For the log-sum-exp smoothing* (126)*, valid gradient and Hessian Lipschitz constants for every pullback F_W are*

$$\ell_{\mathrm{pb}} = 14L + \frac{4L^2}{\tau}, \qquad \rho_{\mathrm{pb}} = 108L + \frac{84L^2}{\tau} + \frac{16L^3}{\tau^2}. \tag{135}$$

Proof. Compose the derivative bounds for softmax and normalization. Write $g_i(W) = a_i(WW^\top)$ for the branch objectives. Their gradients $Dg_i[h] = 2\langle B_iW, h\rangle$ have norm at most $2L$ at a unit factor, their Hessians $h \mapsto 2B_ih$ have operator norm at most $2L$ on $\mathbb{R}^{d\times k}$ with the Frobenius norm, and their third derivatives vanish. At a unit ambient factor, the first three Euclidean derivative norms of $g = \Phi_\tau \circ \mathrm{Gram}$ are therefore bounded by

$$G = 2L, \qquad H = 2L + 4L^2/\tau, \qquad T = 12L^2/\tau + 16L^3/\tau^2,$$

for the following reasons. The gradient of g is the softmax average of the branch gradients. Its Hessian adds to the averaged branch Hessians the covariance of the branch gradients divided by

τ, which is bounded by $(2L)^2/\tau = 4L^2/\tau$. Its third derivative consists of three gradient–Hessian covariance terms, each bounded by $4L^2/\tau$ through Cauchy–Schwarz, and a centered third moment of the branch gradients divided by τ^2. Since $|Dg_i[h]| \le 2L$ for unit h, the centered values are at most $4L$ in absolute value and their variance is at most $4L^2$, so the third centered moment is at most $4L \cdot 4L^2 = 16L^3$, which gives the term $16L^3/\tau^2$. For the normalization map $y \mapsto y/\|y\|_F$, the derivative norms through order three are at most 1, 6, and 36 on inputs of norm at least one, and $\|W + x\|_F \ge 1$ throughout the tangent space because $x \perp W$. The chain rule for the composition then gives $H + 6G$ for the pullback Hessian bound and $T + 18H + 36G$ for the pullback third-derivative bound, which are the constants in (135). □

On the factor sphere, $D = dk - 1$, and Corollary J.6 provides s proportional to k. Theorem J.8 then replaces the local dependence on D by a dependence on D/s, which is of order d, while the alignment failure probability becomes exponentially small in k. This quantifies the advantage that many negative-curvature directions give a gradient-only escape, although it does not remove the matrix-dimension dependence of isotropic noise. The theorem guarantees only exit from a neighborhood, and because a perturbation can increase the objective before the gradient steps begin, a global iteration bound would also need a decrease test and an account of successive perturbations, as in general perturbed-GD analyses (Jin et al., 2017). The local result alone gives no uniform hitting-time theorem for unmodified gradient descent.

K Proofs for overparameterized Gram objectives

We prove the unit-row bound of Theorem 4.3(b), then treat square search rank and analytic descent. All factor derivatives use the ambient Frobenius metric on the product of spheres.

K.1 A representation-specific landscape bound

Recall $\mathcal{E}_N = \{X \succeq 0 : X_{ii} = 1\}$, $\mathcal{M}_{N,k} = (\mathbb{S}^{k-1})^N$, and $F(V) = f(VV^\top)$ for C^2 concave f near $\mathcal{E}_N$. The linear range is

$$R(A) = \max_{Z \in \mathcal{E}_N} \langle A, Z \rangle - \min_{Z \in \mathcal{E}_N} \langle A, Z \rangle.$$

For linear f, the bound is Mei et al. (2017, Theorem 2). We retain the nonlinear curvature in their projected-Gaussian argument (Mei et al., 2017, Section 3.2), producing the additional $C/2$ term.

Fix $V \in \mathcal{M}_{N,k}$ and a comparator $X_* = YY^\top \in \mathcal{E}_N$, with $Y \in \mathbb{R}^{N \times s}$ having rows $y_i^\top$. Draw $G \in \mathbb{R}^{k \times s}$ with independent $\mathcal{N}(0, 1/k)$ entries, set $u_i = (I - v_i v_i^\top) G y_i$, let U have rows $u_i^\top$, and put $H = VU^\top + UV^\top$. The curvature constant $C \ge 0$ in Theorem 4.3(b) satisfies (9), with expectation over G. The constant may depend on V and X_*, while the applications establish bounds uniform over these choices.

K.2 Gaussian tangent moments and the landscape bound

Put $X = VV^\top$ and $P_i = I - v_i v_i^\top$. Since $\mathbb{E}[(Gy_i)(Gy_j)^\top] = (X_*)_{ij} I/k$,

$$\mathbb{E}(UU^\top)_{ij} = (X_*)_{ij}\left(1 - \frac{2}{k} + \frac{X_{ij}^2}{k}\right), \qquad \mathbb{E}\|U\|_F^2 = N(1 - 1/k), \tag{136}$$

$$\mathbb{E}H_{ij}^2 = \frac{2}{k}(1 - X_{ij}^2)(1 - X_{ij}(X_*)_{ij}) \le \frac{4}{k}. \tag{137}$$

The first line follows from $\operatorname{tr}(P_iP_j) = k - 2 + X_{ij}^2$, and for the second we put $a = X_{ij}$ and $b = (X_*)_{ij}$ and note that the two summands $v_i^\top P_j G y_j$ and $v_j^\top P_i G y_i$ of H_{ij} each have variance $(1-a^2)/k$ and covariance $-ab(1-a^2)/k$.

Proof of Theorem 4.3(b). Average the factor Hessian along this probe and use concavity. Write $X = VV^\top$ and $A = \nabla f(X)$, and let Λ be the diagonal matrix with entries $\Lambda_{ii} = (AV)_i \cdot v_i$, so that $\operatorname{tr}\Lambda = \langle A, X\rangle$. Because a sphere geodesic with initial velocity U has row acceleration $-\|u_i\|^2 v_i$, differentiating F twice along it gives

$$\operatorname{Hess} F(V)[U,U] = 2\langle A - \Lambda, UU^\top\rangle + \nabla^2 f(X)[H,H]. \tag{138}$$

Taking expectations over the probe with the moments (136), and noting that the Schur product theorem places $Z = X_* \circ X \circ X$ in the elliptope, we bound the left side by $\eta N(1-1/k)$ and multiply by $k/2$ to obtain

$$(k-2)\langle A, X_*\rangle + \langle A, Z\rangle - (k-1)\langle A, X\rangle \le C/2 + N\eta(k-1)/2,$$

which rearranges to

$$(k-1)\langle A, X_* - X\rangle \le \langle A, X_* - Z\rangle + C/2 + N\eta(k-1)/2.$$

The first term on the right compares two elliptope points and is at most $R(A)$, while concavity of f gives $f(X_*) - f(X) \le \langle A, X_* - X\rangle$, so dividing by $k-1$ yields the bound (10). □

Uniform constants always exist for a fixed N and a fixed C^2 function on a neighborhood of the compact elliptope: if $\|\nabla^2 f(X)\|_{\mathrm{op}} \le M$, then summing the entry bound (137) over all N^2 entries gives $C \le 4MN^2$. That this generic bound grows like N^2 is the reason a dimension-independent rate needs more structure than smoothness alone, and the applications below obtain their constants from the coefficient structure of each objective instead.

K.3 Square search rank and the limits of normalized descent

At square search rank, exact second-order stationarity already certifies optimality.

Proposition K.1 (Exact optimality at square search rank)**.** *Let f be C^2 and concave on a neighborhood of $\mathcal{E}_N$, and let $F(V) = f(VV^\top)$ on $\mathcal{M}_{N,k}$. If $k \ge N$, $\operatorname{grad} F(V) = 0$ and $\operatorname{Hess} F(V) \preceq 0$, then $f(VV^\top) = \max_{Z\in\mathcal{E}_N} f(Z)$.*

Proof. Put $Q = \Lambda - A$ as in (138). Stationarity gives $QV = 0$, because the Riemannian gradient has rows $2((A-\Lambda)V)_i$. If V has full row rank, then $Q = 0$; otherwise $\operatorname{rank} V < N \le k$ supplies a nonzero $z \in \ker V$, and for any $u \in \mathbb{R}^N$ the matrix $U = uz^\top$ is tangent with $H = 0$, so the Hessian condition reads $-2\|z\|^2 u^\top Q u \le 0$. In either case $Q \succeq 0$. For any $Z \in \mathcal{E}_N$, the equal diagonals of Z and X together with $QV = 0$ give $\langle A, Z - X\rangle = -\langle Q, Z\rangle \le 0$, and concavity turns this into $f(Z) \le f(X)$, which is the claim. □

To relate these pointwise certificates to descent, for the loss $L = -F$, write g_i for the sphere-projected gradient of row i and consider the normalized update

$$g_i = (I - v_i v_i^\top)\nabla_{v_i} L(V), \qquad (T_\alpha(V))_i = \frac{v_i - \alpha g_i}{\sqrt{1 + \alpha^2\|g_i\|^2}}, \tag{139}$$

which moves each row against its projected gradient with step α and renormalizes it to the sphere.

Proposition K.2 (Analytic descent limits)**.** *If L is real analytic on $\mathcal{M}_{N,k}$, then for a sufficiently small fixed $\alpha > 0$, every trajectory of (139) converges to a critical point. For any initialization distribution with a density relative to manifold volume, the limiting Hessian is positive semidefinite almost surely.*

Proof. Compactness and a Taylor expansion of the normalization retraction give, for all sufficiently small fixed $\alpha > 0$,

$$L(V) - L(T_\alpha V) \geq \frac{\alpha}{2}\|g(V)\|_F^2, \qquad \|T_\alpha V - V\|_F \leq \alpha\|g(V)\|_F.$$

Extend the loss by $\widetilde{L}(V) = L(\operatorname{rownorm} V)$ to the open set of nonzero rows. It is analytic near the compact trajectory, and $\nabla\widetilde{L} = g$ on the manifold. The displayed estimates give

$$\widetilde{L}(V) - \widetilde{L}(T_\alpha V) \geq \tfrac{1}{2}\|\nabla\widetilde{L}(V)\|_F\|T_\alpha V - V\|_F,$$

with equality of consecutive loss values only at a fixed point. These are the strong-descent conditions of Absil et al. (2005, Definition 3.1 and Theorem 3.2); their local analytic argument at an accumulation point gives convergence. Summing the first estimate shows $g(V_t) \to 0$, so the limit is critical.

Uniformly on the compact manifold, $DT_\alpha = I + O(\alpha)$ as an ambient map on tangent vectors. For small α its differential is injective, hence T_α is a local diffeomorphism. At a critical point, $DT_\alpha = I - \alpha \operatorname{Hess} L$, so a negative Hessian eigenvalue makes that fixed point unstable. Lee et al. (2019, Theorem 2) therefore excludes convergence to such points from a set of positive volume, including when the critical points are nonisolated. □

The proposition supplies a limit statement and no stopping time. Its analyticity hypothesis is stronger than the C^2 assumption of the landscape bound, although the polynomial penalized objectives in Appendix L satisfy it, whereas the positive-part penalty of Appendix L.9 does not.

L Applications and exact feasibility

The finite-GD applications are proved in Appendices H and I. Here we derive the unit-row constants and the required rounding or feasibility repairs. Table 2 collects the constants, and Table 3 identifies the classical results and the precise additions. The labels **R**, **A**, and **E** distinguish recovered guarantees, prescribed-GD realizations, and quantitative extensions of the cited statements.

Table 2: Unit-row applications: Nk stored coordinates and intrinsic dimension $N(k-1)$. The value gap is $(R + C/2)/(k-1) + N\eta/2$ under the appropriate Hessian bound. Here ρ is the penalty weight, B the nonconstant Fourier mass, U_* the cut SDP value, and τ the contrastive temperature. Each objective and its remaining parameters are defined below; neural certificates are proved in Appendix M.

Problem	Rows N / state size	Numerator $R + C/2$	Guarantee before rounding
Normalized MAX-CUT (**R**)	n	U_*	$F \geq (1 - 1/(k-1))U_* - n\eta/2$
Boolean MAX-a-CSP (**E** + **R**)	$\left\lvert \bigcup_C 2^C \right\rvert \leq 1 + m2^a$	$2B + 32\rho 4^a$	Penalized SDP gap and average consistency control
Level-r Boolean Lasserre (**E**)	$N_r = \sum_{j=0}^{r} \binom{n}{j}$	$2B + 32\rho$	Penalized hierarchy gap with mean squared consistency residuals
Lovász theta (**E**)	$n + 1$	$1 + 21\rho/4$	Finite certified interval for $\vartheta(G)/n$ after feasibility repair
Convex pairwise fitting (**E**)	Number of items	$2L_1 + 2L_2$	Average loss within $(2L_1 + 2L_2)/(k-1) + N\eta/2$ of the convex optimum
Contrastive embeddings (**E**)	Number of samples	$4/\tau + 2/\tau^2$	Average training loss within the stated additive bound
K-means SDP (**E**)	$2n$ for n data points	$n + 32\rho$	Penalized SDP gap and average feasibility control
Two-layer Gram training (**E** + **R**)	N samples $h(p+k)$ parameters	As for the Gram loss	Same bound via Corollary 4.5, with full feature rank and nonzero head outputs
Two-layer prediction losses (**R**)	$h(p+1)$ parameters	Separate certificates	Global optimum for convex prediction losses at a stationary output layer with full feature rank, or a local minimum with an unused neuron and matched regularization

Table 3: Classical formulations and the additions proved here. **R**: recovered; **A**: prescribed-GD realization; **E**: quantitative extension of the cited result. These labels concern guarantees, not new optimization models.

Application / status	Closest cited formulation or guarantee	Result here and its scope
Bounded SDP feasibility **A**	Hazan (2008, Theorems 1, 3): trace-ball optimization and smoothed feasibility. Zhang et al. (2023, Proposition 10): factor certificate.	Corollary 4.2: fixed polynomial loss, prescribed GD and a gradient certificate along a uniformly full-rank trajectory.
Nuclear-norm fitting **A**	Vandenberghe and Boyd (1996, Eq. (37)); Fazel et al. (2001, Lemma 1); Jaggi and Sulovský (2010, Lemma 1, Corollary 2, Theorem 3): exact trace lift and certified fitting.	Corollary H.3: prescribed GD for the same convex target, with no planted-matrix recovery claim.
PSD sensing **A**	Jaggi (2013, Theorems 1–2, Section 4.3): trace-ball optimization. Bhojanapalli et al. (2016, Theorem 5): factored GD with suitable initialization and rank.	Corollary H.5: prescribed GD for arbitrary bounded sensing maps. Recovery requires additional measurement assumptions.
Matrix completion **A**	Candès and Recht (2009, Eq. (1.5)): exact-observation convex program. Cai et al. (2010, Theorems 3.1, 4.2): singular-value thresholding.	Corollary H.6: several GD feasibility runs and entry repair give exact observations and an ε norm gap, with no unobserved-entry guarantee.
Peng–Wei K-means SDP **A, E**	Peng and Wei (2007, Section 3.2): the SDP (Eq. (16) in the October 2005 author preprint).	Appendix I: GD threshold certificate. Proposition L.8: $2n$-row penalized lift with numerator $n + 32\rho$. Neither supplies clustering rounding.
MAX-CUT **R**	Mei et al. (2017, Theorem 2, Eq. (8)): Hessian-only rank bound. Goemans and Williamson (1995, Theorem 3.3): hyperplane rounding.	Both guarantees are recovered, including the $n\eta/2$ term.
Boolean CSP / MAX-SAT **E + R**	Yau et al. (2024, Eq. (60), Theorem C.1, Lemma C.3, Corollary 3): penalized moments and robust rounding. Goemans and Williamson (1994, Theorem 5.2): 3/4 rounding.	Exact unit rows at every search rank, numerator $2B + 32\rho 4^a$, defect loss $3\sqrt{D}/2$, and explicit rank in Corollary L.2. The approximation ratio is classical.

Application / status	Closest cited formulation or guarantee	Result here and its scope
Lasserre hierarchy **E**	Lasserre (2001a,b); Laurent (2003, Section 3.3, Eq. (22); Section 6.2.2, Eq. (33)): moment hierarchy and Boolean forms.	Normalized penalty with numerator $2B + 32\rho$ and mean squared residual control. Search rank and hierarchy level remain distinct.
Lovász theta **E**	Lovász (1979, Theorems 4–5); Laurent (2003, Section 6.1, Eq. (27)): theta formulations. Burer and Monteiro (2003, Section 4.1): low-rank optimization.	Proposition L.7: numerator $1 + 21\rho/4$ and explicit repair interval for theta; no independent-set rounding.
Pairwise fitting **E**	Higham (2002a, Eqs. (1.1), (1.3)): correlation fitting. Grubišić and Pietersz (2007, Eqs. (1.1)–(1.2), Section 3): unit-row factors.	Hessian-only gap with numerator $2L_1 + 2L_2$ for the stated smooth convex losses.
Contrastive embeddings **E**	van den Oord et al. (2018, Eq. (4)), Chen et al. (2020, Eq. (1)), Khosla et al. (2020, Eq. (2)): losses. Behnia and Thrampoulidis (2024, Theorem 1): norm-ball result under dimension and temperature assumptions. Vock et al. (2026, Theorem 3.13): exact SupCon geometry for every $\tau > 0$ when the class count is at most the embedding dimension.	Approximate-Hessian gap with numerator $4/\tau + 2/\tau^2$ for arbitrary fixed J_i, q_i and $k \geq 2$. This is not an exact-optimum geometry theorem.
Neural Gram loss **E + R**	Nguyen and Hein (2017, Lemma 3.5): full-feature-rank mechanism. Levin et al. (2025, Proposition 2.12): submersion transfer.	Corollary 4.5: quantitative tolerance $\gamma(\max_i \|z_i\|/\sigma_N(\Psi))^2$, assuming full feature rank and nonzero outputs at the inspected point.
Neural prediction **R**	Nguyen and Hein (2017, Lemma 3.5); Haeffele and Vidal (2015, Theorem 15), later Haeffele and Vidal (2017, Theorem 1).	Full-rank stationary-head and homogeneous regularized unused-neuron certificates. Neither asserts that training reaches its hypotheses.

L.1 MAX-CUT

The simplest instance is the cut objective, for which the nonlinear curvature vanishes. Let $w_{ij} \geq 0$ be edge weights normalized so that $\sum_{i<j} w_{ij} = 1$, and set $f(X) = \frac{1}{2}\sum_{i<j} w_{ij}(1 - X_{ij})$, the normalized cut value of a correlation matrix. Over the elliptope this linear function attains its minimum zero at $X = \mathbf{1}\mathbf{1}^\top$ and its maximum at the cut SDP value U_*, so that $R(\nabla f) = U_*$. Since f is linear, $C = 0$, and the landscape bound recovers Mei et al. (2017, Theorem 2), as recorded in Table 2. Random-hyperplane rounding of any elliptope point X has expected cut value at least $\alpha_{\mathrm{GW}} f(X)$

with the classical $\alpha_{\mathrm{GW}} \simeq 0.87856$ (Goemans and Williamson, 1995), so the additive loss $n\eta/2$ is multiplied by the same factor after rounding.

L.2 Boolean CSP moments

Boolean constraint satisfaction replaces the single cut objective by a weighted predicate score written in clause-local moments, and the curvature now comes from a consistency penalty rather than from the objective. Let $P_C\colon \{-1,+1\}^C \to [0,1]$ be predicates, each on a clause C of at most a variables, with positive weights w_C summing to one. The moment index set $\mathcal{I} = \bigcup_C 2^C$ consists of all subsets of clauses, giving $N = |\mathcal{I}|$ rows, and the moment of a subset T is read from the anchor row as $y_T = X_{\varnothing,T}$. Expanding each predicate in its Fourier series and evaluating it on these moments gives the score and the nonconstant coefficient mass,

$$P_C(x) = \sum_{T\subseteq C} \widehat{P}_C(T) \prod_{i\in T} x_i, \quad S(X) = \sum_C w_C \sum_{T\subseteq C} \widehat{P}_C(T) y_T, \quad B = \sum_C w_C \sum_{\varnothing\neq T\subseteq C} |\widehat{P}_C(T)|,$$

and with $a_0 = \sum_C w_C \widehat{P}_C(\varnothing)$ for the constant term, the score obeys $S(X) \le U := a_0 + B$ on the whole elliptope. Consistency of the clause-local moments is measured over ordered pairs of subsets of each clause. The residuals, their weighted square sum, and the penalized objective with penalty weight ρ are

$$r_{C,A,B} = X_{A,B} - X_{\varnothing,A\triangle B}, \qquad D(X) = \sum_C w_C \sum_{A,B\subseteq C} r^2_{C,A,B}, \qquad f_\rho = S - \rho D,$$

so that $D = 0$ is exact clause moment consistency. For a consistent Gram factor V with rows v_A indexed by $A \in \mathcal{I}$, the quantities

$$\mu_C(x) = \left\| 2^{-|C|} \sum_{A\subseteq C} \left(\prod_{i\in A} x_i \right) v_A \right\|^2, \qquad x \in \{-1,+1\}^C,$$

are nonnegative and sum to one by character orthogonality, so each clause carries a probability law. Consistency makes the moments of that law $\mathbb{E}_{\mu_C} \prod_{i\in T} x_i = y_T$, and the score S is then the expected predicate value under these local laws. Since every Boolean assignment lifts to a consistent factor, $0 \le \mathrm{OPT} \le c_* := \max_{\mathcal{E}_N, D=0} S \le 1$, where c_* is the value of the consistent relaxation.

To apply the landscape bound we need the linear range and the curvature of f_ρ. Each residual has off-diagonal coefficient sum at most two and magnitude at most two, and a clause of arity at most a has at most 4^a ordered pairs of subsets. The gradient of f_ρ therefore has coefficient sum at most $B + 8\rho 4^a$, and $R(\nabla f_\rho) \le 2B + 16\rho 4^a$. For the curvature, the entry bound (137) and the L^2 triangle inequality give $\mathbb{E}(\mathrm{d}r[H])^2 \le 16/k$ for every residual, and since $\nabla^2 f_\rho[H,H] = -2\rho \sum_C w_C \sum_{A,B} (\mathrm{d}r[H])^2$, we may take $C = 32\rho 4^a$. With the resulting defect and the penalized optimum,

$$\Delta = \frac{2B + 32\rho 4^a}{k-1} + \frac{N\eta}{2}, \qquad c_\rho = \max_{X\in\mathcal{E}_N} f_\rho(X),$$

the landscape bound yields

$$0 \le c_\rho - f_\rho(X) \le \Delta, \qquad S(X) \ge c_* - \Delta, \qquad D(X) \le \frac{U+\Delta}{\rho}, \tag{140}$$

where the last inequality uses $c_\rho \ge c_* \ge 0$ and $\rho D = S - f_\rho$. For MAX-SAT, $B \le 1$ and the simpler bound $U = 2$ is valid. When the Hessian bound has been verified at the current point, the same inequalities give the upper certificate $\mathrm{OPT} \le c_* \le f_\rho(X) + \Delta$, even if that point has a nonzero consistency residual.

L.3 Direct rounding with inconsistent moments

The preceding CSP bounds control the weighted sum of squared consistency residuals, so a rounding scheme that needs exact clause laws would first have to repair the moments. This section shows that the classical MAX-SAT marginal rounding of Goemans and Williamson (1994) survives a finite defect without any repair. We use the clause-local MAX-SAT moments and their unnormalized ordered-pair residuals, and for a clause C of size d we write $D_C = \sum_{A,B\subseteq C} r_{C,A,B}^2$ for its sum of squared consistency residuals.

Proposition L.1 (Marginal rounding with inconsistent moments)**.** *Set $p_i = (1+X_{\varnothing,\{i\}})/2$. Independent rounding with probability p_i of setting $x_i = +1$ has expected weighted score at least $\alpha_a(S - \frac{3}{2}\sqrt{D})$, where $\alpha_a = 1-(1-1/a)^a$. Independent rounding with probability $1-4^{-p_i}$ instead has expected score at least $\frac{3}{4}(S - \frac{3}{2}\sqrt{D})$. Both bounds can be derandomized by conditional expectations.*

Proof. Compare the factor marginals with valid clause laws, then apply classical marginal rounding. For every clause define $\mu_C(x) = \|2^{-d}\sum_{A\subseteq C} x_A v_A\|^2$ for $x \in \{-1,+1\}^C$, where $x_A = \prod_{i\in A} x_i$; character orthogonality shows that this is a probability law. Writing $\nu_C(x) = 2^{-d}\sum_{T\subseteq C} x_T y_T$ for the law that exact consistency would produce, expansion gives

$$\mu_C(x) - \nu_C(x) = 2^{-2d} \sum_{A,B\subseteq C} x_{A\triangle B} r_{C,A,B},$$

so Cauchy–Schwarz yields $|\mu_C(x) - \nu_C(x)| \le 2^{-d}\sqrt{D_C}$. The singleton marginals of μ_C satisfy

$$\mu_C(x_i = s) - \frac{1+sy_i}{2} = s2^{-d-1} \sum_{A\triangle B=\{i\}} r_{C,A,B},$$

whose absolute value is at most $2^{-d/2-1}\sqrt{D_C}$ by Cauchy–Schwarz over the 2^d pairs with $A\triangle B = \{i\}$. For the unique violating assignment x^*, the canonical score is $s_C = 1-\nu_C(x^*)$, so $s_C \le 1+2^{-d}\sqrt{D_C}$.

Let p_ℓ be the probability that literal ℓ is true under the untransformed product measure. The law μ_C puts mass $1-\mu_C(x^*)$ on assignments satisfying C, so the union bound under μ_C together with the two discrepancy estimates gives

$$\sum_{\ell\in C} p_\ell \ge s_C - e_C, \qquad e_C = (2^{-d} + d2^{-d/2-1})\sqrt{D_C} \le \sqrt{D_C},$$

where the last scalar inequality follows by checking $d = 1,2,3$ and observing that $d2^{-d/2-1}$ decreases for integer $d \ge 3$. Put $t_C = \min\{1, \max\{0, s_C - e_C\}\}$; then $t_C \le \sum_{\ell\in C} p_\ell$, and $t_C \ge s_C - \frac{3}{2}\sqrt{D_C}$ because clipping at zero can only increase $s_C - e_C$, while clipping at one loses at most $2^{-d}\sqrt{D_C} \le \frac{1}{2}\sqrt{D_C}$.

For the original product measure, the arithmetic–geometric mean inequality gives clause success probability at least $1-(1-t_C/d)^d$, and since this function of t_C is concave on $[0,1]$ and vanishes at zero, it is at least $[1-(1-1/d)^d]t_C \ge \alpha_a t_C$.

For the transformed rounding, a positive literal is false with probability $4^{-p_i} = 4^{-p_\ell}$, whereas a negative literal is false with probability $1-4^{-p_i}$, and for $u = 4^{-p_i} \in [1/4, 1]$,

$$4^{p_i-1} - (1-4^{-p_i}) = \frac{1}{4u} - 1 + u = \frac{(1-2u)^2}{4u} \ge 0,$$

so the failure probability of a negative literal is also at most 4^{-p_ℓ}. Independence then bounds the clause failure probability by $4^{-\sum p_\ell} \le 4^{-t_C}$, and concavity gives $1-4^{-t_C} \ge (3/4)t_C$. Summing either

guarantee over clauses and using $\sum_C w_C\sqrt{D_C} \le \sqrt{\sum_C w_C D_C} = \sqrt{D}$ proves the two expected-score bounds. For the derandomization, expose one variable at a time and select the value with the larger conditional expected score, which never decreases the expectation, so the final assignment attains at least the expected score. □

Combining the defect loss with (140) gives an explicit rank prescription.

Corollary L.2 (Explicit rank for the classical three-quarter factor)**.** *For $0 < \varepsilon \le 1$, choose*

$$\rho = 27\varepsilon^{-2}, \qquad k - 1 \ge 12\varepsilon^{-1} + 5184\, 4^a \varepsilon^{-3}, \qquad \eta \le \frac{\varepsilon}{3N}.$$

Every factor obeying the loss Hessian bound $\mathrm{Hess}(-f_\rho \circ \mathrm{Gram}) \succeq -\eta I$ *rounds to score at least* $\frac{3}{4}(c_* - \varepsilon)$ *in expectation. Conditional expectations yield an assignment with at least that score.*

Proof. The stated choices give $\Delta = (2 + 32\rho 4^a)/(k-1) + N\eta/2 \le \varepsilon/3$, so Equation (140) gives $S \ge c_* - \varepsilon/3$ and $D \le 7\varepsilon^2/81$, and therefore $S - \frac{3}{2}\sqrt{D} \ge c_* - (1/3 + \sqrt{7}/6)\varepsilon \ge c_* - \varepsilon$. Proposition L.1 now gives the expected score and its derandomization. □

Because $c_* \ge \mathrm{OPT}$, the corollary is a classical three-quarter approximation up to the stated additive error, obtained at a finite penalty from a Hessian bound verified at a finite iterate. It does not bound the number of iterations needed to reach that Hessian bound, and since the residual normalization here is clause-local, the same constants must not be transferred unchanged to the normalized Lasserre mean squared consistency residual D_r, defined in the next subsection.

L.4 The complete Boolean Lasserre moment constraints

The clause-local moments used for CSPs and rounding above involve only the subsets of single clauses, and the natural strengthening indexes rows by all small subsets of variables, which is the Lasserre hierarchy. Fix a level $\lceil a/2 \rceil \le r \le n$ and use the index set $\mathcal{I}_r = \{A \subseteq [n] \mid |A| \le r\}$ with $N_r = \sum_{j=0}^{r} \binom{n}{j}$ rows. For each symmetric difference T realized by a pair in $\mathcal{I}_r^2$, choose one representative pair (A_T, B_T) with $A_T \triangle B_T = T$, and measure consistency by the mean squared consistency residual

$$D_r(X) = \frac{1}{N_r^2} \sum_{A,B \in \mathcal{I}_r} (X_{A,B} - X_{A_{A\triangle B}, B_{A\triangle B}})^2. \tag{141}$$

The feasible condition $D_r = 0$ says that $X_{A,B}$ depends only on $A \triangle B$, or equivalently that there are moments y_T for $|T| \le \min(2r, n)$ with $X_{A,B} = y_{A\triangle B}$, and together with $X \succeq 0$ and $y_\varnothing = 1$ this is the Boolean level-r Lasserre moment relaxation (Lasserre, 2001b; Laurent, 2003). For the objective we define $y_T = X_{A_T, B_T}$ through the chosen representative, which is available for every predicate term because $2r \ge a$. When $|T| \le r$ we choose $(A_T, B_T) = (\varnothing, T)$ so as to recover the canonical moment convention, and at exact consistency the objective does not depend on these choices. Boolean lifts still show $c_*^{(r)} \ge \mathrm{OPT} \ge 0$, although for $r < a$ the relaxation can have $c_*^{(r)} > 1$ because it need not supply full clause probability laws; the elliptope bound $S \le a_0 + B$ remains sufficient for the penalty argument.

Because of the normalization in (141), $0 \le D_r \le 4$, and the same coefficient calculation as for the clause-local model yields $R \le 2B + 16\rho$ and $C \le 32\rho$ for $f_\rho = S - \rho D_r$, so that (140) holds with $\Delta_r = (2B + 32\rho)/(k-1) + N_r\eta/2$. Only the mean squared consistency residual is controlled at finite penalty, and a single constraint can require stronger accuracy as N_r grows.

Principal submatrices make the hierarchy values decrease. At $r = n$, the clause-law construction above applies to all variables and gives a probability law reproducing every moment, hence

$$c_*^{(n)} = \mathrm{OPT} \le \cdots \le c_*^{(\lceil a/2 \rceil)}.$$

This exactness is classical (Lasserre, 2001b; Laurent, 2003); it gives no finite running time for reaching the relaxation optimum.

Why all difference classes are necessary. The mean squared consistency residual (141) compares every pair with a representative of its difference class, including classes with $|A \triangle B| > r$ that have no anchor entry $X_{\varnothing,T}$, and the following example shows that the truncated equalities alone would be too weak. For $n = 3$ and $r = 2$, start from the identity matrix indexed by $\varnothing, 1, 2, 3, 12, 13, 23$ and set only $X_{1,23} = X_{23,1} = 1/4$ off the diagonal. Its least eigenvalue is $3/4$, so it belongs to the elliptope, and every equality comparing to $X_{\varnothing,A\triangle B}$ with $|A \triangle B| \le r$ holds, yet the entries $X_{1,23}, X_{2,13}, X_{3,12}$ differ, so those truncated equalities do not define the Lasserre relaxation.

L.5 Penalty limits

The finite-penalty bounds above control aggregate squared consistency residuals, and sending the penalty to infinity recovers the exact relaxation in the limit. For either moment model, fix the instance and suppose that $\rho_j \to \infty$ and $f_{\rho_j}(X_j) \ge c_* - \delta_j$ with $\delta_j \to 0$. Then $D(X_j) \le (U + \delta_j)/\rho_j \to 0$, so every accumulation point is feasible, and since $S(X_j) \ge c_* - \delta_j$, every such point has score c_*. Compactness therefore implies

$$\mathrm{dist}_F(X_j, \mathcal{X}_*) \to 0, \quad S(X_j) \to c_*, \quad f_{\rho_j}(X_j) \to c_*, \quad \rho_j D(X_j) \to 0,$$

where $\mathcal{X}_*$ is the optimal set of the exact relaxation, and the last two limits follow by squeezing $f_{\rho_j}(X_j)$ between $c_* - \delta_j$ and $S(X_j)$. Proposition K.2 and a countable intersection of probability-one events give these limits for separately converged runs with $\rho_j/(k_j - 1) \to 0$, without any independence between the runs, and Proposition K.1 alternatively permits a fixed square search rank $k \ge N$.

L.6 Uniform affine feasibility and refutation

The moment penalties control average residuals. To control every constraint, smooth their maximum and apply the unit-row landscape bound. For $M \ge 1$, consider the affine inequalities

$$a_\ell(X) = b_\ell + \sum_{i<j} c_{\ell,ij} X_{ij} \le 0, \qquad \sum_{i<j} |c_{\ell,ij}| \le L, \qquad 1 \le \ell \le M,$$

where the coefficient bound $L > 0$ bounds every constraint, and where diagonal terms have been absorbed into b_ℓ because $X_{ii} = 1$ on the elliptope. The maximum violation, its minimum over the elliptope, and the log-sum-exp smoothing at temperature $\tau > 0$ are

$$v(X) = \max\{0, a_1(X), \ldots, a_M(X)\}, \qquad \gamma = \min_{X \in \mathcal{E}_N} v(X),$$

$$E_\tau(X) = \tau \log\left(1 + \sum_{\ell=1}^{M} e^{a_\ell(X)/\tau}\right),$$

where compactness guarantees that γ is attained, and the affine system is feasible if and only if $\gamma = 0$.

The first result keeps the temperature free, so that the trade-off between smoothing error and curvature is visible before it is optimized.

Proposition L.3 (Arbitrary smoothing temperature)**.** *For $k \geq 2$, $\eta \geq 0$ and $V \in \mathcal{M}_{N,k}$, suppose* $\mathrm{Hess}(E_\tau \circ \mathrm{Gram})(V) \succeq -\eta I$*. Then*

$$v(VV^\top) \leq \gamma + \frac{2L + 2L^2/\tau}{k-1} + \tau \log(M+1) + \frac{N\eta}{2}. \tag{142}$$

Proof. Apply Theorem 4.3(b) to $f = -E_\tau$. Put $a_0 = 0$ and let $p_\ell = e^{a_\ell(X)/\tau} / \sum_{j=0}^{M} e^{a_j(X)/\tau}$ be the softmax weights. The function $f = -E_\tau$ is analytic and concave on the symmetric matrices. Its gradient coefficients are a negative sub-convex combination of those of the residuals, so their absolute sum is at most L and $R(\nabla f) \leq 2L$.

By the Gaussian entry bound (137) and the L^2 triangle inequality,

$$\mathbb{E}(\mathrm{d}a_\ell[H])^2 \leq \frac{4L^2}{k}.$$

Because the log-sum-exp Hessian is a covariance,

$$-\nabla^2 f(X)[H,H] = \frac{1}{\tau} \operatorname{Var}_{\ell \sim p}(\mathrm{d}a_\ell[H]) \leq \frac{1}{\tau} \sum_{\ell=0}^{M} p_\ell (\mathrm{d}a_\ell[H])^2,$$

and taking expectations gives $-k\mathbb{E}\nabla^2 f(X)[H,H] \leq 4L^2/\tau$. The factor Hessian assumption reads $\mathrm{Hess}(f \circ \mathrm{Gram})(V) \preceq \eta I$ in the sign convention of the theorem, so applying it with a minimizer X_* of v as comparator yields

$$E_\tau(X) - E_\tau(X_*) \leq \frac{2L + 2L^2/\tau}{k-1} + \frac{N\eta}{2}.$$

Finally, the elementary softmax inequalities $v(Z) \leq E_\tau(Z) \leq v(Z) + \tau \log(M+1)$ hold for every Z, and using them at X and at X_* turns the last display into (142). □

Balancing the smoothing and curvature terms gives uniform feasibility.

Corollary L.4 (Uniform feasibility and refutation)**.** *Under the preceding affine-system assumptions, for $k \geq 2$ and $\eta \geq 0$, choose $\tau = L\sqrt{2/((k-1)\log(M+1))}$. If $\mathrm{Hess}(E_\tau \circ \mathrm{Gram})(V) \succeq -\eta I$, then*

$$\begin{aligned} v(VV^\top) &\leq \gamma + B_{k,\eta}, \\ B_{k,\eta} &= \frac{2L}{k-1} + 2\sqrt{2}L\sqrt{\frac{\log(M+1)}{k-1}} + \frac{N\eta}{2}. \end{aligned} \tag{143}$$

Proof of Corollary L.4. The terms of (142) that depend on τ are $2L^2/((k-1)\tau) + \tau \log(M+1)$, which are minimized at $\tau = L\sqrt{2/((k-1)\log(M+1))}$, where their sum equals $2\sqrt{2}L\sqrt{\log(M+1)/(k-1)}$; substituting this value into (142) gives (143). □

The corollary says that a factor whose loss Hessian has no eigenvalue below $-\eta$ violates no constraint by more than $B_{k,\eta}$ beyond the best possible violation γ. When the system is feasible, $\gamma = 0$ and the bound controls every constraint simultaneously, so a search rank $k = O(L/\varepsilon + L^2 \log(M+1)/\varepsilon^2)$ together with $\eta = O(\varepsilon/N)$ gives uniform $O(\varepsilon)$ violation. Read in the other direction, at a point where the Hessian bound has been verified the inequality supplies the lower bound $\gamma \geq v(VV^\top) - B_{k,\eta}$, so that $v(VV^\top) > B_{k,\eta}$ proves infeasibility, whereas positive violation at a candidate point or a small gradient on their own prove nothing.

L.6.1 The SAT specialization

The refutation test applies to the clause-local SAT relaxation once its constraints are written as an affine system with a coefficient bound. Consider a CNF formula whose nonempty clauses contain distinct variables, after repeated literals and tautological clauses have been removed; an empty clause already proves unsatisfiability. Use the moment indices $\mathcal{I} = \bigcup_C 2^C$ with $N = |\mathcal{I}|$ rows, and for a literal on variable i in clause C write $\sigma_{C,i} \in \{-1, +1\}$ for its satisfying sign and $\sigma_{C,T} = \prod_{i \in T} \sigma_{C,i}$ for the sign of a subset. For $X \in \mathcal{E}_N$, set $y_T = X_{\varnothing,T}$ and let

$$s_C(X) = 1 - 2^{-|C|} \sum_{T \subseteq C} (-1)^{|T|} \sigma_{C,T} y_T$$

be the clause score, which at exact consistency is the probability under the clause-local law that C is satisfied. The affine system consists of the two-sided consistency constraints and the requirement that every clause be satisfied,

$$\begin{aligned} \frac{1}{2}(X_{A,B} - X_{\varnothing,A\triangle B}) &\le 0, \\ -\frac{1}{2}(X_{A,B} - X_{\varnothing,A\triangle B}) &\le 0 \quad (A, B \subseteq C), \\ 1 - s_C(X) &\le 0, \end{aligned}$$

whose off-diagonal coefficient sums are at most one, so that $L = 1$ is a valid coefficient bound. For any satisfying assignment $x \in \{-1, +1\}^n$, the Boolean lift $v_A = (\prod_{i \in A} x_i)e$ with a fixed unit vector e has exact moment consistency and $s_C = 1$ for every clause, so satisfiability implies $\gamma = 0$. A verified Hessian bound together with $v(VV^\top) > B_{k,\eta}$ therefore proves unsatisfiability. An unsatisfiable formula can still have a feasible clause-local relaxation, in which case this test cannot refute it, and larger moment relaxations can be handled by the same affine theorem once their complete constraint system and coefficient bound are specified.

L.7 Lovász theta and a finite certified interval

Penalizing the theta-body equations gives an upper certificate; an explicit feasibility repair supplies the lower endpoint. Let $G = ([n], E)$ be a simple graph with $n \ge 2$ vertices and $m = |E| > 0$ edges, and write $\alpha(G)$ for its maximum independent-set size; the edgeless case has $\alpha(G) = \vartheta(G) = n$ and needs no optimization. We use the standard theta-body formulation (Gaar and Rendl, 2020, Section 2.2),

$$\begin{aligned} \vartheta(G) = \max & \sum_{i=1}^{n} \|u_i\|^2 \\ \text{subject to} \quad & \|u_0\| = 1, \quad \langle u_0, u_i \rangle = \|u_i\|^2 \quad (1 \le i \le n), \\ & \langle u_i, u_j \rangle = 0 \quad (ij \in E), \end{aligned} \tag{144}$$

in which the maximum ranges over finite-dimensional vector families whose Gram matrix makes the problem a semidefinite program.

To place this program on the elliptope, let $\mathcal{E}_{n+1} = \{X = X^\top \succeq 0 \mid X_{ii} = 1\}$ have rows and columns indexed by $0, 1, \ldots, n$, where index zero is an optimized unit anchor rather than a fixed vector. For $X \in \mathcal{E}_{n+1}$ we define the normalized score, the edge residuals, and the theta feasible set

$$\begin{aligned} S_G(X) &= \frac{1}{n} \sum_{i=1}^{n} \frac{1 + X_{0i}}{2}, \qquad r_{ij}(X) = \frac{1 + X_{0i} + X_{0j} + X_{ij}}{4} \quad (ij \in E), \\ \mathcal{T}_G &= \{X \in \mathcal{E}_{n+1} \mid r_{ij}(X) = 0 \ (ij \in E)\}. \end{aligned} \tag{145}$$

The lemma below checks that this feasible set reproduces the theta body, so that every later bound on S_G over $\mathcal{T}_G$ is a bound on $\vartheta(G)/n$.

Lemma L.5 (Theta as an affine slice of the elliptope)**.** *The set $\mathcal{T}_G$ is nonempty and compact, and*

$$\theta_* := \max_{X \in \mathcal{T}_G} S_G(X) = \frac{\vartheta(G)}{n}, \qquad \frac{\alpha(G)}{n} \le \theta_* \le 1.$$

Proof. The two formulations are related by the affine change of vectors $u_i = (v_0 + v_i)/2$, which we check in both directions. Factor $X \in \mathcal{T}_G$ as $X = VV^\top$ with unit rows v_i and set $u_0 = v_0$ and $u_i = (v_0 + v_i)/2$; then

$$\langle u_0, u_i \rangle = \|u_i\|^2 = (1 + X_{0i})/2, \qquad \langle u_i, u_j \rangle = r_{ij}(X) = 0 \quad (ij \in E),$$

which is a theta-body representation with score $nS_G(X)$. Conversely, for vectors satisfying (144), put $v_0 = u_0$ and $v_i = 2u_i - u_0$, whose squared norms are $4\|u_i\|^2 - 4\langle u_0, u_i \rangle + 1 = 1$, so that their Gram matrix lies in $\mathcal{T}_G$ and preserves the score; the two optima therefore agree. For an independent set I, taking $v_0 = e$, $v_i = e$ for $i \in I$, and $v_i = -e$ otherwise makes every residual vanish with $nS_G = |I|$, which proves the lower bound, and the empty set shows that $\mathcal{T}_G$ is nonempty. The upper bound follows from $0 \le S_G(X) \le 1$ on the whole elliptope, and $\mathcal{T}_G$ is compact as a closed subset of a compact set. □

Rather than enforce the edge equations, we penalize them. For a penalty weight $\rho > 0$, define the mean squared edge residual, the penalized score, its factor version, and the penalized optimum

$$\begin{aligned} D_G(X) &= \frac{1}{m} \sum_{ij \in E} r_{ij}(X)^2, \qquad f_\rho^G(X) = S_G(X) - \rho D_G(X), \\ F_\rho^G(V) &= f_\rho^G(VV^\top), \qquad c_\rho^G = \max_{X \in \mathcal{E}_{n+1}} f_\rho^G(X). \end{aligned} \tag{146}$$

The function f_ρ^G is concave and polynomial in X, its factor domain is $\mathcal{M}_{n+1,k} = (\mathbb{S}^{k-1})^{n+1}$, and all Hessians below use the Frobenius metric of this manifold.

Proposition L.6 (Landscape bound and upper certificate)**.** *For $k \ge 2$, $\eta \ge 0$ and $V \in \mathcal{M}_{n+1,k}$, suppose* $\operatorname{Hess} F_\rho^G(V) \preceq \eta I$*. Define*

$$\Delta_G = \frac{1 + 21\rho/4}{k - 1} + \frac{(n+1)\eta}{2}.$$

With $X = VV^\top$, one has

$$\begin{aligned} 0 \le c_\rho^G - F_\rho^G(V) \le \Delta_G, \\ S_G(X) \ge \theta_* - \Delta_G, \qquad D_G(X) \le \frac{1 + \Delta_G}{\rho}. \end{aligned} \tag{147}$$

In particular, $\alpha(G)/n \le \vartheta(G)/n \le F_\rho^G(V) + \Delta_G$.

Proof. We compute the two constants of Theorem 4.3(b). The sum of the absolute unordered off-diagonal coefficients of S_G is $1/2$, and each edge residual has coefficient sum $3/4$ with $|r_{ij}(X)| \le 1$ on the elliptope. The gradient of f_ρ^G therefore has coefficient sum at most $1/2 + 3\rho/2$, and its linear range in the theorem is at most $R = 1 + 3\rho$.

For the curvature, (137) and the L^2 triangle inequality give

$$\mathbb{E}(\mathrm{d}r_{ij}[H])^2 \le \frac{4}{k}\left(\frac{3}{4}\right)^2 = \frac{9}{4k},$$

and since

$$\nabla^2 f_\rho^G(X)[H,H] = -\frac{2\rho}{m}\sum_{ij\in E}(\mathrm{d}r_{ij}[H])^2,$$

the curvature constant is $C \le 9\rho/2$. Theorem 4.3(b) applied with a penalized maximizer as comparator gives the first bound in (147), with $R + C/2 \le 1 + 3\rho + 9\rho/4 = 1 + 21\rho/4$. Because feasible points have zero penalty, $c_\rho^G \ge \theta_* \ge 0$, and combining this with $F_\rho^G \le S_G \le 1$ and $\rho D_G = S_G - F_\rho^G$ gives the remaining bounds and, through Lemma L.5, the upper certificate. □

We now repair the same factor to obtain a lower endpoint.

Proposition L.7 (Explicit repair and a finite theta interval)**.** *Let $\mu = 1/[n(n+1)]$ and $\rho > 0$. For every $X \in \mathcal{E}_{n+1}$, one can construct $Z \in \mathcal{T}_G$ satisfying*

$$f_\rho^G(X) - S_G(Z) \le \frac{8m}{\mu^2\rho}. \tag{148}$$

If $X = VV^\top$ also satisfies Proposition L.6, then

$$\begin{aligned} S_G(Z) &\le \frac{\vartheta(G)}{n} \le F_\rho^G(V) + \Delta_G, \\ F_\rho^G(V) + \Delta_G - S_G(Z) &\le \Delta_G + \frac{8m}{\mu^2\rho}. \end{aligned} \tag{149}$$

Proof. The repair subtracts the edge residuals from X, which restores the edge equations but may destroy positive semidefiniteness. That matrix is then mixed with a positive definite feasible matrix W whose smallest eigenvalue μ controls how much mixing is needed. We construct W by mixing the independent-set Boolean lifts used in Lemma L.5, with weight $1/2$ on the empty set and $1/(2n)$ on each singleton, which gives

$$W_{ii} = 1, \qquad W_{0i} = 1/n - 1, \qquad W_{ij} = 1 - 2/n \quad (i \ne j,\ i,j \ge 1), \qquad S_G(W) = 1/(2n),$$

with vanishing residual on every edge. On the vertex subspace orthogonal to the all-ones vector its eigenvalue is $2/n$, and on the orthogonal two-dimensional complement it acts by the matrix

$$\begin{pmatrix} 1 & \sqrt{n}(1/n-1) \\ \sqrt{n}(1/n-1) & n-2+2/n \end{pmatrix},$$

which has positive trace at most $n+1$ and determinant $1/n$. The smaller eigenvalue of a positive definite 2×2 matrix is at least its determinant divided by its trace, so $W \succeq \mu I$.

Now define a symmetric correction matrix E by $\mathsf{E}_{ij} = \mathsf{E}_{ji} = 4r_{ij}(X)$ for $ij \in E$, with all other entries zero, including the anchor row and the diagonal. The matrix $Y = X - \mathsf{E}$ has diagonal one, satisfies every edge equation, and obeys $S_G(Y) = S_G(X)$ because the score reads only the anchor row. Put $d = \|\mathsf{E}\|_F = \sqrt{32mD_G(X)}$; since $X \succeq 0$ and $\|\mathsf{E}\|_{\mathrm{op}} \le d$, we have $Y \succeq -dI$, so that

$$t = d/\mu, \qquad Z = \frac{Y + tW}{1+t}$$

is positive semidefinite while retaining diagonal one and all edge equations, and hence $Z \in \mathcal{T}_G$. Because the score is affine,

$$S_G(X) - S_G(Z) = \frac{t}{1+t}(S_G(X) - S_G(W)) \le t,$$

and substituting the value of t gives

$$f_\rho^G(X) - S_G(Z) \le \frac{\sqrt{32m}}{\mu}\sqrt{D_G(X)} - \rho D_G(X) \le \frac{8m}{\mu^2\rho},$$

where the last step completes the square in $\sqrt{D_G(X)}$; this is (148). Feasibility of Z gives $S_G(Z) \le \vartheta(G)/n$, Proposition L.6 supplies the upper endpoint, and adding Δ_G to (148) proves the interval-width bound. □

To see the interval width as a function of the target accuracy, let $B_G = 8m/\mu^2$ and fix $0 < \varepsilon < 1$; the choices

$$\rho = 2B_G/\varepsilon, \qquad \eta \le \frac{\varepsilon}{2(n+1)}, \qquad k - 1 \ge \frac{4(1 + 21\rho/4)}{\varepsilon}$$

give an interval of width at most ε whenever the Hessian condition has been verified. Since $\mu^{-2} = n^2(n+1)^2$, these choices make the example's search rank $k-1 = \Theta(mn^4/\varepsilon^2)$, so the constants are instance-dependent and large, and the repaired matrix may have full rank. The certificate bounds the theta relaxation and supplies no iteration bound for reaching the Hessian condition, while extracting an independent set requires a separate procedure; low-rank theta optimization itself is classical (Burer and Monteiro, 2003, Section 4.1).

L.8 Continuous optimization examples

Convex correlation losses need nonlinear-curvature control without a consistency penalty. For normalized nonnegative weights w_{ij}, consider the pairwise loss $h(X) = \sum_{i<j} w_{ij}\ell_{ij}(X_{ij})$ with convex C^2 losses obeying $|\ell'_{ij}| \le L_1$ and $0 \le \ell''_{ij} \le L_2$ on $[-1, 1]$, and apply the landscape bound to $f = -h$. The gradient coefficient sum is at most L_1, giving $R \le 2L_1$, and the entry bound (137) gives $-k\mathbb{E}\nabla^2 f[H, H] \le 4L_2$, so the entry in Table 2 holds whenever $\mathrm{Hess}(h \circ \mathrm{Gram}) \succeq -\eta I$. Squared fitting to supplied correlations and logistic pairwise losses are examples, and in each case the guarantee concerns the specified average loss.

For a contrastive loss, fix nonempty comparison sets J_i and probability vectors q_i supported on them, and define the temperature-τ objective

$$h_\tau(X) = \frac{1}{N}\sum_i \left[\log \sum_{j \in J_i} e^{X_{ij}/\tau} - \frac{1}{\tau}\sum_{j \in J_i} q_{ij} X_{ij}\right].$$

Writing p_i for the softmax probabilities of row i, the directed differential coefficients are $(p_{ij} - q_{ij})/(N\tau)$, whose absolute sum is at most $2/\tau$, so that $R \le 4/\tau$. The Hessian is a sum of softmax variances,

$$\nabla^2 h_\tau(X)[H, H] = \frac{1}{N\tau^2}\sum_i \mathrm{Var}_{j \sim p_i}(H_{ij}),$$

and bounding each variance by the second moment gives $C \le 4/\tau^2$, which is the contrastive entry of the table; the resulting guarantee compares in-sample loss with the convex Gram optimum.

L.9 A penalized semidefinite relaxation of K-means

A doubled set of unit rows represents the trace-constrained Peng–Wei relaxation, with the remaining constraints enforced by a penalty. Here K is the number of clusters and k remains the search rank of the Gram factor. Let $x_1, \dots, x_n$ be data points with $n \geq 2$ and $1 \leq K \leq n$, and rescale the squared distances so that $d_{ij} = \|x_i - x_j\|^2 \leq 1$, with $d_{ii} = 0$. The Peng–Wei semidefinite relaxation is

$$h_*^{\mathrm{PW}} = \min_{Z \succeq 0,\ Z\mathbf{1}=\mathbf{1},\ \operatorname{tr} Z = K,\ Z \geq 0} \frac{1}{2n} \sum_{i,j=1}^{n} d_{ij} Z_{ij}, \tag{150}$$

where $Z \geq 0$ is entrywise (Peng and Wei, 2007). For a partition into nonempty clusters, setting Z_{ij} to the reciprocal of the cluster size when i, j share a cluster and to zero otherwise gives a feasible matrix whose objective equals the average squared distance to the cluster means.

An elliptope lift. To represent the relaxation with unit rows, use $2n$ rows, two per data point, and read off the averaged block

$$Z_{ij}(X) = \frac{X_{i,j} + X_{i,n+j} + X_{n+i,j} + X_{n+i,n+j}}{4}, \qquad h(X) = \frac{1}{2n} \sum_{i,j} d_{ij} Z_{ij}(X) \tag{151}$$

for $X \in \mathcal{E}_{2n}$. If the unit rows representing X are $v_1, \dots, v_{2n}$, then $Z(X)$ is the Gram matrix of the midpoints $(v_i + v_{n+i})/2$, so $Z(X) \succeq 0$ and $Z_{ii}(X) \leq 1$. Conversely, let $Z \succeq 0$ have diagonal at most one, choose vectors w_i with Gram matrix Z, and in additional orthogonal coordinates choose mutually orthogonal unit vectors e_i; the vectors

$$v_i = w_i + \sqrt{1 - \|w_i\|^2}\, e_i, \qquad v_{n+i} = w_i - \sqrt{1 - \|w_i\|^2}\, e_i$$

are then unit and reproduce Z through (151). Since every feasible matrix in (150) has diagonal at most one, by nonnegativity and its row sums, the lift represents the entire Peng–Wei feasible set.

The constraints become the affine residuals

$$a_i(X) = \frac{\sum_j Z_{ij}(X) - 1}{n+1}, \quad a_{\mathrm{tr}}(X) = \frac{\operatorname{tr} Z(X) - K}{2n}, \quad b_{ij}(X) = -Z_{ij}(X) \quad (i < j), \tag{152}$$

which encode the row-sum, trace, and sign constraints, and with $M = n + 1 + \binom{n}{2}$ residuals in all and $t_+ = \max\{t, 0\}$ we form the penalty and the penalized objective

$$\mathcal{D}(X) = \frac{1}{M} \left(\sum_i a_i(X)^2 + a_{\mathrm{tr}}(X)^2 + \sum_{i<j} b_{ij}(X)_+^4 \right), \qquad h_\rho(X) = h(X) + \rho \mathcal{D}(X). \tag{153}$$

The penalty is convex and C^3, and its zero set maps onto the feasible set of (150). Let $h_{\rho,*} = \min_{X \in \mathcal{E}_{2n}} h_\rho(X)$ be the penalized optimum.

Proposition L.8 (A K-means relaxation certificate)**.** *Let $\rho > 0$, $k \geq 2$, $\eta \geq 0$, and $V \in \mathcal{M}_{2n,k}$. Suppose*

$$\mathrm{Hess}(h_\rho \circ \mathrm{Gram})(V) \succeq -\eta I, \qquad \delta = \frac{n + 32\rho}{k-1} + n\eta.$$

Then, with $X = VV^\top$,

$$0 \leq h_\rho(X) - h_{\rho,*} \leq \delta, \tag{154}$$

$$h(X) \leq h_*^{\mathrm{PW}} + \delta, \qquad \mathcal{D}(X) \leq \frac{n/2 + \delta}{\rho}, \tag{155}$$

$$h_\rho(X) - \delta \leq h_*^{\mathrm{PW}}. \tag{156}$$

The factor has $2nk$ stored coordinates and intrinsic dimension $2n(k-1)$.

Proof. Count coefficients and compare the penalized optimum with the feasible Peng–Wei value. Written as a constant plus unordered off-diagonal coordinates of X, each affine residual has absolute coefficient sum at most one, the values being $(n-1/2)/(n+1)$ for a_i, $1/4$ for a_{tr}, and one for b_{ij}. Every residual also has magnitude at most one on $\mathcal{E}_{2n}$, because $|Z_{ij}| \le 1$, $0 \le \operatorname{tr} Z \le n$, and $|\sum_j Z_{ij} - 1| \le n$.

For $|t| \le 1$, the first two derivatives of t^2 have magnitudes at most 2 and 2, while those of t_+^4 have magnitudes at most 4 and 12, so the off-diagonal coefficient sum of $\nabla\mathcal{D}$ is at most four. The coefficient sum of h is at most

$$L_h = \frac{1}{n}\sum_{i<j} d_{ij} \le \frac{n-1}{2},$$

and the gradient range therefore satisfies $R(\nabla f) \le 2L_h + 8\rho \le n - 1 + 8\rho$. For the Gaussian tangent probe of Theorem 4.3(b), each affine residual a obeys $\mathbb{E}(da[H])^2 \le 4/k$ by its coefficient sum, so that

$$-k\mathbb{E}\nabla^2 f(X)[H,H] = k\rho\mathbb{E}\nabla^2\mathcal{D}(X)[H,H] \le 48\rho,$$

and the landscape bound gives (154), even with $n - 1 + 32\rho$ in place of $n + 32\rho$.

Since every feasible Peng–Wei matrix admits a lift with zero penalty, $h_{\rho,*} \le h_*^{\mathrm{PW}}$, which proves the objective bound and (156). For the residual bound, note that $h(X) \ge -(n-1)/2$ on the elliptope. Any partition into K nonempty clusters has objective at most $(n-K)/(2n) \le 1/2$, because its within-cluster distances are at most one, and hence

$$\rho\mathcal{D}(X) = h_\rho(X) - h(X) \le h_*^{\mathrm{PW}} + \delta - h(X) \le n/2 + \delta,$$

which is the second half of (155). □

The proposition certifies the penalized relaxation at a point that may still violate the Peng–Wei constraints. Equation (156) lower-bounds the optimal clustering cost, whereas an upper bound on that cost needs a feasible partition, and neither Lloyd's algorithm nor a rounding of the penalized output is analyzed here. Because the positive-part penalty is C^3 and semialgebraic rather than analytic, Proposition K.2 does not apply to it directly, and the certificate rests on the displayed Hessian condition alone.

M Neural parameterizations and their hypotheses

We prove the normalized-head transfer in Section 4.3, give a sufficient condition for feature rank to persist, and record two classical prediction-loss certificates.

M.1 The normalized-head transfer

For inputs $x_i \in \mathbb{R}^p$, two-layer features $\psi_i = (\sigma(w_j^\top x_i))_{j=1}^h$, and head $B \in \mathbb{R}^{k\times h}$, the parameter count is $h(p+k)$. The transfer requires $\operatorname{rank}\Psi = N$ and $B\psi_i \neq 0$ at the inspected point.

Proof of Corollary 4.5. Full feature rank lets the head realize every sphere-tangent direction. For $U \in T_V\mathcal{M}_{N,k}$, let $Z_U = [\|z_i\|u_i]_i$ be the tangent direction rescaled row by row and put $\delta B = Z_U\Psi^+$; since full column rank gives $\Psi^+\Psi = I$, the perturbed head satisfies $\delta B\psi_i = \|z_i\|u_i \perp z_i$. Along the straight head perturbation $B + t\,\delta B$, row i of the normalized factor becomes

$$v_i(t) = \frac{v_i + tu_i}{\sqrt{1+t^2\|u_i\|^2}},$$

whose first two derivatives at $t=0$ are u_i and $-\|u_i\|^2 v_i$ and agree with those of the sphere geodesic, so that

$$\nabla_B^2[-F(V(B))][\delta B, \delta B] = -\operatorname{Hess} F(V)[U,U].$$

Since $\|\Psi^+\|_{\mathrm{op}} = 1/\sigma_N(\Psi)$, the perturbation obeys $\|\delta B\|_F \le \max_i \|z_i\|\|U\|_F/\sigma_N(\Psi)$, and inserting this into the head Hessian bound gives the stated tolerance η. The first-derivative chain rule with the same arbitrary U shows that an exact stationary parameter point gives an exact stationary factor point, which completes the proof. □

Because only head directions are used, the backbone may be trained jointly, although the theorem remains a statement at the inspected parameter point. Its full-rank hypothesis fails whenever two feature columns coincide, and width alone does not repair this. An anonymous message-passing encoder with identical initial node features produces equal features on the cells of the coarsest equitable partition at every width. Distinct inputs do not suffice either, since a bias-free tanh backbone maps the inputs x and $-x$ to feature columns that are negatives of each other.

Some rows may also be optimized freely on their spheres while the others come from the encoder. The joint Hessian bound over head and free-row directions then gives the defect

$$\frac{N_{\mathrm{enc}}\gamma}{2}\left(\frac{\max_i \|z_i\|}{\sigma_{N_{\mathrm{enc}}}(\Psi)}\right)^2 + \frac{N_{\mathrm{free}}\gamma}{2},$$

in which only the encoder rows enter the full-rank feature condition. A bound on the separate diagonal Hessian blocks would be insufficient when mixed curvature is present.

M.2 A sufficient finite-horizon condition

A balanced pair of linear layers can preserve feature rank for a conditional finite horizon. Suppose that the features factor as $\Psi = B_1\Phi(\theta)$ through an inner linear layer $B_1 \in \mathbb{R}^{d\times d'}$ with $d \ge d' \ge N$ applied to backbone features $\Phi(\theta) \in \mathbb{R}^{d'\times N}$, and that a head B_2, playing the role of the head B of Corollary 4.5, acts on Ψ. Assume that the loss depends on B_2, B_1 only through the product B_2B_1

and that all outputs remain nonzero on the trajectory. Under joint Euclidean gradient flow the balance matrix

$$K = B_1 B_1^\top - B_2^\top B_2$$

is constant (Du et al., 2018, Theorem 2.2), because writing $Q = \partial \ell / \partial (B_2 B_1)$ gives $\nabla_{B_2} \ell = Q B_1^\top$ and $\nabla_{B_1} \ell = B_2^\top Q$, so that the derivatives of $B_1 B_1^\top$ and $B_2^\top B_2$ coincide. If $\lambda_{d'}(K(0)) = \lambda_0 > 0$, then the identity $B_1 B_1^\top = K + B_2^\top B_2 \succeq K$ and Weyl monotonicity give $\sigma_{d'}(B_1(t)) \geq \sqrt{\lambda_0}$ for all t.

Under plain gradient descent the balance matrix is no longer conserved. For a step α satisfying the descent inequality $\ell_s - \ell_{s+1} \geq (\alpha/2)\|\nabla \ell_s\|^2$, direct expansion gives

$$K_{s+1} - K_s = \alpha^2 (\nabla_{B_1} \ell_s \nabla_{B_1} \ell_s^\top - \nabla_{B_2} \ell_s^\top \nabla_{B_2} \ell_s),$$

so if $\ell_0 - \inf \ell \leq D_0$, then summing the descent inequality and using the same Weyl step yields $\sigma_{d'}(B_{1,s})^2 \geq \lambda_0 - 2\alpha D_0$. For an L_Φ-Lipschitz backbone and $T = \alpha s$, Cauchy–Schwarz and the same descent inequality bound the drift of the backbone features by

$$\sigma_N(\Phi_s) \geq \sigma_N(\Phi_0) - L_\Phi \sqrt{2TD_0},$$

and therefore, if $\sigma_N(\Phi_0) > 0$, $2\alpha D_0 < \lambda_0$, and $L_\Phi \sqrt{2TD_0} \leq \sigma_N(\Phi_0)/2$,

$$\sigma_N(\Psi_s) \geq \tfrac{1}{2} \sqrt{\lambda_0 - 2\alpha D_0}\, \sigma_N(\Phi_0) > 0.$$

This is a conditional horizon guarantee for plain gradient descent. It keeps the feature matrix well conditioned but does not ensure that the required Hessian tolerance is reached within the horizon, and a perturbed method would need additional control of the perturbations in both recursions.

M.3 Unnormalized convex losses

A scalar-output two-layer network has $h(p+1)$ parameters, with biases absorbed into the inputs if needed. At full feature rank, its linear head can realize every perturbation in prediction space. Let $Z = W\Psi$ be the prediction vector and let $h(Z)$ be differentiable and convex. If Ψ has full column rank, every output perturbation A is realized by the head perturbation $\delta W = A\Psi^+$, so that

$$\|\nabla_Z h(Z)\|_F \leq \frac{\|\nabla_W (h \circ Z)\|_F}{\sigma_N(\Psi)},$$

and a stationary head is therefore a global minimizer in prediction space. For a minimizer Z_*, convexity gives the quantitative version

$$h(Z) - h(Z_*) \leq \frac{\|\nabla_W (h \circ Z)\|_F \|Z - Z_*\|_F}{\sigma_N(\Psi)},$$

which is the familiar full-rank wide-layer mechanism of Nguyen and Hein (2017).

M.4 A classical certificate for regularized two-layer networks

The second classical certificate replaces the full-rank condition by positive homogeneity and a matched regularizer. Let $\phi(w) = (\sigma(w^\top x_i))_{i=1}^N$ be the response of one hidden unit on the training set, where σ is continuous and positively homogeneous of degree one, such as ReLU, and for a differentiable convex loss $h\colon \mathbb{R}^N \to \mathbb{R}$ and a regularization weight $\lambda > 0$ consider

$$J(a, w) = h\left(\sum_{j=1}^m a_j \phi(w_j)\right) + \frac{\lambda}{2} \sum_{j=1}^m (a_j^2 + \|w_j\|^2).$$

The proposition below specializes the homogeneous-network certificate of Haeffele and Vidal (2017) to this loss.

Proposition M.1 (Unused-neuron certificate)**.** *If a local minimum of J contains a neuron with $a_j = 0, w_j = 0$, then its value is globally optimal among networks of every finite width.*

Proof. Put $z = \sum_j a_j \phi(w_j)$ and $g = \nabla h(z)$. Perturbing the unused neuron to $(a, w) = (\pm t, tu)$ for unit u gives $|\langle g, \phi(u)\rangle| \le \lambda$. Rescaling each existing (a_j, w_j) by $\sqrt{1+s}$ gives $\langle g, z\rangle + R = 0$, where $R = \frac{\lambda}{2}\sum_j (a_j^2 + \|w_j\|^2)$. For any finite-width competitor with output $\widetilde{z}$ and regularizer $\widetilde{R}$, homogeneity and $2|a|\|w\| \le a^2 + \|w\|^2$ give $\langle g, \widetilde{z}\rangle \ge -\widetilde{R}$. Convexity then yields

$$h(\widetilde{z}) + \widetilde{R} \ge h(z) + \langle g, \widetilde{z} - z\rangle + \widetilde{R} \ge h(z) - \langle g, z\rangle = h(z) + R.$$

□

The certificate depends on the unused neuron, on positive homogeneity, and on the matched regularizer, and it is separate from the C^2 Gram theorem, since its ReLU form need not have a classical Hessian at the unused neuron.